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Aligning Aviation Safety Reporting System Anomaly Codes for Crew Communication and Coordination with Human Factors Taxonomies

Jennifer Rose Herr

Dissertation Submitted to the College of Aviation in Partial Fulfillment of the Requirements for the Degree of Doctor of Philosophy in Aviation

Embry-Riddle Aeronautical University

Daytona Beach, Florida

August 2025

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Aligning Aviation Safety Reporting System Anomaly Codes for Crew Communication and Coordination with Human Factors Taxonomies

By

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This dissertation was prepared under the direction of the candidate's Dissertation Committee Chair, Alan J. Stolzer, Ph.D. , and has been approved by the members of the dissertation committee. It was submitted to the College of Aviation and was accepted in partial fulfillment of the requirements for the Degree of Doctor of Philosophy in Aviation.

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Abstract

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This study addresses the lack of theoretical grounding and definitions for the Aviation Safety Reporting System (ASRS) anomaly codes, particularly those related to pilot communication and coordination errors. To improve the conceptual consistency and analytical usability of ASRS data, the research developed a framework that maps ASRS codes to three established theory-based human factors taxonomies: the Human Factors Analysis and Classification System (HFACS), Threat and Error Management (TEM), and Aviation Causal Contributors for Error Reporting Systems (ACCERS). The study used qualitative text mining to systematically code communication and coordination errors submitted by Part 121 pilots for a stratified sample of 365 ASRS reports from 2007 to 2022. Additional analysis examined trends related to changes in pilot training and safety programs, while demographic data ensured sample representativeness.

The resulting taxonomy framework revealed both overlapping and distinct emphases in how the taxonomies classify events, with each taxonomy offering unique strengths, TEM emphasizing error detail, ACCERS capturing influencing factors, HFACS focusing on human error, and ASRS documenting event types. Despite these differences, substantial alignment was found, with approximately 85% of interactively coded taxonomy results matching those derived through the framework, supporting its validity for cross-database analysis. Trends also showed

increases in taxonomy coding rates and communication-related codes following the introduction of key safety initiatives like AQP and SMS, suggesting some influence on reporting practices.

This framework contributes a structured, theory-informed method for aligning safety data across disparate taxonomies, enhancing the accuracy, efficiency, and depth of aviation safety analysis. It provides practical tools for analysts conducting cross-database searches and offers a foundation for developing a standardized industry taxonomy. The study recommends further expansion to include more error types and taxonomies, ultimately supporting future automation and machine learning applications for predictive safety analysis.

Keywords: communication and coordination, taxonomy framework, qualitative research, error classification, aviation safety, pilot reporting, cross-database analysis, ASRS

Dedication

To my husband,

Your belief in me never wavered,

even when mine did.

I am forever grateful.

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Completing this dissertation has been one of the most challenging and rewarding journeys of my life, and I could not have reached this milestone without the support, guidance, and encouragement of many remarkable people.

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Chapter I: Introduction

ASRS Background

On December 1, 1974, Trans World Airlines (TWA) Flight 514 crashed near Washington Dulles International Airport due to a breakdown in communication with Air Traffic Control, resulting in a controlled flight into terrain. During the investigation of this crash, it was discovered that a similar miscommunication had happened a few weeks prior with a United Airlines flight (National Transportation Safety Board [NTSB], 1975). United Airlines had an anonymous safety awareness program that allowed the information about this miscommunication to be reported and shared internally with United Airlines flight crews. But the information went no further because there were no systems yet available to share this type of information between airlines. The establishment of the Aviation Safety Reporting System (ASRS) in 1976 marked a significant advancement in the systematic collection and analysis of safety-related data within the aviation industry (NASA Aviation Safety Reporting System, 2001). Designed to facilitate the voluntary reporting of safety concerns, ASRS enabled aviation stakeholders—including pilots, air traffic controllers, dispatchers, and cabin crew—to confidentially share information about unsafe events and hazardous conditions. In addition to promoting industry-wide information sharing, the system provided regulatory bodies with a valuable tool for identifying recurring hazards and informing safety improvements. Since its inception, ASRS has accumulated more than 1.7 million reports, offering a rich repository of data to support ongoing efforts in hazard identification, risk mitigation, and the enhancement of overall aviation safety (NASA, 2021).

The ASRS database contains narratives of crew-reported unsafe occurrences and hazardous situations. The ASRS database is governed and run by NASA, which acts as a neutral party and, therefore, has no authority to enforce or penalize individuals submitting reports. This

process allows for sensitive safety information to be shared without fear of reprisal (ASRS, 2021a).

All ASRS reports are submitted voluntarily by a crew member or aviation professional involved in the event. While reporting programs such as this one offer invaluable information that might not otherwise be shared for fear of consequences (ASRS, 2001), there are concerns that the reports may be submitted by people without a full understanding of the event, thereby offering inaccurate or incomplete information. Additionally, the human factors of the user interface can impact reporting rate and quality (Gong, 2011). There are also concerns that less serious events are not reported through these programs. (Drupsteen & Guldenmund, 2014). According to the FAA, voluntarily submitted safety data has several limitations: uncertainty regarding its completeness; potential issues with quality control, and subjectivity of the data. Reports are often based on individual perceptions, which can lead to inconsistent or inaccurate event descriptions because different pilots may interpret and report the same incident in varying ways (Government Accountability Office [GAO], 2010). However, voluntary reports are still important as a confidential way to share knowledge and thought processes, which can provide critical information to help prevent future incidents (ASRS, 2001). Additionally, safety programs that rely on these voluntary reports, such as the Aviation Safety Action Program (ASAP) and ASRS, offer unique insights that are often unattainable through other data collection methods. These programs contribute to a more comprehensive understanding of the underlying causes of safety events, not merely the outcomes (GAO, 2010). ASAP, as defined in AC120-66B, is a program similar to ASRS, safety incidents or errors are reported voluntarily with no threat of punitive action. Because ASAP data is gathered voluntarily, with safety as the primary objective, its data would otherwise be unknown or unreported. These reports can be used to develop

corrective actions for identified safety concerns and to educate pilots and flight staff to prevent reoccurrences (FAA, 2023).

Upon submission, each report is reviewed and coded by ASRS analysts, who apply standardized classifications based on the nature of the event and the contributing factors involved. These analysts bring diverse backgrounds in aviation operations, which support a nuanced and informed interpretation of the reported incidents (ASRS, 2021a). This process ensures the reliability and utility of the ASRS database as a key resource for data-driven safety research and hazard identification.

Since the ASRS database's inception, its reports have been instrumental in analyzing safety trends and identifying contributing factors behind aviation events. Researchers have used this data to investigate issues such as automation-related errors, alarm mismanagement, disruptive passenger behavior, inflight safety concerns, pilot performance and decision-making, and numerous other operational challenges (Bell, 2022; Bliss et al., 1999; Taylor et al., 2020).

Structuring the ASRS database, and similar repositories, in a way that facilitates efficient and comprehensive analysis is essential for uncovering critical insights into aviation safety. A well-organized taxonomy and coding system enables researchers, safety analysts, and other users to effectively navigate the database, ensuring relevant reports can be identified and examined with accuracy and completeness.

One key element used by ASRS analysts is the anomaly code, which categorizes the nature of the reported event into a distinct type, such as operational deviations, procedural discrepancies, adverse weather encounters, traffic conflicts, aircraft malfunctions, or issues related to Air Traffic Control (ATC) (ASRS, 2021a). There is disagreement about these codes, however, because they are not tied to a human factors taxonomy, and they were initially

developed by subject matter experts without input from pilots, who primarily report flight operation-related events (Boesser, 2020; Stolzer et al., 2023).

While there are other contributors to the ASRS reports, this research focused on U.S. commercial and regional airline pilots, referred to as Part 121 pilots based on the set of regulations they follow. While pilots are not responsible for assigning ASRS codes, research indicates that pilots are more consistent in this type of categorizing than human factors experts, so it is important to have their input (Boesser, 2020). Similarly, studies suggest that pilots were not significantly involved in the development of the codes, so the codes may not accurately represent reported information or capture the significance of that information (Boesser, 2020; Stolzer et al., 2023). For example, the codes may only have broad categories where subcategories would be beneficial. Additionally, studies from Boesser (2020) and Stolzer, et al. (2023) note that it is difficult to determine how the codes were developed and that codes are not grounded in human factors theory. Beaubien & Baker (2002) and Tanguy et al. (2016) also note that detailed human factors information on the event and contributing factors is stored almost entirely within the text narrative. Much of the prior research explores how the narrative sections of these reports can be analyzed to simplify the extraction of the human factors information, as this is not always coded in the ASRS anomaly code (Boesser, 2020; Rose et al., 2020; Tanguy et al., 2016; Zhang, 2012). The ASRS database includes a separate human factors classification system, distinct from the anomaly codes, in which narratives may be assigned one or more of 12 categories. However, the current classification system is limited, as it may not fully capture all human factors described in the reports. Also, the human factors categories lack precise definitions and are not consistently grounded in an established theoretical framework, hindering the depth and consistency of analysis (ASRS, 2021a).

Since the development of the ASRS database, the airline industry has undergone many changes, including training methods, new safety programs and systems and regulations, many of which include new ways to think and talk about safety. Two of the most notable changes were the introduction of AQP in 2005 and the requirement to use Safety Management Systems (SMS) in 2018 (FAA, 2020a; Rosenkrans, 2011). AQP, a training program, allows airlines to adjust the frequency and method of delivering training requirements (FAA, 2017). Data collected during training and during line events are analyzed to find trends or commonalities in errors (FAA, 2017). SMS is comprised of four major focuses: safety policy, safety risk management, safety promotion, and safety assurance (FAA, 2020a).

Taxonomy Overview

During this time, a shift also occurred in both the types of data used and their applications in safety analysis. In a 2009 survey, the FAA found that FOQA data offered a more robust source of safety information and advocated for its increased use, noting more accurate rates of safety occurrences and the capturing of multiple operational parameters in each event, enabling more precise and data-driven assessments (GAO, 2010). While new training methods and safety systems ushered in new processes and applications for analyzing trends and reports, each uses a different taxonomy. Some of these include: the Human Factors Analysis and Classification System (HFACS), Threat and Error Management Model (TEM), and Aviation Causal Contributors for Event Reporting System (ACCERS) (Baker & Krokos, 2007; Brennan et al., 2020; Wiegmann & Shappell, 2016). Although all three of these systems have been used for some time, they have changed and expanded over the years. Airlines have also adapted these classification systems for their own purposes. By modeling how different taxonomies are applied to the ASRS reports, changes can be made to the ASRS anomaly codes to create a more effective

analysis of the ASRS database and simpler integration with internal airline databases. As Boesser (2020) states, it is important to remember that these data are being collected to make a meaningful change. While finding better ways to tabulate and classify the data is important, making the data more meaningful is the ideal goal.

The focus of this research was on the HFACS, TEM, and ACCERS taxonomies and their alignment to the ASRS anomaly codes. TEM and HFACS were selected because of their strong foundations with Reason's model for human error (Brennan et al., 2020, Wiegmann & Shappell, 2016). They are widely used within the aviation industry and have detailed development. Reason's four-layer model includes organizational influences, unsafe supervision, preconditions for unsafe acts, and unsafe acts (Reason, 2000). These are almost directly translated into the HFACS model.

TEM gives a broader picture of events and looks at the threats and errors that can lead to an undesired aircraft state.

ACCERS was included as it is very comparable to the ASRS anomaly codes. ASRS and ASAP have very similar reporting methods and analysis goals. Over 60% of ASAP reports are included in the ASRS database, so the coding should be relatively similar in structure (GAO, 2010). ACCERS was specifically developed for ASAP reports to identify and categorize event errors and causal factors.

These three taxonomies were selected because they have strong theoretical foundations, are widely used and accepted in the aviation industry, and are publicly available. Some airlines might not use these taxonomies or might have adapted these taxonomies to their own needs, so this research might not be as relevant to them. However, this research could be expanded to include additional taxonomies and event types.

Building on this framework, it is also critical to examine how pilots acquire and internalize this knowledge along with other emerging concepts. A comprehensive understanding of these learning processes is essential, as they can significantly influence pilots' responses to operational events and shape the way they communicate their experiences in safety reports. Educational and cognitive learning models offer valuable insights into how individuals assimilate new information and procedures, developing declarative knowledge and understanding rules, concepts, and protocols. This declarative foundation supports the progression to more advanced skill acquisition and informed decision-making within the complex and dynamic aviation environment (Taylor, 2005). After pilots are taught new safety concepts and trainings, it is likely they will change what they report on and how they report safety events. As such, it is important to consider changes in reporting that might be a result of changes in training.

An additional consideration in this analysis is the fact that airline training and safety programs have changed over time; thus, pilots have been exposed to and trained to use multiple taxonomies and classification systems. As such, they will likely use different systems to write ASAP and ASRS reports. Moreover, pilots and airlines apply variable taxonomies when characterizing events, which can result in multiple descriptions for the same incident or an increased emphasis on particular types of events. For instance, events that receive focused attention during training are more likely to be recognized by pilots, increasing the likelihood of reporting. When organizations distribute information, differences in terminology or interpretation may lead to misrepresented findings or inaccurate reports of the frequency of occurrences. Nevertheless, given that the ASRS database aggregates reports from various airlines

and this study employs a representative random sample spanning more than 15 years with a narrowly defined scope, the impact of these potential inconsistencies is expected to be minimal.

Anomaly Codes

There is very little information on the creation of the anomaly codes, how they are assigned to the reports, and their definitions. A variety of studies have sought to develop methods for analyzing the text narratives, showing that these anomaly codes do not fully capture or categorize the contents of the reports (Boesser, 2020; Rose et al., 2020; Tanguy et al., 2016; Zhang, 2012). This lack of information causes analysts to make their own judgments about using and interpreting these codes. These judgements can result in misinterpreted data or decisions made from incorrect information. Additionally, if the reports are not analyzed uniformly within or among airlines, aviation employees can focus on the incorrect areas to correct safety concerns.

Given the absence of documentation about anomaly code development, definitions, and theoretical foundations, NASA ASRS was contacted directly to determine if there was any information not publicly available that might help in this research. From this communication it was established that the anomaly codes were part of the ASRS database from its start; and that many of the founders and creators of the codes were no longer with the program to offer first-hand accounts to their original thoughts and theories. However, communication with the program manager for the ASRS contract with the NASA Ames Research Center provided some background information about the development of the codes—they were not based on a different taxonomy, but were created organically during the program's development in the mid to late 1970s. Analysts reading and analyzing the reports then refined the codes during the program's first year. Documentation of the development of the ASRS program provides an example of this process. At the program's start, it was not apparent that altitude deviations were a significant

reporting category, so there was no code for this anomaly. However, based on the first year reports, that anomaly code was added (Reynard et al., 1986). Since that first year, there have been only two code changes in the past 28 years. When a code is changed, all the old fields are remapped with the new codes so there is consistency in coding and data retrieval. The most recent change was in 2021 with the introduction of codes surrounding Unmanned Aircraft System (UAS) related activities and operators who began reporting to the ASRS program (E. Marfise, personal communication, December 14, 2023).

Because of the absence of clear definitions or an established theoretical framework for these anomaly codes, analysts must exercise individual judgment when interpreting and applying the codes. Although it is known that the codes were developed organically based on emergent patterns in early reports, and later refined through analyst input, the lack of formal documentation or publicly available guidance limits consistency in their application. Even if theoretical considerations initially informed the code development—a possibility that remains speculative, this foundational knowledge is no longer accessible to current analysts. As a result, the interpretation and classification of events may vary depending on the analyst's understanding, experience, and contextual judgment. This lack of standardized reference material introduces an element of subjectivity into the coding process, potentially influencing both the reliability and comparability of data across time and between different users of the ASRS database.

Statement of the Problem

The problem addressed in this research is the current absence of a structured framework that aligns ASRS anomaly codes related to communication and coordination with established, theory-based human factors taxonomies. The lack of such a framework limits researchers' ability

to compare ASRS data to other safety databases, constrains cross-system learning, and reduces the effectiveness of human factors interventions targeting pilot error.

By developing a framework that maps ASRS anomaly codes to key human factors models, specifically HFACS, TEM, and ACCERS, this study aims to enhance conceptual clarity, coding consistency, and cross-database compatibility of pilot error data. This theoretical and operational foundation enables a more precise analysis of crew communication and coordination, among the most frequent and preventable contributors to aviation incidents and accidents.

The current ASRS anomaly codes lack theoretical grounding, particularly in categories related to pilot communication and coordination errors. Moreover, the absence of clear definitions limits the consistency and precision of error identification, coding, and analysis. This gap diminishes the interpretability of reported data and the effectiveness of subsequent safety interventions. Due to the extensive scope of the ASRS database, this study focuses specifically on communication and coordination, given their critical role in the causal chain of aviation incidents.

Prior literature has highlighted several concerns with the ASRS anomaly codes: that they were developed primarily by subject matter experts rather than from pilots or other reporters. This disconnect can result in misinterpretation or omission of certain scenarios, leading to improper classifications. Also, research indicates that pilots demonstrate greater consistency than human factors experts in categorizing ASRS reports, underscoring the importance of involving pilots in code development (Boesser, 2020).

Additional critiques of the codes include the failure to differentiate between event types and contributing factors, as well as the lack of theoretical foundations and definitions (Stolzer et al., 2023). Without clear definitions or a theoretical basis, there may be inconsistent

interpretations of terms during coding and analysis, undermining data reliability and safety outcomes.

Focusing on pilot communication and coordination is essential, as these factors consistently contribute to adverse outcomes in aviation safety (Helmreich et al., 1999; Wiegmann & Shappell, 2016). Improving the theoretical clarity and coding accuracy directly supports the advancement of safety interventions and risk mitigation strategies.

Purpose Statement

The purpose of this study was to develop a framework that systematically maps each anomaly code associated with pilot error—specifically those related to communication and coordination—to terms in established taxonomies. This framework integrates ASRS anomaly codes with these well-recognized taxonomies, enabling researchers to link ASRS reports with other safety databases and to enhance the clarity and definitions of the anomaly codes. This study focused on three selected human factors-based taxonomies: HFACS, TEM, and ACCERS and their relation to the ASRS anomaly codes. To limit the scope of the study, ASRS reports include only those where a breakdown of crew coordination and communication was a primary factor. By establishing a framework that associates taxonomy items to anomaly codes, the strict definitions that are used as the foundations for the taxonomies can be applied to the anomaly codes.

Communication and coordination continue to be leading contributors to aviation accidents and incidents. Human error accounts for the majority of mishaps in commercial aviation, and within that category, miscommunication and poor coordination among crew members are especially common and preventable (Wiegmann & Shappell, 2016). By targeting these errors, the study addressed a high-risk area with significant potential for improvement.

Additionally, communication and coordination are central to crew resource management (CRM), making them ideal focal points for aligning ASRS reports with taxonomies that emphasize human performance and organizational factors.

The objective of this study was to identify gaps within the existing ASRS coding system and propose potential enhancements. The framework was validated using additional ASRS reports to confirm its accuracy and alignment with other taxonomy terminology.

Significance of the Study

Developing a framework to align ASRS anomaly codes with established human factors taxonomies addressed several criticisms associated with the current ASRS coding structure. By mapping anomaly codes to taxonomies grounded in theoretical foundations, researchers can better justify the application and interpretation of these codes within incident reports. Establishing these connections also facilitates the integration of ASRS data with other datasets that utilize HFACS, TEM, or ACCERS, thereby enabling more comprehensive and quantifiable analyses. It is important to note, however, that this study is limited in scope to incidents and accidents in which crew communication and coordination were primary contributing factors, and only includes three taxonomies. As such, the generalizability of the findings will be confined to these parameters.

This study contributes to aviation knowledge by improving the usability and theoretical validity of ASRS data, grounding the ASRS anomaly codes in established theory, something the anomaly codes currently lack. In doing this, this study establishes a stronger conceptual foundation, enhancing the codes' interpretability and relevance for research and safety analysis. The framework enhances clarity, supports more consistent interpretation of pilot error data and enables cross-dataset comparisons. In doing so, the study highlights gaps in current coding

practices, revealing that existing ASRS classifications are limited in scope and also lack a cohesive theoretical framework. By identifying these deficiencies, the research underscores the need for a more robust and theory-driven approach to anomaly classification. This contribution advances the theoretical foundation of aviation safety data analysis, demonstrating how integrating established human factors taxonomies can systematically refine and expand ASRS classifications. As a result, the study's framework bridges the gap between empirical data coding and human factors theory, enhancing both the conceptual rigor and the practical utility of ASRS data for future research and safety interventions. Ultimately, these improvements can inform more effective pilot training, crew resource management strategies, and system design to support safer airline operations.

Airlines that utilize TEM, HFACS, or ACCERS to classify pilot-related errors in their internal safety databases may benefit from this model, as it aligns ASRS reports and anomaly codes with their preferred taxonomy framework. Allowing analysts to use one taxonomy for both internal databases and the ASRS database ensures data are aligned correctly, and relevant reports are accounted for. This study also determined how well-tested and theory-based taxonomy and classification systems sorted these same reports.

Another significant contribution of this study was the revealing of issues or contributing factors previously unaddressed by ASRS anomaly codes. By mapping these codes to established human factors taxonomies, the study exposed gaps and limitations within the existing ASRS classification system and demonstrated how these shortcomings were addressed within other taxonomies. Given that not all elements, or nodes, within the three selected taxonomies corresponded directly to existing ASRS anomaly codes, this analysis identified areas where the

ASRS coding structure could be expanded or refined to achieve greater comprehensiveness and analytical utility.

Focusing specifically on communication and coordination was critical due to their prevalence and impact on aviation safety outcomes. These types of errors frequently contribute to the causal chain in accidents and incidents, often stemming from breakdowns in crew resource management, situational awareness, and decision-making processes (Helmreich et al., 1999; Wiegmann & Shappell, 2016). By concentrating on communication and coordination, the study addressed an area where improvements in taxonomy alignment and reporting can have tangible safety benefits. Communication and coordination are complex human factors inherently difficult to capture consistently without robust theoretical grounding. Aligning ASRS anomaly codes with established taxonomies ensured that these errors were analyzed with greater precision and context. This focus not only enhances the validity of pilot error data but also supports the development of targeted interventions such as enhanced training programs and CRM initiatives, ultimately contributing to safer flight operations.

This framework also considered whether training or new programs influences how TEM, HFACS, and ACCERS are used in pilot reporting. If there was a significant change, this information could be used to set a basis for future changes and the impacts they might have on pilot reporting.

Research Questions

RQ_1

What alignments exist between the selected taxonomies (HFACS, TEM, and ACCERS) and the ASRS anomaly codes related to pilot communication and coordination errors?

RQ2

How do the results of each taxonomy coded through the created framework relate to the anomaly codes given by the ASRS analysts?

RQ3

From 2007 to 2022, what are the changes in reporting of TEM, HFACS, and ACCERS elements in ASRS pilot reports?

Delimitations

This study focused on the past 15 years of ASRS reports. This allowed for the inclusion of several significant safety programs: the Advanced Qualification Program (AQP), Fatigue Risk Management System (FRMS), and Safety Management System (SMS). AQP was codified as 14 CFR Part 121 Subpart Y in 2005; Delta Air Lines was the first to apply for the training program in 2006 for their inflight crews (Rosenkrans, 2011). The FRMS was first introduced in AC120-103 in 2010 with an update in 2013 (FAA, 2013). And SMS became mandatory for airlines to have in place by 2018 (FAA, 2020a). Additionally, Leadership and Command training was recently introduced in 2020 and will be required for both Pilot in Command (PIC) and Second in Command (SIC) pilots (FAA, 2020b).

While other databases and reports, such as ASAP reports, contain similar data to the ASRS database, these are held within airlines and are proprietary, so they were excluded. Also, since this study addresses industry-wide practices, it was necessary to collect comparable data across multiple airlines to mitigate the potential influence of training programs unique to any carrier.

The study focused on only Part 121 pilot reports. While other work groups might report similar errors or issues as pilots—such as ATC, dispatch, mechanics, or flight attendants—these

groups have their own safety programs and, thus, their own way of writing. As this study is focused on the airline pilot group and the language used by pilots, including reports from other working groups would add confusion and irrelevant findings to this study.

The study focused exclusively on ASRS reports that documented breakdown of crew communication and coordination. For the purposes of this research, these incidents or accidents were defined as events arising from pilot miscommunication or non-communication, deficiencies in crew coordination, or failures in effective team management. These errors can be related to communication errors within the cockpit or with ATC and other external resources; the lack of a coordinated plan; or the lack of defined roles, for example. Incidents excluded from this study include those primarily attributable to mechanical failures, external factors such as adverse weather or actions by third-party personnel, as well as events resulting solely from time pressure, stress, or fatigue, without a clear component of crew communication or coordination failure. Adding this delimitation ensures that the reports used in the study are ones where pilots were significant influences on the event rather than just reacting to a situation, such as a mechanical failure. Because the pilots are significantly involved in events with a communication and coordination breakdown, their reports offer detailed information, background and secondary factors.

Limitations and Assumptions

This study is limited by the information contained within ASRS reports. Pilots are human, so their reports vary, with some including much more detail than others. This limits the study because sparsely detailed reports may not be able to be classified fully, based on the taxonomies examined. These reports will be classified with only the information given and no assumptions will be made about the scenario that led to the event. The data examined here is

based on the narrative provided, without factoring in pilot age, experience, seat, or recency of training.

It was assumed that the ASRS reports are subjective rather than based on hard data like FOQA or flight data monitoring. It was also assumed that the pilots making the reports are from Part 121 carriers who are professional pilots. These pilots were assumed to be current, qualified, and maintained the reporting standards expected by their airline and the FAA. There was no consideration given to age, gender, race, or pilot background, as these are not identifiable from the ASRS database.

As these are professional pilots who go through annual training with, it was assumed that pilots change their reporting practices after new training and safety programs, as required by their airline. This assumption relates to how learners take in information and make changes based on it. Learning models show how an individual processes information and adopts new theories (Illeris, 2004). The process of assimilating new information and effecting behavioral change can require considerable time (Illeris, 2004). Furthermore, pilot training within an airline is typically staggered, with intervals of up to one year before the entire pilot cadre completes training programs (FAA, 2017). Consequently, observable changes in incident reporting may not emerge until a majority of pilots have undergone these trainings, which introduce new safety initiatives and concepts. Therefore, it was reasonable to assume that the impact of training modifications on reporting behaviors may take up to one year to become evident.

When ASRS reports are received, two analysts independently study and classify them by event type, yet no information is given on the agreement of these analysts, nor is there much detail on how they are classified (Beaubien & Baker, 2002). It is assumed that the classification

given by the ASRS analyst is an accurate reflection of the report. As such, the codes given by the ASRS analyst were not further validated in this research.

Definitions of Terms

Air Carrier	A person or organization who starts directly by lease, or other arrangement, to engage in air transportation (Definitions and Abbreviations, 1994).
Aircraft Accident	An occurrence where a person suffers serious injury or death or the aircraft receives substantial damage, during the period of time from when a person boards the aircraft with the intention of flight until all persons have disembarked (Definitions, 2022).
FAR Part 121	Regulations governing regularly scheduled air carriers, consisting of both passenger and cargo carriers (FAA, 2021).
Flight Crew Member	A pilot assigned to duty in an aircraft during flight time (Definitions and Abbreviations, 1994).
Incident	An occurrence related to the operation of an aircraft, which does not meet the criteria for an accident, but still impacts the safety of the operations (Definitions, 2022).
Pilot Error	Errors in judgment, decision-making, perception, or action by the flight crew that contribute to undesired events (ICAO, 2017)
Taxonomy	A system of classification (Jackson & Groves, 2015).

Text Mining	A process of extracting implicit knowledge from textual data (Zhang, 2012).
Violation	Willful disregard of known regulations and procedures (Wiegmann & Shappell, 2016).

List of Acronyms

ACCERS	Aviation Causal Contributors for Error Reporting Systems
AQP	Advanced Qualification Program
ASAP	Aviation Safety Action Program
ASRS	Aviation Safety Reporting System
ATC	Air Traffic Control
BASIS	British Airways Safety Information System
CRM	Crew Resource Management
FAA	Federal Aviation Administration
FAR	Federal Aviation Regulation
FRMS	Fatigue Risk Management System
HFACS	Human Factors Analysis and Classification System
ICAO	International Civil Aviation Organization
LOSA	Line Operation Safety Audit
MCDU	Multi-Function Control and Display Unit
NASA	National Air and Space Administration
PF	Pilot Flying
PIC	Pilot in Command

PM	Pilot Monitoring
PNF	Pilot Not Flying
SIC	Second in Command
SMS	Safety Management Systems
SRM	Safety Risk Management
TEM	Threat and Error Management

Chapter II: Review of the Relevant Literature

The ASRS database contains aviation safety data collected from frontline personnel, such as pilots, flight attendants, dispatchers, ATC, mechanics, etc. Each report contains both structured data such as time, location, and other identifying information, along with a text narrative of the event. These reports offer an excellent source for trending recurrent events and analyzing causes. However, much of the information about the type of event and the cause is contained within the narrative section (Zhang, 2012). Quickly sorting through these reports is critical to efficiently analyzing them.

Because these reports are narrative, text queries can extract event type and causal information. However, key word searches can be cumbersome and miss relevant reports with alternate phrasing or misspellings (Zhang, 2012). To address these issues, ASRS analysts code each report with an anomaly code, to help sort through relevant reports (ASRS, 2021a). However, the anomaly codes can be challenging to apply, some are overly broad and there is a notable lack of coding related to human factors. In addition to these anomaly codes, additional coded fields indicate the primary problem and contributing factors. These do include a generic code labeled *human factor*; however, there is no further breakdown within this code. Another element assigned to each report, separate from the anomaly code, contains 12 different coding options for human factors. However, these human factors codes are associated with the *person* category of coding rather than the event anomaly code (ASRS, 2021b).

Although these anomaly codes contain limited coding related to human factors, existing literature suggests that their effectiveness and validity could be enhanced by grounding them in an established human factors taxonomy or another robust theoretical framework, which is not done currently (Stolzer et al., 2023). There are no available definitions for the codes used

(ASRS, 2021; Stolzer et al., 2023). Conversely, when the ACCERS taxonomy was developed to classify ASAP reports, taxonomy items were rigorously pulled from previous human factors research and analyzed for their inclusion, so there were clear definitions of each term and details on why they were included. The ACCERS taxonomy was also tested and validated by pilots who would be using it (Baker & Krokos, 2007). Even with these strict definitions, there was only a 71% agreement on category and just above 50% agreement at a factor level (Baker & Krokos, 2007). HFACS was tested in a similar manner and resulted in a higher factor-level agreement than ACCERS but, again, there are strict definitions for the factors included in HFACS, and they can be traced to Reason's Swiss Cheese model for errors. This model is a basis for many error-based taxonomies (Baker & Krokos, 2007; Reason, 2000).

ACCERS is similar to ASRS anomaly codes in that it categorizes reports submitted through the voluntary reporting systems of ASAP and ASRS. These programs function similarly but have different scopes. ASRS contains reports submitted from multiple airlines and aviation professionals, whereas ASAP is internal to an airline. However, ACCERS was developed to capture both the error and the contributing factors, distinguished by their parent categories (Baker & Krokos, 2007). The ASRS anomaly codes, however, lack this distinction between event type and contributing factor, creating a challenge during analysis (Stolzer et al., 2023). Similarly, the ACCERS taxonomy was tested and validated by pilots trained in its use. Pilots' familiarity with the taxonomy enables accurate classifications, ensuring the most pertinent factors are systematically identified and captured. (Baker & Krokos, 2007). Conversely, the ASRS anomaly codes were developed by subject matter experts without pilot input, which could cause the primary factors of the report to be misidentified (ASRS, 2021). Boesser (2020) determined that during the classification of the ASRS reports, pilots more consistently

categorized their reports than human factors experts. This indicates there may be inconsistencies with the classification of ASRS reports using ASRS anomaly codes.

Much of the current literature involving ASRS report analysis is focused on the narrative section of the reports. Studies highlight the need for analyzing the narrative section because of limited functionality of ASRS coding, including the anomaly codes (Rose et al., 2020; Tanguy et al., 2016; Zhang, 2012). Other studies focus on whether anomaly codes in these reports are correct and repeatable (Boesser, 2020).

Existing literature suggests that the ASRS anomaly code system is insufficient for fully capturing the complexity of pilot communication and coordination errors. Revisions to the system should be based on empirical research that identifies its gaps and limitations. Using established, well-documented, and theoretically grounded taxonomies and classification systems to analyze ASRS reports could address these deficiencies and help refine the coding framework.

There is a lack of research connecting the ACCERS, HFACS, and TEM taxonomies in a way that enables consistent cross-database analysis. Each of these taxonomies serves a distinct purpose: HFACS focuses on identifying human error across multiple layers of organizational and operational contexts (Wiegmann & Shappell, 2016); TEM emphasizes proactive threat and error management in real-time operations (CASA, 2021); and ACCERS focuses on capturing causal contributors and errors reported in aviation events, providing a detailed classification system for understanding human factors (Baker & Krokos, 2007). However, these frameworks are often applied in isolation, and databases coded using one taxonomy are not easily comparable to those using another. This lack of integration poses a significant barrier for researchers seeking to perform cross-dataset analysis, limiting their ability to synthesize findings, identify broader trends, or develop unified models of human performance in aviation contexts. Bridging this gap

through a systematic mapping or alignment of taxonomy elements would support more robust and scalable human factors research.

Taxonomies

Three taxonomies were selected for this research: TEM, HFACS, and ACCERS. These taxonomies were selected based on their strong theoretical foundation, use in the aviation industry, and applicability to the specific types of events being studied. The taxonomies were developed, in part, to classify human factor related events, and the reports selected for this research are limited to events that involved crew communication and coordination breakdowns.

TEM

The Swiss Cheese Model of accident causation, developed by James Reason, was the foundation for TEM (Brennan et al., 2020). In the Swiss Cheese Model, there are layers of protections that help to prevent failures; however, these layers can have holes, like Swiss cheese, and when these holes align, failures can occur (Reason, 2020). Reason (2000) states that past research has been focused on minimizing unsafe acts rather than creating organizations that are able to tolerate errors and still have safe outcomes.

TEM is a proactive framework employed by flight crews and aviation professionals to anticipate and mitigate potential threats and errors, thereby preventing their escalation into accidents, incidents, or operational deviations. The goal of TEM is to break the chain of events, which can be visualized with the Swiss Cheese Model, and give crews the resources they need to manage these situations (Brennan et al., 2020). As a model, TEM studies the inter-relationships between safety and human performance in the complex and demanding operational environment of the flight deck (Maurino, 2012). The TEM model can be applied in several manners. It can be used to study a single event to understand the errors and threats, how they contributed to the

event and how they could have been prevented. The model also uncovers systemic patterns within a larger database of information and events. Additionally, the TEM model can be used to pinpoint the strengths and weaknesses of individuals, exemplifying human performance needs and providing definitions of competencies in a safety management context. Similarly, the TEM model can be used to train flight crew members and other aviation personnel to spot weaknesses and improve the effectiveness of training and organizational safeguards (Maurino, 2012).

There are three major components to the TEM model: threats, errors, and undesired aircraft states. Threats are consistently defined as "events or errors that occur beyond the influence of the flight crew, increase operational complexity, and which must be managed to maintain the margins of safety" (Maurino, 2012, pg.14). Threats include items such as: weather, terrain, airspace, aircraft malfunctions, impaired judgement, errors caused by other individuals, etc. Because these items have the potential to negatively impact the crew's performance, they are considered threats. Threats can be divided into a few categories. First, the threat can either be internal or external. Internal threats impact the physiological and psychological state of the crew member, such as fatigue, lack of proficiency, or overconfidence. External threats are outside of the crew member's control, such as poor weather, unfamiliar terrain, or aircraft malfunctions (Civil Aviation Safety Authority [CASA], 2021). Additionally, threats can be either anticipated or unanticipated. Anticipated threats are issues a pilot can expect and plan for, like thunderstorms, busy airspace, or nearby mountains. Unanticipated threats happen without warning, such as sudden turbulence, unexpected equipment problems, or system errors (CASA, 2021). The last category of threats are latent threats. These may not be directly observed by the flight crew and may need a formal safety analysis to recognize. Latent threats include such things

as organizational culture, incomplete or inaccurate manuals, equipment design, or organizational pressure from delays or other irregular operations (CASA, 2021).

Errors are the second major component of the TEM model. Errors comprise any action or inaction by the flight crew that is a deviation from operational policy or procedure, reduces safety margins, or increases the likelihood of an adverse event. Adverse events can include programming or handling errors, communication errors, or procedural errors (CASA, 2021). Errors can be either intended or unintended actions. Intended actions include knowingly violating operational policies and procedures as a shortcut or to try to increase efficiency or intentional non-compliance. Unintended actions could include slips, lapses, or mistakes. These are generally a result of attention or memory failures or lack of knowledge about policies and procedures (CASA, 2021).

The third component of the TEM model is undesired aircraft states, when the aircraft is not in the required configuration, location, or trajectory because of an action made by a crew member. These states include speed or altitude deviations, misapplication of flight controls, inaccurate system or programming inputs, incorrect ground location, and take-offs or landings on the incorrect runway (CASA, 2021; Maurino, 2012).

The application of TEM has expanded to a broader spectrum of aviation, including maintenance and ATC and the concepts have been applied in medical practices (Hickey et al., 2015; Researchers at Queen Alexandra Hospital Target Medical Errors, 2020; Zhao et al., 2017). TEM has been validated through years of use in practice and has been used as a model for additional research. The International Civil Aviation Organization (ICAO) includes TEM training as a critical foundation for all flight crew training regardless of the size or scope of the operation (CASA, 2021). Examples of use of the TEM model in research include studies from air

traffic management, general aviation and military operations, along with commercial airlines. In research by Zhao et al. (2017), the TEM model was used to establish a Bayesian Network Model based on irregular ATC events in 2011. This network model allowed researchers to determine how ATC errors were related to undesired aircraft states. This study found that threats around the airspace were the major contributor to ATC communication errors, which were the primary factor of undesired aircraft states for aircraft on the ground (Zhao et al., 2017). A study by Lee et al. (2017) looked at the application of TEM in Australian aviation training, surveying 59 general aviation pilots to find a decrease from previous studies in their confidences about the perceived effectiveness of TEM. In previous studies, 90% of respondents said TEM would increase safety; however, only 2/3 of respondents in this study felt this was true. However, a majority of the respondents did report that TEM improved their non-technical skills (Lee et al., 2017). A report by Spare (2006) looked at several approach and landing accidents and applied the TEM model to these events to determine their root causes and provide an analysis for military training. While not providing any significant findings, this report strongly makes the case for using TEM in the flight deck because it can reduce accident occurrences in approaches and landings by making pilots more aware of potential threats, mitigating their effect, and managing the outcomes (Spare, 2006). A study by Thomas (2009) looked at commercial operations and used the TEM model to analyze crew performance during normal operations based on a set of contextual factors and non-technical skills. Researchers measured the crew's ability to identify threats and errors based on these factors and created a predictive model, which showed how this type of analysis could be used to show organizational strengths and weaknesses of crew performance, as well as inform future scenario-based training (Thomas, 2009).

HFACS

HFACS is based on Reason's Swiss Cheese Model of accident causation, which says accidents are the result of multiple, layered system failures, the interaction of latent failures and active failures. Much like TEM, HFACS classifies these failures so they can be identified and negated. Unlike TEM's broad look at threats and errors in crew performance, HFACS is used once an event has already occurred (Wiegmann & Shappell, 2016). Because HFACS is based on Reason's model, it defines four levels of failure that represent the four layers of Reason's model: unsafe acts, preconditions for unsafe acts, unsafe supervision, and organizational influences.

Unsafe acts are classified into two categories: errors and violations. Errors are unintentional and can be classified as either skill-based, decision, or perceptual. Skill-based errors are related to the physical act of flying the aircraft and can include automation programming, aircraft handling or a breakdown of normal flying skills, like visual scans, miss-ordering procedures, or task fixation. Decision errors happen when a crew member makes an inappropriate or inadequate plan. Without intending to, they choose the incorrect path or lack the knowledge to recognize that their plan will not work. Perceptual errors occur when a crew members' interpretation differs from the reality of the situation, and can arise because of various factors including sensory degradation, visual illusions, or spatial disorientation (Wiegmann & Shappell, 2016).

Violations can be classified as either routine or exceptional, with routine violations occurring on a regular basis, perhaps *bending the rules*: rushing through checklists, skipping preflight steps, or not following all procedures. Exceptional violations are isolated departures from standard operating procedure and not indicative of the crew member's regular behavior, making them more difficult to predict and reduce (Wiegmann & Shappell, 2016).

Preconditions for unsafe acts are the underlying conditions that facilitate unsafe behaviors and are commonly classified into three categories: environmental factors, personnel factors, and the physical and psychological state of the operators. Environmental factors include the physical and technological environment; personnel factors include crew resource management and personnel readiness; and condition of operator includes adverse mental states, adverse physiological states, and physical or mental limitations (Wiegmann & Shappell, 2016).

Reason's Model highlights the influence of a supervisor's actions on the operational environment and how these actions might contribution to unsafe acts. The HFACS framework also incorporates the concept of unsafe supervision and categorizes it into four distinct types: inadequate supervision, planned inappropriate operations (when operations do not meet the needs of crew members, not allowing needed rest or overwhelming crew members with work) failure to correct known problems (when supervisors know of unsafe behaviors but ignore them), and supervisory violations (when supervisors ignore policies and procedures, leading to unsafe conditions) (Wiegmann & Shappell, 2016).

The final component of the HFACS model addresses organizational influences, which can establish environments that breed unsafe behaviors and system vulnerabilities. These influences center around three areas:

- Resource management—If physical and personnel assets and resources are not properly distributed, unsafe acts may result. Cost cutting or poor allocation can leave crew members without needed resources or cause them to look for alternative, potentially unsafe, ways to complete the work.

- Organizational climate—If there is a culture of non-compliance or animosity within the organization, crew members may be less willing to follow safety directives.
- Organizational process—If daily processes and activities are poorly defined or suffer under an excessive amount of external pressure, safety might be compromised as crews seek to find more efficient ways to complete tasks (Wiegmann & Shappell, 2016).

HFACS is well established and used by many organizations to analyze accidents and incidents and reduce reoccurrences. HFACS has also been translated for use in other high-risk operational environments, such as maritime, railway, and chemical operations (Chauvin et al., 2013; Wang et al., 2020; Zhan et al., 2017).

Research focused on the HFACS framework has shown it to be consistent and reliable in identifying patters of human factors that contribute to aviation accidents. Research by Shappell et al. (2006) expanded this concept by investigating HFACS in Part 121 and Part 135 accidents. Researchers studied aircrew, environmental, supervisory, and organizational factors related to these accidents, with Shappell, et al. (2006) finding that the majority of accidents were related to aircrew and the environment rather than supervisory or organizational causes.

Additional research adapts the HFACS model for use in applications outside of aviation. Chauvin et al. (2013) studied the use of HFACS in about 30 maritime accidents, finding that most of the collisions were a result of decision error and that leadership was a contributing factor as a result of planning inappropriate operations and non-compliance with SMS.

Zhan et al. (2006) and Wang et al. (2020) studied how HFACS can be adapted for use in railway and chemical accidents, using the newly defined HFACS-RA for railway accidents and HFACS-CSME for chemical accidents in small and medium-sized enterprises.

ACCERS

ACCERS was developed by the American Institutes for Research (at the request of the FAA) as a classification system to identify and analyze pilot error in ASAP reports. In both ASAP and ASRS, pilots and aviation personnel voluntarily report errors or safety situations they encounter. ASAP data are typically kept within the airline while ASRS data are shared publicly by NASA (Baker & Krokos, 2007). However, about 62% of ASAP reports end up as ASRS reports when airlines share their data with ASRS or users file both an ASAP and an ASRS report (GAO, 2010). ACCERS was developed because the FAA needed a taxonomy pilots could use to classify the causes of errors, knowing that these classifications would help organizations improve human performance. ACCERS was developed and validated through three studies conducted by Baker and Krokos (2007), two involving pilot interviews, comprehensive literature reviews, and a classification exercise and one that validated the findings from earlier studies (Baker & Krokos, 2007). System developers identified 300 factors that caused aviation incidents and accidents in literature, then condensed those to 94, eliminating duplicates and similar items. Pilots sorted these 94 items into clusters by themes, deciding on seven categories and 91 items in the taxonomy. Pilots validated the taxonomy by reviewing reports and classifying major contributing factors (Baker & Krokos, 2007). The full ACCERS taxonomy is presented in Appendix A.

One of the few limitations of ACCERS is the differentiation between human error and human factors as a parent category. These categories are relatively similar: human factors refers

to the state of the pilot or environment around the pilot that contributes to the error, while human error is the error itself. Baker and Krokos (2007) suggest future research is needed to determine whether pilots identifying the error and then classifying its human factors would lead to greater accuracy.

Because ACCERS was designed for use with ASAP databases, typically contained within airlines and including proprietary information, few studies use ACCERS as a classification method. However, the study outlined above by Baker and Krokos (2007) contains detailed information on the validation of ACCERS and its use for classification of ASRS reports.

ASRS Reporting

The user interface plays a major role on reporting quality and quantity Gong (2011). Interface design in medical incident reporting systems influences the rate, completeness, and accuracy of reports and researchers found that systems that do not align their categories with reporters' nomenclature often suffer from underreporting or misclassification, with many incidents logged under broad, non-specific categories such as "miscellaneous" or "other." This mismatch reduces the usefulness of the reports for analysis or patient safety research. Reporters, who are typically healthcare professionals under intense pressure, may choose safe but vague options because of workload or insufficient time. Additionally, limitations of the reporting system, such as limiting the number of categories per report, can cause confusion when incidents span multiple types. User interfaces that do not support detailed, flexible data entry contribute to incomplete and inconsistent documentation. Applying human-centered design principles, such as intelligent category suggestions, multiple classifications, and alignment with reporters' cognitive models, can improve user acceptance, accuracy, and overall quality of the data collected (Kalsy et al., 2024; Gong, 2011).

Similar challenges exist in aviation safety reporting—frontline personnel operate in high-stress, multitasking environments, so poorly designed interfaces may lead to underreporting or vague categorization of safety incidents. Improving the interface by incorporating human factors principles can enhance reporting rates and data quality, which is essential for identifying safety risks, analyzing incident trends, and developing effective mitigation strategies to prevent future accidents. As such, it is important to understand the user interface and collection form when analyzing data obtained from this system.

Figure 1 is the form that a pilot or aviation professional fills out when submitting an ASRS form. Airlines can use their own forms and user interfaces when collecting ASAP reports, which make up a little more than 60% of ASRS reports. Because these forms are internal to the airline they are not included, however, they contain similar fields, so the information can be reported correctly to the ASRS database (GAO, 2010).

The ASRS reporting form starts with demographic information, with reporters either typing in information or selecting from pre-defined menus. None of this information can be used to identify the pilot, although the report does ask for a name, address and phone number. A copy of the report is mailed to the reporter after submission. Some reports will trigger a follow-up call with the reporter to obtain additional information. All these fields are stripped from the report once it goes into the database.

Next, the ASRS requests information about the environment of the event, the aircraft, phase of flight, route, mission, and location including information about a conflict if one occurred. These values are a combination of text boxes, dropdown menus, and check boxes.

Finally, there is a section a narrative of the event with the following instructions:

Keeping in mind the topics shown below, discuss those which you feel are relevant and anything else you think is important. Include what you believe really caused the problem, and what can be done to prevent a recurrence, or correct the situation.

Chain of Events

- *How the problem arose*
- *How it was discovered*
- *Contributing factors*
- *Corrective actions*

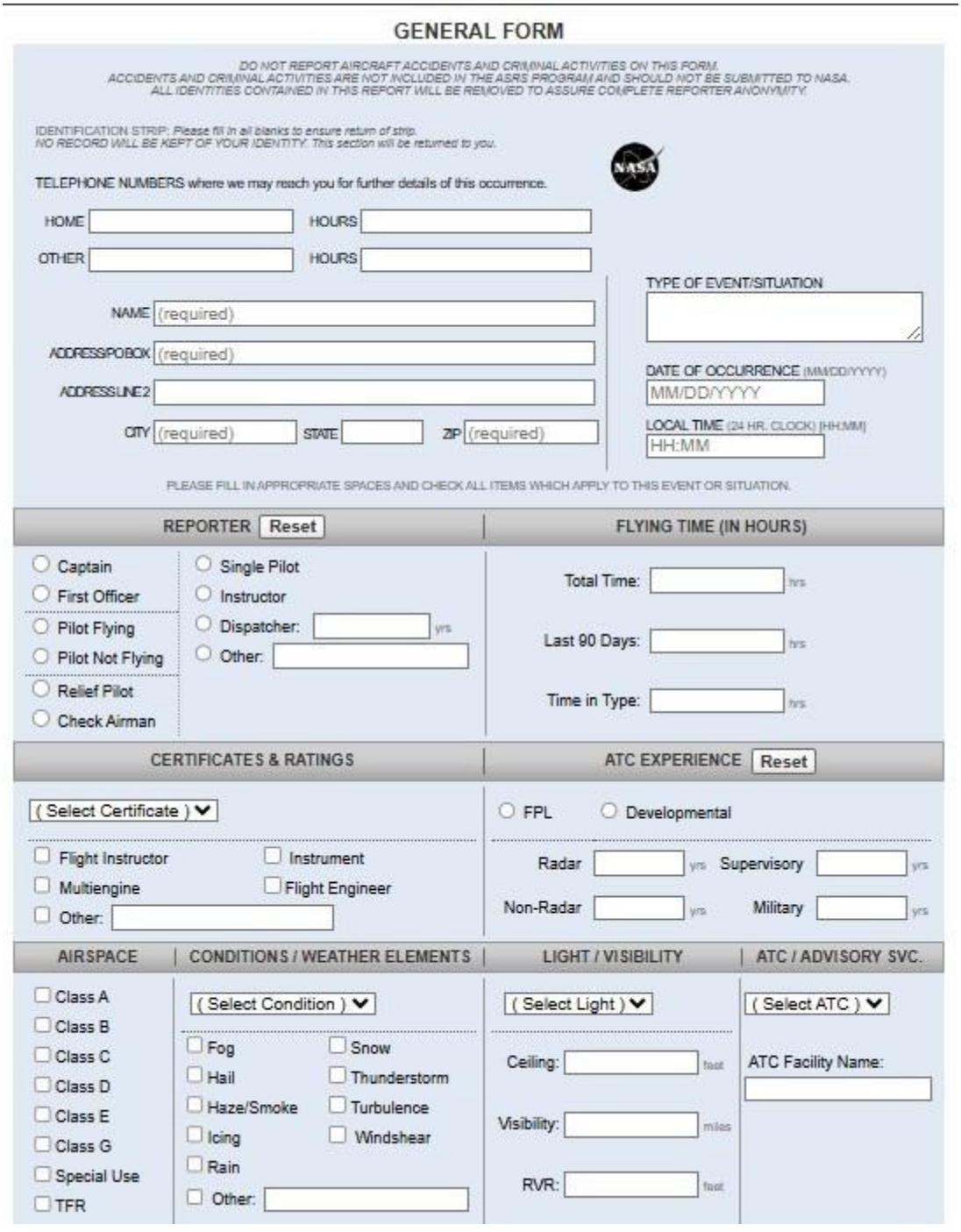
Human Performance Considerations

- *Perceptions, judgements, decisions*
- *Factors affecting the quality of human performance*
- *Actions or inactions*

Structured instructions and prompts in these narrative sections can enhance the quality, completeness, and usability of incident reports and reduce uncertainty for less experienced or infrequent reporters. When thoughtfully designed, these narratives provide important information about the sequence of events, contributing factors, and outcomes (Curtis et al., 2025; Thota et al., 2025). This is especially valuable in complex, high-stress environments where individuals may not recall key details because of stress or cognitive overload (Kalsy et al., 2024; Gong, 2011). Well-crafted prompts also make for consistent reports, enabling researchers to identify patterns, conduct comparative analyses, and derive meaningful safety insights. Clear guidance supports narrative organization, ensuring coherence and relevance in both scholarly and professional contexts (Curtis et al., 2025; Thota et al., 2025). Structured approaches also facilitate standardization, allowing reports to be compared and supporting longitudinal or collaborative studies (Curtis et al., 2025; Thota et al., 2025). Informative prompts also reduce ambiguity and foster clearer communication between frontline personnel and safety investigators, strengthening safety management systems (Reason, 1997). There are concerns, however, that overly directive questions may bias the account, constrain interpretation, or limit the narrative by focusing too narrowly on predefined elements (Reason, 1997).

Figure 1

ASRS Reporting Form



GENERAL FORM

DO NOT REPORT AIRCRAFT ACCIDENTS AND CRIMINAL ACTIVITIES ON THIS FORM.
ACCIDENTS AND CRIMINAL ACTIVITIES ARE NOT INCLUDED IN THE ASRS PROGRAM AND SHOULD NOT BE SUBMITTED TO NASA.
ALL IDENTITIES CONTAINED IN THIS REPORT WILL BE REMOVED TO ASSURE COMPLETE REPORTER ANONYMITY.

IDENTIFICATION STRIP: Please fill in all blanks to ensure return of strip.
NO RECORD WILL BE KEPT OF YOUR IDENTITY. This section will be returned to you.

TELEPHONE NUMBERS where we may reach you for further details of this occurrence.

HOME HOURS
OTHER HOURS

NAME (required)
ADDRESS/POBOX (required)
ADDRESS LINE2
CITY (required) STATE ZIP (required)

TYPE OF EVENT/SITUATION
DATE OF OCCURRENCE (MM/DD/YYYY) MM/DD/YYYY
LOCAL TIME (24 HR. CLOCK) [HH:MM] HH:MM

PLEASE FILL IN APPROPRIATE SPACES AND CHECK ALL ITEMS WHICH APPLY TO THIS EVENT OR SITUATION.

REPORTER Reset
Captain
First Officer
Pilot Flying
Pilot Not Flying
Relief Pilot
Check Airman
Single Pilot
Instructor
Dispatcher: yrs
Other:

FLYING TIME (IN HOURS)
Total Time: hrs
Last 90 Days: hrs
Time in Type: hrs

CERTIFICATES & RATINGS
(Select Certificate)
Flight Instructor
Instrument
Multiengine
Flight Engineer
Other:

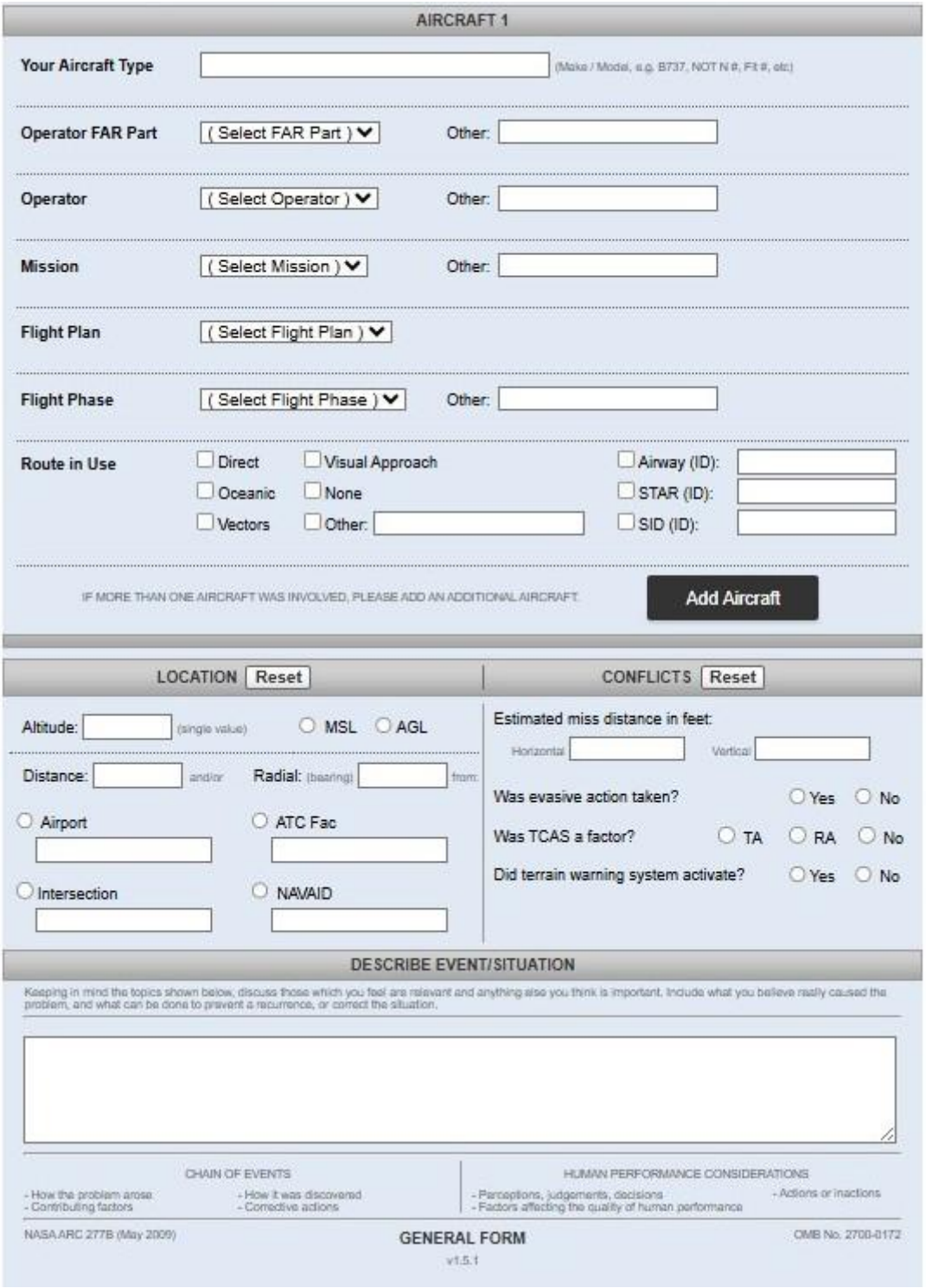
ATC EXPERIENCE Reset
FPL
Developmental
Radar yrs Supervisory yrs
Non-Radar yrs Military yrs

AIRSPACE
Class A
Class B
Class C
Class D
Class E
Class G
Special Use
TFR

CONDITIONS / WEATHER ELEMENTS
(Select Condition)
Fog
Snow
Hail
Thunderstorm
Haze/Smoke
Turbulence
Icing
Windshear
Rain
Other:

LIGHT / VISIBILITY
(Select Light)
Ceiling: feet
Visibility: miles
RVR: feet

ATC / ADVISORY SVC.
(Select ATC)
ATC Facility Name:



Note. Adapted from "ASRS General Report Form" by NASA, 2009.

Use of Multiple Taxonomies

Variations in the application of taxonomies across airlines and between pilots can introduce complexity and ambiguity. Airlines often modify taxonomies and classification systems to align with their operational requirements (Stolzer et al., 2023), which could change how these systems are used within ASRS. Because ASRS includes reports from multiple airlines and the original system (rather than airline-specific systems) is used to analyze each taxonomy, these changes may be minimized. Also, changes in safety training may move the focus to one taxonomy rather than another, changing how a pilot writes their ASRS reports.

While research dedicated to aviation taxonomy development and error coding remains limited, literature does address the adoption and integration of new taxonomies within various aviation contexts. For example, Lanicci et al. (2020) proposed a taxonomy focused on general aviation weather education and training. This taxonomy evaluates pilots' knowledge of aviation weather and assesses their training for both real-world scenarios and knowledge-based examinations. This taxonomy is consistent across dispatchers, pilots, and air traffic controllers and can be used in training and testing. Providing a taxonomy that pilots, controllers, and dispatchers use in the same way reduces the likelihood that information will be lost between work groups (Lanicci et al., 2020). While this study focused on weather reporting, the same concept can be applied to safety reporting. A common theory-based taxonomy that can be applied and interpreted the same way by different airlines and regulators allows for efficient information exchange, lessens the chance of misinterpretation and ensures the preservation of critical safety information.

Outside of aviation, taxonomies are also used to describe safety-related events. Pham et al. (2016) developed a common taxonomy to study adverse events for patients using ventilators,

compiling reports from three different databases to find commonalities and develop mitigation strategies. However, each of these databases used different reporting methods and coded the events differently, so the researchers had to develop a common taxonomy that allowed them to calculate accurate numbers for the ventilator events (Pham et al., 2016). In a similar manner, airlines have their own event databases along with the ASRS database. Because there is not a singular taxonomy used to analyze these separate databases, events can be misrepresented, leading to gaps in safety.

The same concept can be applied to the classification of animals and plants, as Thomson et al. (2021) describes in their study of bird species. There are multiple bird classification lists, many using slightly different scientific names for the same birds, leading to inaccurate reports of and bird counts. Similarly, without well-defined taxonomies, diversification rates may be under reported for animal species, which is explored in a study by Muggia et al. (2020) related to lichens. Researchers tested lichen DNA against established types of lichens and found there was significantly more diversification than originally thought. Due to the lack of a well-defined taxonomy, this had never been recorded. Researchers concluded that creating a taxonomy with the additional characteristics would provide a full understanding of lichen types. In much the same way, if ASRS codes do not offer enough detail, events could be grouped into a general category, but more information could be gained from the reports if the taxonomy allowed for it.

Crew Communication and Coordination

Since the ASRS database contains more than 1.7 million reports (NASA, 2021), the scope of this study was limited to incident reports in which crew communication and coordination errors were the primary contributing factor in order to make the study manageable and feasible.

The study focused on crew communication and coordination to ensure the role of the flight crew remained central to the identification and understanding of the contributing error. Because the study examined human factors taxonomies, narratives selected from the ASRS database were heavily human factors based, remembering that 60 to 80 percent of all accidents have human error as a contributing factor (FAA, 2004).

Historically, training has focused on the technical aspects of flying, but recently training has shifted to address crew coordination, communication, resource management, decision making, and leadership (FAA, 2004), aspects contained in what is now Crew Resource Management (CRM). According to AC 120-51E, crew communication and coordination are primary components of CRM (FAA, 2004) — "Crew communication and coordination are the foundation of safety during normal and emergency operations" (FAA, 2020c, pg. 3). As of 1998, all flight crew members are required to have CRM training conducted by air carriers (Compliance Dates, 1996).

Crew communication and coordination are critical aspects of flight safety (Vieira & dos Santos, 2010). Shappell et al. (2007) found that CRM failures were found in 20% of air carrier accidents they examined and 60% of the CRM failures were associated with inflight crew coordination, communication, and monitoring activities. A study of military aviation accidents by Merket et al. (1999) found that 68% included a breakdown of team performance in crew coordination, decision making, situational awareness, and communication.

Crew communication and coordination is not limited to the cockpit, but also includes communication with ATC, dispatch, flight attendants, and maintenance. For this study, the focus will primarily fall on communication within the cockpit and with ATC. Several notable crashes resulted from pilot controller miscommunication, including Avianca Flight 052, American

Airlines Flight 956, and the deadliest accident in history at Tenerife. Between 1976 and 2000, more than 1,100 passengers and crew members were killed in accidents where pilot communication to ATC was a contributing factor (Krivonos, 2007). Clear and concise language is critical for pilot-controller interactions and incidents can result from miscommunications and different interpretations of directions (Hamzah & Wong, 2018; Wu et al., 2019). Language barriers between pilots and controllers can also play a role in some cases.

A study of ASRS reports by Krivonos (2007) highlighted the importance of crew communication and coordination to safety, including communication within the cockpit and with flight attendants or ATC. Crew communication and coordination were selected as the primary focus of this taxonomy analysis because of their critical role in aviation safety and their strong interrelation with a broad range of human factors domains. Nevile (2006) notes that communication is especially important in aviation because it is how other human factors are realized or made possible. It is the backbone to planning, leadership, decision-making, information gathering and sharing, and the identification and management of errors.

While this study primarily focused on identifying and coding communication and coordination errors, it also acknowledges that successful pilot communications and their associated hazard controls are critical counterpoints. Effective communication within the cockpit allows for shared situational awareness, timely decision-making, and coordinated responses to emerging threats, serving as a barrier against errors and incidents (Salas et al., 2006). This successful communication involves standardized language, assertiveness, and confirmation protocols that help reduce hazards before they escalate (Helmreich & Merritt, 1998). In contrast, ineffective communication can cause misunderstandings, incomplete information exchange, or

breakdowns in coordination, which can directly contribute to errors and incidents (Sexton et al., 2000).

Effective crew member communication is the clear, concise, and timely exchange of information that gives the crew shared situational awareness and coordinated decision-making (O'Connor & Flin, 2003). It involves both accurate transmission by the sender and correct reception and interpretation by the receiver, ensuring mutual understanding and minimizing ambiguity or misinterpretation (Helmreich, Merritt, & Wilhelm, 1999). Key components are the use of standardized language, confirmation of messages through readbacks, and active listening to reduce errors and enhance effectiveness (Sexton, Helmreich, & Merritt, 2000). This two-way communication is fundamental to safe and efficient cockpit operations.

ATC Communication

The 2025 mid-air collision involving American Airlines Flight 5342 and a Black Hawk helicopter over the Potomac River underscores the critical role of air traffic control (ATC) communication in flight safety. Investigations revealed that helicopter crew members may have missed a crucial directive from ATC due to a muffled radio transmission, leading to a tragic outcome. This incident highlights how dangerous lapses or misinterpretations of ATC communication can be for situational awareness and coordination among flight crews (NTSB, 2025a).

This is not an isolated case. Less than one month after this crash, a Southwest Airlines flight narrowly avoided another tragedy when it performed a go-around to avoid a corporate jet crossing its landing runway (NTSB, 2025b). This incident was a result of failures in both ATC and crew member communication and coordination. ATC had cleared the Southwest flight to land, but the corporate jet had not been cleared to cross. There was a breakdown in crew

communication and coordination within the flight deck that led to the corporate jet crossing the runway without clearance influenced by miscommunications with ATC. The Southwest first officer communicated effectively with the captain, calling for a go-around when he noticed the corporate jet, and the captain executed that call without hesitation, passing just 200 feet behind the other aircraft (NTSB, 2025b).

Incidents such as these two underscore the pivotal role that ATC plays not only in airspace management and traffic flow but also in crew communication and coordination within the cockpit. In both cases, lapses or misinterpretations of ATC communications contributed to serious safety risks—highlighting how ATC communications must be effectively processed, interpreted, and acted upon by flight crews (NTSB 2025a; NTSB, 2025b). These situations require a high level of shared situational awareness and mutual understanding among crew members, which can be disrupted when ATC instructions are unclear, delayed, or contradictory. Even when technical systems function correctly, communication breakdowns—either between ATC and flight crews or among the crew — can lead to near-miss or tragic outcomes. Because this communication is so vital for safe and effective crew performance, robust communication protocols, CRM training, and systemic safeguards are key to mitigating human errors (Endsley, 2017).

Introduction of New Safety and Training Programs

When considering what changes should be suggested within this study, it is essential to recognize that aviation is constantly developing new safety programs, training methods, and systems. Even longstanding programs and taxonomies have evolved as new information is discovered. Two of the most significant changes in pilot training and airline safety are the

Advanced Qualification Program (AQP) and the Safety Management Systems (SMS). These programs changed the ways that hazards, risks, and other data are classified and analyzed.

AQP was first introduced in 1991 in AC 120-54, but it was not until 2005 (when this AC was updated and codified as 14 CFR Part 121 Subpart Y in federal regulations) that airlines started to adopt this training method. Delta Air Lines was the first to adopt AQP in 2006 in their inflight program (Rosenkrans, 2011). AQP is a data-driving training approach that allows airlines to adjust the frequency and method for delivering training programs, with the AC suggesting that airlines will see the reduction of crew related errors, higher standardization, and more efficient training (FAA, 2017). With the introduction of AQP came new taxonomies, requiring airlines to create reason codes for every pilot error captured during training. Since these reason codes categorize the errors, it allows for more efficient data analysis (FAA, 2017).

AQP also integrates CRM training throughout the curriculum, with CRM tasks and procedures carrying the same weight as technical procedures. Not only are CRM aspects included, but activities that are a part of maneuver-specific events are also evaluated, such as callouts for normal take off, approach, and landing phases, as well as management through non-normal events (FAA, 2017). According to the FAA, 90% of large Part 121 carriers — those with more than 1000 pilots—currently use AQP (FAA, 2021). Over the last 10 years, a system to simplify and analyze AQP data has been created, along with a renewed and extensive focus on CRM training and the concepts behind this training. Although CRM has been around since 1979, it has evolved into the form currently trained in AQP (Helmreich et al., 1999).

The introduction of SMS also added taxonomies, or ways of classifying and analyzing events. SMS was first introduced by the FAA in 2008 with Order 8000.369 and was eventually

required by 2018. At a basic level, SMS is a way to manage safety risk and ensure the effectiveness of safety risk controls (FAA, 2020a).

Effective Safety Risk Management (SRM) relies on a common taxonomy so that the measures of risk can be repeatable and reliable (Stolzer et al., 2023). Stolzer et al. (2023) presented several options for aviation taxonomies, highlighting the benefits of each, but it ultimately recommended the TEM Model because it closely follows the language of SMS and offers simple understanding for front line employees. When airlines creating their SMS, they are not required to select any specific taxonomy for their SRM. TEM can also be found in ASAP and LOSA programs along with its use in SMS (Stolzer et al., 2023).

Other taxonomies that could be used for SMS include HFACS, ACCERS, and British Airways Safety Information System (BASIS). While there are others, these are the most recognized withing the U.S. airline industry (Duke, 2001; Krokos & Baker, 2005; Wiegmann & Shappell, 2001). For this research, TEM, HFACS, and ACCERS are included in the study of the taxonomy and classification systems within the ASRS database.

As both Zhang (2012) and Tanguy et al. (2016) state, aviation safety reports pose a unique challenge for analysts because while they describe complex events, they can include varying levels of detail. Zhang (2012) discusses some of the difficulties pulling critical pieces of information from narrative reports, specifically the ASRS database, and connecting those to other reports. For example, causal factors tend to be buried within the narrative text, so in order to connect these factors and find connections between reports, an analyst would have to read through each report and manually code these factors for each report. Similarly, Tanguy et al. (2016) noted that the analysis of safety reports is complex due to the amount of information and detail contained in each report.

Beaubien and Baker (2002) discuss similar complications in analyzing ASAP reports. For pilots, ASAP has similar reporting to ASRS, with some data collected on the specific flight and the pilot, but the majority collected in narrative style, written by the pilot. While ASAP reports are submitted by employees within a single airline, ASRS reports are submitted by employees of multiple airlines. Because of their narratives, analyzing ASAP reports is time-consuming and costly for an airline. In their analysis of human factors taxonomies, reporting systems, and data collection tools, Beaubien and Baker (2002) propose a more structured form, with pull-down menus and check boxes to supplement the narrative reports and make analysis simpler (Beaubien & Baker, 2002).

Theoretical Framework

One important consideration when analyzing the ASRS database is that the lack of standard taxonomies or event descriptions can cause confusion and missed connections between reports. Thomson et al. (2021) noted this problem when studying the classification of bird species, finding that multiple ways of classifying the same species resulted in competing lists (Thomson, 2021). At the same time, when changes that are made to a classification system do not reach all users, more complications can occur. Similarly, users also interpret taxonomies differently and have varying knowledge (Neat-Clegg et al., 2021). Not only do multiple taxonomies influence reports, they can also impact how analysts read and understand these reports. In the case of bird species, when experts use different taxonomies or methods, diversification rates and other such statistics may be over or under reported (Faurby et al., 2016). This same concept applies to the use of multiple taxonomies within aviation reporting—taxonomies can be based on varied theories and contain unique requirements regarding detail, causing overlap or gaps during analysis. Studying how multiple taxonomies are applied to the

ASRS database can help identity these overlaps and gaps. Researchers can use definitions from well-established theory-based taxonomies to clarify taxonomies that lack detail, resulting in fewer errors in taxonomy assignment and interpretation. Similarly, consistent taxonomies allow for more standardized analysis when researchers combine sets of data that use different classification taxonomies.

Bosser (2020) started developing a taxonomy framework for the ASRS database by studying ASRS report classifications by both humans (pilots and human factors experts) and machine learning systems. The study found that pilots offered more reliable and consistent classifications (Boesser, 2020). This finding, along with information from Stolzer et al. (2023) that pilots were not used in the development of ASRS anomaly codes, suggests there could be missing detail in these codes. Boesser's study (2020) also suggests the creation of a human factors taxonomy to better classify ASRS reports, tying this taxonomy to the information that an airline or analyst wants to obtain from the reports. This taxonomy can be created by developing a well-researched framework of how theory-based taxonomies are already applied to ASRS reports by airlines.

When examining the impact of multiple taxonomies on report analysis, it is necessary to apply a learning model to explore how pilots discover these taxonomies and apply them in practice. Illeris (2004) details such a model for individual learning, also factoring in the influence of coworkers and the workplace environment. Illeris' (2004) concept explains how a pilot learns a new taxonomy or safety system and how the way that they apply and use it might change how the system, taxonomy, or classification is understood. This model shows that individual learning is influenced by both the technical-organizational learning environment and with the social-cultural learning environment. Working practice informs the individuals' work identity and how

they use knowledge gained through their work environment. Any factors present in the learning space—from influences tied to training, performance, or the workplace—impact the learner's experience and working practice, becoming part of the learning process and the knowledge and experience gained (Illeris, 2004). For this reason, it is important to consider how theory-based taxonomies and classification systems are used to analyze reports, but also how pilots use these taxonomies and classification systems within the reports.

According to Edkins (2002), there has been a shift toward behavioral training within aviation. This method allows for training to a specific skill or concept and offers a distinct way to measure effectiveness. At the time of this research, behavioral training was not fully implemented across airlines, but Edkins (2002) notes that the idea is spreading and being used more frequently, allowing researchers to study how pilots learn as new programs are introduced.

Taylor (2005) conducted a comprehensive review of 117 studies related to behavioral training, finding that behavioral training was most impactful in the areas of declarative knowledge and procedural knowledge-skills. Declarative knowledge is the understanding of facts, events, or processes and the relationships among them, typically regarded as static knowledge. In contrast, procedural knowledge involves the comprehension of how skills or tasks are executed, with an emphasis on methods and processes (Saks et al., 2021). The study by Taylor (2005) shows the relationship between training and declarative knowledge; the understanding of new taxonomies, systems, and programs; and the word choices used in reports. As a pilot learns a new system, the way they store this information in their memory changes and their event descriptions may also change, highlighting the importance of developing the proposed framework while also considering the timing for introduction of new training and

safety systems. A pilot's focus when writing a report might evolve in relation to significant changes in training or safety systems.

Summary

A comprehensive understanding of the TEM, HFACS, and ACCERS frameworks was essential for the effective design and execution of this study. The literature review examined the challenges associated with the use of multiple taxonomies to classify similar types of events, noting how such variation can complicate data analysis. Investigating the application of multiple taxonomies to the ASRS database enables the researcher to identify areas where certain taxonomies offer greater specificity or, conversely, where they may lack completeness or coverage. Researchers can use the definitions from well-established theory-based taxonomies to clarify taxonomies that lack detail, resulting in fewer errors in assignment and interpretation. Similarly, consistent taxonomies allow for more standardized analysis when a researcher combines information from data that use different taxonomies.

The literature review also explored the ways that pilots acquire knowledge and adapt their behaviors, particularly in response to training interventions, a key factor in this study. The application of these changes could influence the language pilots use when writing ASRS reports, which could influence how the taxonomies are applied by analysts assigning ASRS anomaly codes.

Chapter III: Methodology

Research Method Selection

This research used qualitative text coding methods to collect and analyze reports from the NASA ASRS database. Additional analysis supplemented the qualitative analysis, validating the framework, and further answering the research questions.

The research used text coding and analysis, with NVivio software, to focus on narrative text within the ASRS reports. Coded fields were also considered for demographic information that could provide additional insights. The research used common words and phrases found within the selected taxonomies and classification systems to analyze each selected report. Each report was classified using the three selected taxonomies based on the event that took place and the contributing factors. For example, a report classified with HFACS in which a crew fails to descend at the appropriate time because the first officer's MCDU was malfunctioning and the captain failed to call for descent, would be classified as an unsafe act of a skill-based error. The precondition for the unsafe act would be crew resource mismanagement. This same event would be classified within TEM as the aircraft system (MCDU inaccurate) being the threat and automation, incorrect altitude, labeled as the error. (The coding process is further described in the following sections with models for HFACS, ACCERS, and TEM shown in Appendix A.)

Since coding of the reports used these three taxonomies, a framework was developed to show how the taxonomies relate to each other within each report and how they relate to the assigned anomaly codes. To validate the developed framework, it was applied to a set of additional reports categorized by their assigned anomaly codes. These reports were also manually coded through an interactive process. The classifications produced by the framework

were then compared with the interactive coding results to evaluate the misclassification rate and assess the framework's accuracy.

Changes in training and new safety programs were also analyzed in relation to the shift in focus of pilot reporting. A descriptive analysis was used to determine the trend of each coded taxonomy through these changes. The analysis noted any significant changes in the focus of the reports after the introduction of new training or safety programs, providing a simple foundation for the way changes in training impact pilot reporting. If there were significant impacts from training, further research can explore the extent of these relationships and determine causation.

The conceptual foundation for this research was adapted from the methodology outlined by Bazeley (2013), which details the systematic development of frameworks for grounded theory using qualitative text coding techniques. The process began with basic descriptive statistics, progressing to a cross-case analysis, and included methods for presenting the findings in a visual form. Each step in this process built more depth and understanding of the concept being studied and developed a data-based theory. This theory, and the connections created through the various steps, were then developed into a visual representation applied to the problem. Bazeley (2013) says these steps create patterns for how established groups of data interact and where these interactions are more or less likely to occur. Initial work was completed through coding queries to develop connections between data groupings and determine where they occur. The cross-case analysis was used to group cases into families based on their characteristics and to then synthesize interpretations across these cases. Jackson and Bazeley (2007) describe this method and how it can be accomplished with NVivo.

Hutchison et al. (2010) validated the method presented by Jackson and Bazeley (2007) through a worked example, concluding that when the method is applied correctly, it can aid the

development of a theoretical concept from early design and sampling procedures to presentation of findings and analysis of data. The theoretical model developed needs to be grounded in relevant data, so the method presented by Jackson and Bazeley (2007) steps through a process to provide evidence of the theoretical basis for the final model.

Within the scope of this study, the terms *relationship, alignment,* and *connection*—as used to describe the links between the selected taxonomies and the ASRS anomaly codes—should be understood as qualitative, conceptual associations, not quantitative or statistical correlations. These associations are not based on numerical indicators of co-occurrence or statistical significance, but reflect linkages grounded in the thematic content of the ASRS narrative reports. The intent is to elucidate how constructs embedded within each taxonomy conceptually map onto the emergent themes and categories derived from the narrative data, particularly those pertaining to communication and coordination errors.

Population and Sample

This research used reports submitted to the ASRS database by Part 121 pilots employed by domestic U.S. operators from January 2007 through December 2022. This selection allowed for the introduction of AQP to be included, even though most of the significant changes associated with AQP took place several years after 2007. Only reports submitted by flight crew members were included, as some of these programs and taxonomies are specific to pilot training and some of the taxonomies, like TEM, are based on the language pilots use.

Stratified sampling was used to obtain a selection of reports for this study. There are approximately 6,800 ASRS reports meeting the qualifications of this study from 2007 to 2022. Using Cochran's formula with a 95% confidence level and a 5% margin of error, a sample of 365 reports was required (Qualtrics, 2022). Dividing this over the years included in the study

calculates to 23 reports required for each year. However, as there are years where more reports are submitted, a proportional amount of the 365 reports based on the number of total reports for each year were used for the study. If reports lacked sufficient information for analysis or needed to be excluded for other reasons, additional reports were randomly selected to ensure the sample size was maintained.

Data Collection Process

The ASRS database filters reduced the number of results the initial search returned, with filters also selecting reports where human factors were the primary problem type. Filters were also added to exclude aircraft equipment malfunctions, illness and injury of crew or passengers, and fume and fire events. Reports were read for inclusion in this study so only reports of crew communication and coordination errors were used. The filter for primary problem of "human factors" along with the exclusion of specific events were the most detailed filters for this research. Contributing factors and specific human factors were not used as filters.

The study employed an exploratory method using text mining along with a descriptive analysis to answer the research questions. Text mining was used to find patterns within large data sets, analyzing data to solve problems and find new insights (Shmueli, Bruce, & Patel, 2016; Witten, Frank, & Hall, 2011). A subset of data mining, text mining looks for patterns in text, in this case the narrative text section of the ASRS reports. In this way, patterns in this text can be observed and used to form conclusions (SAS® Institute, 2020; Žižka et al., 2020). This research looked for patterns involving how the selected taxonomies relate to each other though the narratives. The conclusions relate to how the selected taxonomies are applied to the ASRS database and allowed the researcher to identify areas in each taxonomy that had more detail in their classification terms or lacked classification terms.

Descriptive research was used to explore how changes in pilot training and safety system introductions impacted how pilots used these taxonomies when writing reports. Descriptive research allows for the study of events or conditions as they occur, without variable manipulation (Siedlecki, 2020). Using descriptive research allowed analysis of taxonomy use over time based on changes in training and safety systems.

Demographic information from the ASRS database was analyzed including: position (captain, first officer, check pilot, etc.), role (pilot flying, pilot monitoring, or relief pilot), qualification, and experience. This information shows the representativeness of the sample but was not used for any assumptions when coding the reports.

The reports were filtered using the tools available on the ASRS website to capture time frame and identity of the reporter. Data from 2007 through 2022 were downloaded from the ASRS website. Filters were set to only keep reports made by Part 121 pilots. Each report was read and only reports that meet the criteria of crew communication and coordination were kept for further analysis.

Ethical Consideration

ASRS reports are submitted voluntarily by individuals who made an error or were involved in an event. The reports are de-identified so they cannot be traced back to the reporter (ASRS, 2021a). When submitting a report, submitters must acknowledge that their report will be released and could be used for research purposes (ASRS, 2021a). No data for this study was used for discriminatory purposes. The study did not consider race, age, religion, or sex. None of this information is included in the ASRS database and it cannot be determined from the reports.

Additionally, as the ASRS database is made publicly available, an IRB application and approval was not required for this research, even though the reports are from human subjects.

Measurement Instrument

The instrument for this research was related to the constructs found within the selected taxonomies, TEM, HFACS, and ACCERS.

TEM and HFACS are the first selected taxonomies. These were included in this research because they both have strong foundations based in Reason's (2000) model for human error, or the Swiss Cheese model. This model is made of four layers: organizational influences, unsafe supervision, preconditions for unsafe acts, and unsafe acts (Reason, 2000). These are almost directly translated into the HFACS model. TEM offers a broader picture of events and looks at threats and errors that can lead to an undesired aircraft state.

ACCERS was included in this research because it is very comparable to the ASRS anomaly codes. ASRS and ASAP have very similar reporting methods and analysis goals. Over 60% of ASAP reports are included in the ASRS database so, the coding should be relatively similar in structure (GAO, 2010). ACCERS was specifically developed for ASAP reports to identify and categorize event errors and causal factors.

The main constructs of the three selected taxonomies included:

- TEM: threats, errors, and undesired aircraft states
- HFACS: organizational influences, unsafe supervision, preconditions for unsafe acts, and unsafe acts
- ACCERS: policies and procedures, human error, human factors, organizational factors, hardware, weather or environment, and airspace or ATC

These formed the parent nodes during the coding process, which are explained in detail in the next section. Under these parent nodes are all the smaller components of the taxonomies. Each of

these taxonomies are presented in Appendix A which includes the full list of the child nodes that will be used when coding.

Coder Qualification and Training

To ensure consistency and validity in the coding process, specific qualifications for individuals responsible for coding were established. A related study conducted by Sarab and Farsani (2023) reviewed the strengths and weaknesses of studies in applied linguistics journals, finding that more reliable data collection and analysis could be achieved by paying closer attention to coder qualification and training. A majority of the 90 studies that Sarab and Farsani (2023) included in their research did not report any information regarding coder qualification or training.

As the ASRS reports being considered are written by pilots, it was important that coders have some level of aviation experience to be able to understand common abbreviations and terminology. As such, each coder had at minimum, a Private Pilot Certificate, to ensure a basic understanding of aviation language. Additionally, because the reports relate to airline operations, coders needed to possess a robust foundation of airline-specific expertise, either from direct professional experience or from an extensive academic understanding of airline operations and procedures. Furthermore, since the reports were aviation safety and human-factors based, coders needed formal training in these areas, either through higher education in aviation safety or human factors; courses offered through professional organizations; military training focused in these areas; or equivalent formal training.

While limited studies offer qualifications for coders with respect to ASRS or aviation studies, these qualifications are designed to ensure each coder has a similar experience and knowledge of the material. They also exceed the qualifications expected for individuals applying

for positions as ASAP analysts, and other analyst positions within the airline industry (United Airlines, 2023; jetBlue, 2023; Department of Transportation, 2023). Given that the ASAP is analogous to the ASRS and shares reports with it, analysts involved in ASAP are likely to possess comparable backgrounds and expertise. This level of experience and training is reflected in the coders for this study, who were required to read and analyze the same reports as ASAP and ASRS analysts. Although prior literature does not specify explicit qualifications for coders, several studies have outlined effective training practices. Because the selected coders had not used NVivo and were not well-versed in qualitative coding, it was necessary to establish a process for training and checking coding. Studies by Giesen and Roeser (2020), Vaughn and Turner (2015), and Yang et al. (2017) establish a similar process for training and checking.

First, the coders were introduced to the code book, which in this case comprised the three established taxonomies used to code the reports. To begin the training process, an example report was reviewed collectively by the team. Working through the example together allowed the coders to become familiar with both the coding software and the process of assigning codes to textual data. During this step, special attention was paid to calibrating the granularity of coding and ensuring that only the most relevant phrases were coded, rather than larger, less-specific sections of text.

Based on the recommendations of Yang et al. (2017) the coders used collaborative discussions and worked to establish a shared understanding of the coding categories. Although the taxonomies employed in the study already offered well-defined codes, this phase was crucial for ensuring that all coders interpreted the definitions consistently. This reduced the risk of subjective bias or misapplication of the codes, thereby enhancing the reliability of the analysis.

Coders were then provided with another example report, which they coded independently. Upon completion, the team reconvened to review how each coder analyzed the report. The aim of this step was not to alter individual coding decisions, but rather to assess and reinforce consistency in how codes were applied and to ensure uniformity in the amount of text selected for coding. This peer review process also offered an opportunity to resolve any lingering ambiguities, clarify decision-making rationales, and reinforce the coding guidelines established earlier in the training.

Giesen and Roeser (2020) offer additional insight into establishing an effective and reliable team coding process. Central to their approach is the emphasis on open communication, where no coder should feel hesitant to ask questions or seek clarification at any stage of the process. To support this, both group and individual check-ins were implemented at regular intervals to monitor progress, address any uncertainties, and ensure that coders felt supported throughout the training and coding process. These check-ins served not only as a mechanism for problem-solving but also as an opportunity to reinforce consistency and shared understanding across the team.

Importantly, coding was recognized as a skill that develops over time, and coders were encouraged to view the training process as iterative and developmental (Giesen & Roeser, 2020). Depending on prior experience and familiarity with qualitative analysis, some coders may have required only one or two reports to reach competency, while others may have needed more extensive practice to become confident in using the software and applying the codes correctly. The pace of training was therefore adapted to meet the needs of individual coders, with the overarching goal of achieving a high level of inter-coder reliability.

The training phase was critical to ensuring that all codes were applied in a consistent and systematic manner, thereby minimizing discrepancies in interpretation during the later stages of analysis. To facilitate this, detailed reference materials were provided to all coders, including comprehensive documentation on the three taxonomies used in the study, complete with definitions, coding criteria, and illustrative examples. These materials served as essential tools to support decision-making during the coding process and to reduce variability in how codes were applied. In addition to these resources, coders retained access to the example reports completed during training. These reports functioned as practical benchmarks and reference points, allowing coders to revisit specific instances of coding as needed to guide their work and maintain alignment with the established coding framework.

Data Analysis Approach

NASA grants public access to the ASRS database reports that were used for this research (NASA, 2021). The reports, downloaded from the ASRS website, were uploaded into Microsoft Excel, where they could be organized and formatted to be uploaded into the NVivo software. The reports were filtered to include only those where human factors were identified as the primary problem. Additional filters excluded incidents involving aircraft equipment malfunctions, crew or passenger illness or injury, and fume or fire events. Each report was reviewed to ensure that only those involving crew communication and coordination errors were included in the study. This set of filtered reports was then analyzed using NVivo based on the selected taxonomies.

Each selected report was classified within each of the three taxonomies: TEM, HFACS, and ACCERS, then analyzed with respect to the model of the taxonomy and assigned classifications for each applicable construct within that taxonomy. These classifications were

also compared to the given anomaly codes to uncover potential gaps in the current anomaly code system and overlaps in taxonomies.

The coding process in NVivo was interactive or manual—coders highlighted words or phrases relating to a specific theme so they could be compared to other highlighted text. To begin, a node hierarchy was developed, which came directly from the taxonomies. As an example, TEM was coded as the theme. Threats, errors, and undesired aircraft states were coded as parent nodes and the breakdown of each of these items (found in Appendix A), were coded as child nodes. Typically, these node hierarchies undergo validation by additional subject matter experts. However, since the taxonomies employed in this study have already been validated in prior research and were used in their original form, further expert validation was unnecessary.

Each report was reviewed and coded based on each of the three taxonomies selected: TEM, HFACS, and ACCERS. The relevant information for each child/parent node was highlighted and assigned to the relevant code. This process was completed three times for each report, once for each of the taxonomies.

Reliability and Validity

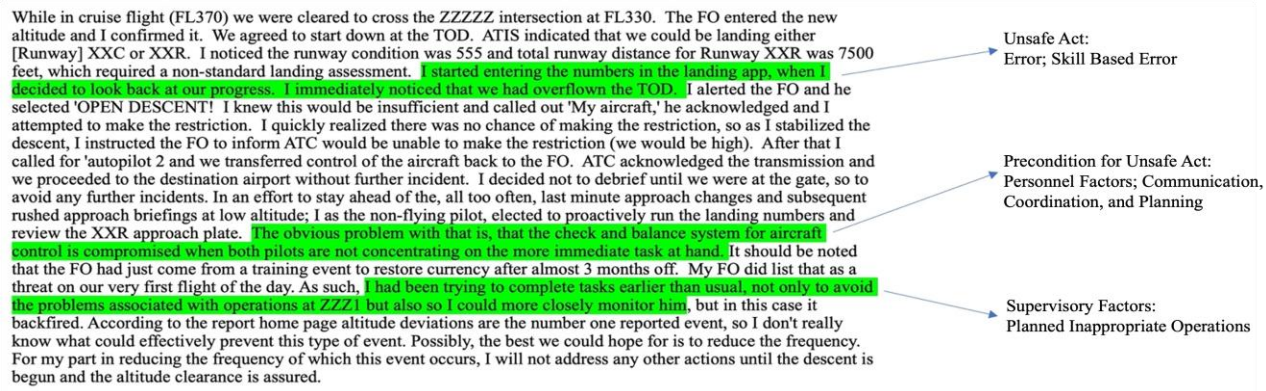
As interactive coding is based on an individual's perception and interpretation, it is important to check for reliability and validity. A random selection of 50% of the reports was coded in the same manner by two additional researchers with expertise in these taxonomies and ASRS reports, as described previously. Half of these sample reports overlapped between all three researchers and the other half for the two additional coders were different from each other. This means that 25% of all the reports were coded by all three coders and 12.5% of reports completed by the two additional coders only overlapped with the primary researcher. A coding comparison was completed for these three sets of coded data to evaluate Kappa and the level of agreement

between all researchers. If significant differences were found, the coding would be revised and another test of reliability and validity for the codes would have been conducted. If the three researchers had similar codes, then it can be concluded the codes are reliable.

An example of this coding process is shown in Figure 2, an ASRS report coded with HFACS. In this example, the green text is coded with the parent nodes Unsafe Act, Precondition for Unsafe Act, and Supervisory Factors, along with their stated child nodes as defined in the HFACS taxonomy. As HFACS has three levels of taxonomy, there are three coded layers in the node hierarchy. Because no statement in the report fits the definition of Organizational Influences, it is excluded. Coders strictly follow the taxonomy definitions when assigning codes to the corresponding nodes. All influencing factors and error details that matched the taxonomy definitions were included during coding. While some reports contained detailed narratives that allowed the identification of multiple terms, others offered only limited information. Coders were not responsible for assigning responsibility or determining causality; their role was to identify and code textual elements in the reports that matched the predefined taxonomy criteria.

Figure 2

Coding Example



Once the reports were coded, a series of steps was followed to analyze the results and form a framework for how each taxonomy was used to help answer the research questions. First, the descriptive statistics were considered, looking at the statistics of the reports and identifying common attributes for further examination. For the ASRS reports, these statistics focused on the frequency that each word/element in the taxonomy was used, providing a basis for determining which taxonomy might be the most prevalent. Demographics of the pilots were also collected and analyzed to ensure there was a representative sample of Part 121 pilots and that the study could be generalized to the larger pilot population. If the demographics revealed significantly more reports from captains or first officers or more high versus low time pilot reports, an additional sample of reports could be used to address this divergence.

Coding queries were used to explore emerging patterns, test ideas within those patterns, and gain insight into the connections between common themes. As these themes emerged, matrix coding was used to observe the connection between two areas or themes. Queries were used to examine all the coded references at a particular node or multiple nodes, allowing the researcher to read them together and develop deeper understanding. These queries were used to examine how different taxonomies overlapped in a report, even when the taxonomies had dissimilar wording for that reference.

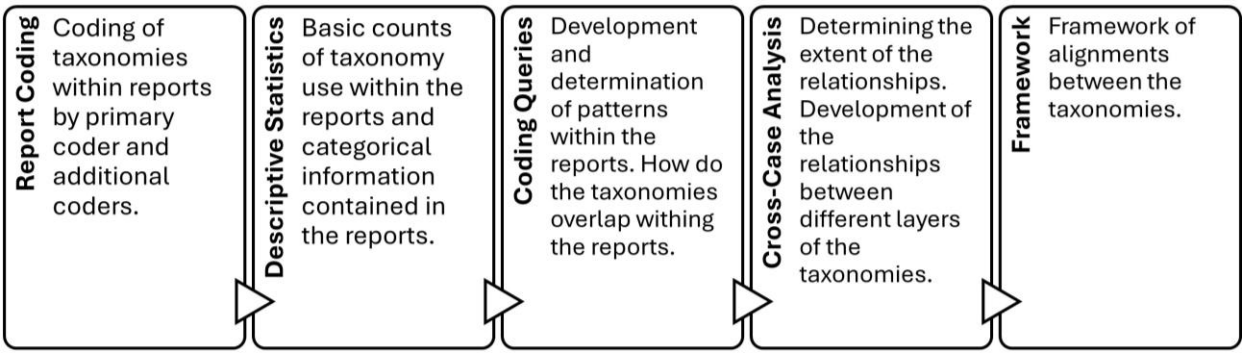
Next, cross-case analysis was used to consider the extent of each association and observe similarities and differences within connections. As some taxonomies use a layered approach, such as TEM, a cross-case analysis showed how coded sections were related to each other and to the other taxonomies. For example, since TEM is a model where a threat can cause a pilot to make an error leading to an undesired aircraft state, each layer of this event each might align differently to another taxonomy which does not use a layered concept.

A new framework was then developed showing the connections of each taxonomy-coded section within the reports, considering how each report was analyzed and coded with each taxonomy (Bazeley, 2013). Through coding queries and cross-case analysis, a series of connections and relations between the taxonomies was determined. By analyzing overlaps, as well as sections covered by only one taxonomy, a detailed framework was developed to show these interactions. This framework answered the first research question: what significant alignments exist between the taxonomies and showed these alignments based on the coded words for each taxonomy. Because each report was also classified with the taxonomies selected for this research, these classifications are also related through the new framework, which enabled identification of how each classification relates to the anomaly codes assigned by ASRS analysts – the second research question. By having a detailed framework of the connections and overlaps between the selected taxonomies, along with the classifications of each report, the anomaly codes the analysts assigned to the framework could be assessed to determine how they relate to the taxonomies. This comparison allowed gaps or missing details in the anomaly codes to be examined and used for recommendations to create more detailed anomaly codes.

The process described above for analyzing ASRS reports is depicted in Figure 3.

Figure 3

Text Analysis Process and Description



Note. Figure created by the author from methodology adapted from Bazeley, 2013.

A simple descriptive analysis was used to assess the third research question: what are the changes in reporting of TEM, HFACS, and ACCERS elements in ASRS pilot reports? A count of the use of terms and categories of the selected taxonomies through each year was done using the coded reports to determine any significant changes in reporting that occurred because of training or safety program changes. Significant changes that were considered included the introduction of SMS in 2018, the introduction of AQP in 2007, and a focus on CRM training that occurred with the introduction of AQP. The coded sections were also analyzed to determine if there were any shifts in taxonomy use outside of these larger events. As this is a descriptive design, there was no manipulation of the variables and, as such, causation cannot be determined. However, examining these changes did offer insight into the impacts these training and safety program changes might have had on a pilot's use of the selected taxonomies.

The taxonomy framework developed formed the foundation for a more encompassing theory and user-based classification system, which addresses the concern about the lack of pilot input during development of the anomaly code system and the question about the system not

being based on previous theories of taxonomies. By using the pilot reports and the most common elements of each taxonomy, the framework aligns with these reports and the wording the pilots use. Additionally, accounting for changes in training and safety systems developed a basis for how these changes impact taxonomies used and how they might change with future introductions. Lastly, comparing the framework and the classified reports to current anomaly codes revealed where there were potential gaps in the codes, where additions could be made, and how the anomaly codes relate to the theories found in the other taxonomies.

Analysis and Framework Validation

The new framework identifies and proposes novel connections between ASRS anomaly codes, and the nodes derived from three selected taxonomies. The validity of the framework was evaluated to determine whether the instrument effectively measured its intended constructs and provided accurate results (Hair et al., 2010). To test the framework validity, ASRS reports that met the same criteria as the reports used in framework development were sampled over the same period. These reports were selected to have different ASRS anomaly codes through stratified sampling. The reports were then tested through the framework to determine the classifications from each of the three taxonomies, TEM, HFACS, and ACCERS, assigned based on their given anomaly codes. The reports were then interactively coded in the same process as the original reports, highlighting the parent and child nodes of each taxonomy.

Agreement between the results from the classification of the reports through the framework and the results from the interactive coding were calculated. A strong agreement between the outputs of the framework and the interactively coded taxonomies, based on Cohen's kappa, confirmed the connections found in the framework development. This process was completed independently for each taxonomy.

Chapter IV: Results

The methodology discussed in Chapter III was used to develop the following framework and results. The framework presents the overlap of the coded reports for the studied taxonomies and their relation to the ASRS taxonomy, using reports of communication and coordination errors from the ASRS database from 2007 to 2022. Descriptive statistics, coding queries, and cross-case analysis were used in the process to develop and validate the frameworks.

Demographics Results

As ASRS reports are deidentified, there are limited demographics that can be studied; however, crew positions and experience are available. Since some reports did not contain information for every category, demographic data is limited to reports with the relevant information. Available information varied across reports; the absence of data for one category did not necessarily indicate missing information in other categories.

Crew position was analyzed across the reports: 229 were submitted by captains, 125 by first officers, and 6 by check pilots or instructors. These results are shown in Table 1. As this study focused on communication and coordination, it was assumed that captains and first officers would be equally responsible for these events. However, it is typically the captain who is responsible for making any reports of deviations or discrepancies that would result from a breakdown in communication or coordination. In the scope of this study, the reports are submitted by captains 64% of the time, which aligns with most airlines, where the captain is responsible for reporting when there is a non-standard occurrence during the flight. So, if the communication or coordination error led to an altitude deviation, it would likely be the captain reporting this event.

Table 1

Pilot Position

Pilot Position	Number of Reports	% of Total
Captain	229	63.6%
First Officer	125	34.4%
Check Pilot	3	0.8%
Relief Pilot	3	0.8%

The role of the reporter is broken down into Pilot Flying (PF), Pilot Not Flying (PNF), and Relief Pilot. In a 2003 Advisory Circular, the FAA recommended the term Pilot Not Flying should be replaced with the term Pilot Monitoring (PM) (AC120-71A). The FAA emphasized that the pilot should be characterized by what they are doing rather than what they are not doing. The PF is the pilot controlling the path of the aircraft, in the air or on the ground. The PM is responsible for monitoring the aircraft state and the status of the systems, acknowledging any potential deviation and intervening when required (AC 120-71B). As this study considers communication and coordination errors, which typically involve both pilots, and there is no way of knowing if the PF or PM was the captain or first officer, it is reasonable to conclude that these reports should be relatively equally divided between the two. The demographics of the reports, found in Table 2, confirm this, with 165 reports from PF, 154 from PNF.

Table 2

Pilot Role

Pilot Role	Number of Reports	% of Total
Pilot Not Flying	154	48.3%
Pilot Flying	165	51.7%

Table 3

Flight Crew Hours

	Hours	Number of Reports
Flight Crew in Past 90 Days	179.4	154
Flight Crew Total	11,850	107
Flight Crew Type	4,011	140

Another fact included with demographics in ASRS reports is time experience for the crew, reported in time total in the last 90 days and in the type of aircraft. However, only one of the two pilots is filling out the ASRS form and may not know the other pilot's time experience; so, it is possible these numbers are not completely accurate, and there is no way to validate them. Again, some of the values are left blank, so the following averages are of those that were reported. The average total time for the crews submitting the reports included in this study was 11,850 hours. Time in the last 90 days was 179 hours per crew. Time in type for each crew was an average of 4,011 hours. Time in the last 90 days would result in an average of about 30 hours per crew member per month. These averages are presented in Table 3. While times vary based on airline and fleet type, this average is lower than expected. There are no exact averages for what a commercial pilot flies in one month, but ranges are suggested between 60-75 hours (Chamerlik, 2024).

Descriptive Statistics

The initial assessment of the ASRS database showed there were about 6,800 reports that met the qualifications of the study. It was calculated that a total of 365 reports would be needed to meet the required confidence level and margin of error for this study. These reports were taken proportionally from each year from 2007 to 2022.

Table 4

Sample Size Per Year

Year	Available Reports	Sample Size
2007	517	28
2008	646	35
2009	397	21
2010	291	16
2011	282	15
2012	259	14
2013	266	14
2014	343	18
2015	703	38
2016	516	28
2017	425	23
2018	579	31
2019	484	26
2020	434	23
2021	377	20
2022	289	15
Total	6808	365

Table 4 details how many reports were submitted each year that met the criteria of the study and how many reports from that year were sampled for this study.

Of the 356 reports selected, 11 did not provide enough narrative detail to be effectively analyzed, so they were excluded. Each of these reports was replaced with a different randomly selected report from the same year.

Additional descriptive statistics were computed for variables including lighting conditions, flight conditions, local time of day, and phase of flight, as presented in Tables 5 through 8. Accounting for these factors enhances the generalizability of the study's findings.

Table 5

Environment – Light

Environment Light	Number of Reports	% of Total
Night	54	25.5%
Dusk	9	4.3%
Daylight	141	66.5%
Dawn	8	3.8%

Table 6

Environment – Flight Conditions

Environment Flight Conditions	Number of Reports	% of Total
VMC	173	80.8%
Mixed	7	3.3%
Marginal	5	2.3%
IMC	29	13.6%

Table 7

Local Time

Local Time	Number of Reports	% of Total
0001-0600	43	13.0%
0601-1200	100	30.1%
1201-1800	116	34.9%
1801-2400	73	22.0%

Table 8

Phase of Flight

Phase of Flight	Number of Reports	% of Total
Parked	86	23.6%
Taxi	79	21.7%
Takeoff / Launch	31	8.5%
Initial Climb	6	1.7%
Climb	27	7.4%
Cruise	23	6.3%
Initial Approach	29	8.0%
Final Approach	22	6.0%
Descent	35	9.6%
Landing	17	4.7%
Other	9	2.5%

Coding Result Statistics

Each of the 365 reports was coded using the three selected taxonomies. Codes from ACCERS, HFACS, and TEM were assigned to the report sections that fit the code definitions. Coding was only assigned when there were clear and unambiguous examples meeting the code's definition, so not all codes from each taxonomy were applied. Codes from ACCERS were assigned 1,315 times, HFACS codes were assigned 1,240 times, and TEM codes were assigned 1,106 times. The values above indicate that each report had an average of 3 to 4 codes assigned per taxonomy.

Reliability and Validity Testing Results

To test for reliability, two additional coders coded 50% of the reports and then compared their results to the original coding. These additional reports were selected in the same manner as the original sample. NVivo allows for the calculation of Cohen's kappa coefficient (κ) and the percentage agreement between the coders. Typically, κ is used to determine if there is a good

measure of inter-rater reliability because it accounts for the probability that researchers selected the same codes by chance. However, if most of a source remains uncoded to a node, but the coders have coded different small sections to a node, κ will be small, as this is likely to occur by chance.

Cohen's κ in NVivo is calculated based on the degree of agreement between two coders who apply the same node to the same segments of text. NVivo generates a contingency table that compares coding assignments across the same dataset. For example, if both Coder A and Coder B assign a node to the exact same segment of text, calculated at the character level, this is treated as a full agreement, contributing positively to the observed agreement score. Conversely, if the coders apply the same node to entirely different segments of text, even within the same report, this is recorded as a disagreement, and the observed agreement is effectively zero for that unit.

The observed agreement is then adjusted to produce κ by accounting for the agreement that might occur by chance, producing a more conservative and meaningful measure of inter-coder reliability. However, in practice, when coders select different (non-overlapping) portions of text for the same node—especially in lengthy, narrative-rich data like the ASRS reports—this can result in a low observed agreement, thereby lowering the overall κ value. This sensitivity to exact text overlap highlights a limitation of κ in some qualitative contexts, where differences in segment boundaries may not reflect true conceptual disagreement, but still negatively affect the reliability coefficient. As such, for this study, κ is considered but also supplemented by the percentage of coder agreement.

There is a calculated agreement between the coders of $\kappa = 0.34$. Agreement was also calculated by taxonomy, with ACCERS having the highest at $\kappa = 0.36$, HFACS with the lowest agreement at $\kappa = 0.32$, and TEM with an agreement of $\kappa = 0.34$. Although the observed κ value

is lower than what is typically strong agreement, this result was expected given the nature of the data.

NVivo also calculates the percentage agreement between coders. For this study, the agreement was 89.5% between the two secondary coders and the primary coder for nodes where text was assigned. A value of almost 90% represents very good agreement. Nodes without any assigned text are calculated as 100% and artificially inflate the agreement. Using all codes, the agreement is 93%. The breakdown of percentage agreement for each node can be found in Appendix C.

To ensure coding validity, the process was grounded in the established, theory-based taxonomies: HFACS, TEM, and ACCERS. These taxonomies have been extensively validated in aviation human factors research, providing a solid foundation that enhances construct validity by ensuring the codes accurately represent key dimensions of pilot communication and coordination errors. Additionally, coders were trained on the taxonomy frameworks and coding procedures to ensure a common understanding of definitions. During the training, coders independently coded sample reports and then met to discuss discrepancies and align their interpretations. This iterative training and calibration process ensured that the coding was applied accurately and in accordance with the intended definitions, reducing subjective bias and increasing the validity of the results.

To ensure coding reliability, NVivo allows for a detailed study of the coding comparison content. This tool reviews codes that have a significantly lower κ or low percentage agreement value. In most cases of lower κ, researchers had chosen different text to code. While percentage agreement looks at how much was coded to a node, κ looks specifically at what was coded, resulting in the low κ but the high percentage agreement.

Overall, there is fair reliability of the codes. While κ is lower than what is generally considered good agreement, this was expected based on the types of reports. Areas that were identified as significantly lower were examined and showed that the disagreement was because of the length and detail of the reports, leading coders to select different content to code. The high percentage agreement leads to the conclusion that the codes are reliable and valid.

Additionally, the reliability of qualitative research depends significantly on the consistency and transparency of the data analysis process. In this study, several deliberate methodological steps were taken to enhance the reliability of the coding process and, by extension, the overall findings.

Firstly, coders were thoroughly trained using a structured approach that involved exposure to a detailed code book, developed from three established taxonomies. This initial phase included a collaborative walkthrough of an example report, which served to familiarize coders with the software and ensure agreement on the scope of coding—focusing on coding only the most relevant and meaningful phrases. This shared exercise laid the groundwork for consistency in code application across the team.

Building on this foundation, a collaborative process was used to refine and clarify the definitions of each code. While the selected taxonomies already provided strong conceptual frameworks, this phase allowed the team to eliminate potential ambiguities and ensure mutual understanding of each coding category (Yang et al., 2017). As a result, coders entered the analysis phase with a unified interpretation of the coding schema, reducing the likelihood of discrepancies in code assignment.

To further reinforce reliability, coders individually coded a second example report and then engaged in a peer review process. This step was not intended to alter coding judgments but

to ensure uniformity in the application of codes and in the extent of the narrative being analyzed. Such calibration exercises are crucial for promoting inter-coder agreement, a key indicator of reliability in qualitative analysis.

In alignment with the recommendations of Giesen and Roeser (2020), the study also emphasized the importance of ongoing communication throughout the coding process. Coders were encouraged to raise questions and seek clarification when needed, fostering an open and collaborative environment. Regular group and individual check-ins were implemented to monitor progress, address challenges, and ensure that no coder worked in isolation. This continuous feedback loop contributed to consistency in both process and output.

Moreover, the training process was intentionally adaptive, allowing coders to build their skills at a pace suited to their prior experience and learning needs. Whether a coder required minimal or extensive practice, the primary objective remained the same: to ensure accurate and consistent application of codes across the dataset.

Finally, coders were provided with comprehensive reference materials, including detailed definitions of the three taxonomies and annotated examples from training sessions. These resources served as crucial points of reference throughout the coding process, supporting uniform decision-making and reducing subjectivity.

Together, these strategies significantly enhanced the reliability of the study by promoting systematic, replicable, and consistent coding practices. Through rigorous training, collaborative clarification, structured peer review, and continuous support, the coding process was designed to minimize individual bias and maximize inter-coder consistency—key components in producing trustworthy qualitative research.

Coder Qualifications

As described previously, two additional coders worked on these reports to provide a reliability and validity assessment. Qualifications of these coders are detailed in Chapter III. The individuals that were selected have the following qualifications: Coder 1 is a pilot for a major Part 121 U.S. carrier and is involved in the airline's test flight program. This individual was previously a military pilot and safety officer and attended multiple formal training courses in human factors and safety training. Coder 2 is a check airman for a major Part 121 U.S. carrier. Coder 2 has a civilian background and was a flight instructor before transitioning to business aircraft operations and then commercial aircraft operations. Coder 2 also holds a master's degree in human factors.

Coding Results

As described in the methodology, the first step in analysis was to examine how the codes were distributed and what codes were assigned most frequently, as shown in Figure 4. Communication errors were the most frequently coded texts for each taxonomy, which was expected because the reports were focused on communication and coordination errors. Next-highest coded items related to the state of the pilot and external influences. While Figure 4 is a simple and broad overview, understanding the order of items coded provides a starting point for further analysis. Figure 4 also provides a depiction of the number and distribution between the taxonomies of the coded information, providing the first step in answering the first research question.

Figure 4

Hierarchy Chart of Categorical Codes

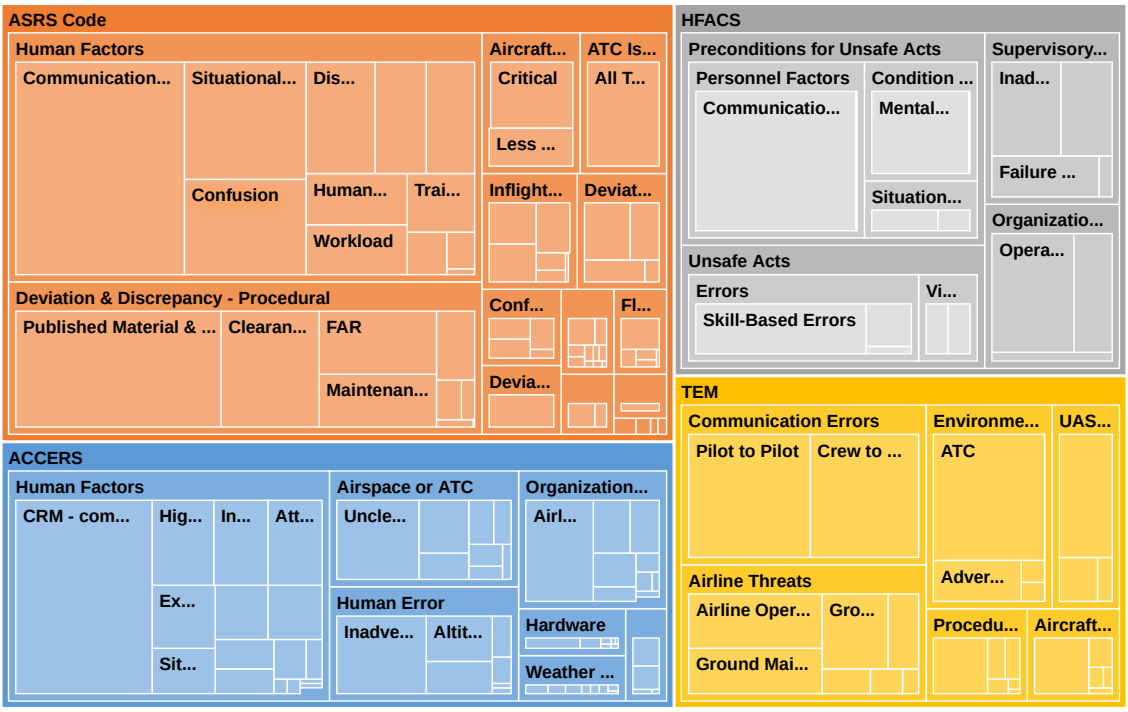


Figure 4 was created by NVivo, and is interactive within the software, but this does not translate well to the static depiction shown here. However, it does provide a good comparative depiction of how many items are coded to each node. Tables 9 through 12 show the exact numbers that make up what is shown in Figure 4. These tables are simplified, removing any node that had less than 10 coded references. Full tables for these taxonomies can be found in Appendix B.

Table 9

ACCERS Coded References

Codes	Number of coding references	Aggregate number of coding references
ACCERS	**0**	**1315**
Airspace or ATC	**0**	**190**
Unclear ATC clearance	89	89
Conflicting ATC clearance	36	36
Frequency congestion	18	18
Hear back & read back	15	15
Late ATC clearance	11	11
Expected vs. assigned ATC clearance	10	10
Hardware	**0**	**44**
Aircraft equipment malfunction	26	26
Aircraft damage	10	10
Human Error	**0**	**188**
Inadvertent or intentional disregard for policy or procedure	97	97
Altitude planning & awareness	42	42
Improper use of autopilot, FMS, FMC navigational equipment	29	29
MEL & AML error	11	11
Human Factors	**0**	**640**
CRM - communication	286	286
High workload & task saturation	57	57
Attention to detail	50	50
Interruption - distraction	50	50
Experience level	43	43
Fatigue	31	31
Personal attitude(s) toward safety	31	31
Situational awareness	31	31
Complacency	19	19
CRM - leadership and command	15	15
Self-induced time pressure	14	14
Organizational Factors	**0**	**177**
Airline's safety culture	90	90
Inadequate training	25	25
Inadequate ground management	21	21
On-time performance pressure	19	19
Policies or Procedures	**0**	**34**
Lack of policy or procedure	17	17
Confusing policies and procedures	14	14
Weather or Environment	**0**	**42**
Thunderstorm or severe weather	10	10

Table 10

ASRS Coded References

Codes	Number of coding references	Aggregate number of coding references
ASRS Code	**0**	**2167**
Aircraft Equipment	**1**	**114**
Critical	72	72
Less Severe	41	41
ATC Issues	**0**	**100**
All Types	100	100
Conflict	**0**	**49**
Ground Conflict, Critical	16	16
Airborne Conflict	15	15
Ground Conflict, Less Severe	14	14
Deviation - Altitude	**0**	**85**
Overshoot	37	37
Crossing Restriction Not Met	25	25
Excursion from Assigned Altitude	19	19
Deviation - Speed	**0**	**19**
All Types	19	19
Deviation - Track & Heading	**0**	**45**
All Types	45	45
Deviation & Discrepancy - Procedural	**0**	**590**
Published Material & Policy	264	264
Clearance	126	126
FAR	82	82
Maintenance	68	68
MEL & CDL	29	29
Weight and Balance	11	11
Flight Deck & Cabin & Aircraft Event	**0**	**37**
Other or Unknown	23	23
Ground Event & Encounter	**0**	**37**
Other or Unknown	14	14
Ground Incursion	**0**	**26**
Runway	18	18
Human Factors	**0**	**968**
Communication Breakdown	355	355
Situational Awareness	141	141
Confusion	116	116
Distraction	76	76
Troubleshooting	57	57
Time Pressure	53	53
Human-Machine Interface	51	51
Fatigue	17	17
Workload	50	50
Training and Qualification	39	39
Other or Unknown	10	10
Inflight Event & Encounter	**0**	**90**
Weather or Turbulence	27	27
Unstabilized Approach	26	26
CFTT & CFIT	24	24

Table 11

HFACS Coded References

Codes	Number of coding references	Aggregate number of coding references
HFACS	**0**	**1240**
Organizational Influences	**0**	**185**
Operational Process	117	117
Organizational Culture	56	56
Resource Management	12	12
Preconditions for Unsafe Acts	**0**	**544**
Condition of Operators	0	143
Mental State	141	141
Personnel Factors	0	332
Communication, Coordination & Planning	330	330
Situational Factors	0	69
Physical Environment	46	46
Tools & Technology	23	23
Supervisory Factors	**0**	**195**
Inadequate Supervision	77	77
Planned Inappropriate Operations	59	59
Failure to Correct Known Problem	53	53
Unsafe Acts	**0**	**316**
Errors	0	252
Skill-Based Errors	198	198
Decision Errors	45	45
Violations	0	64
Routine Violations	33	33
Exceptional Violations	31	31

Table 12

TEM Coded References

Codes	Number of coding references	Aggregate number of coding references
TEM	**0**	**1106**
Aircraft Handling Errors	**0**	**74**
Automation	51	51
Flight Control	12	12
Airline Threats	**0**	**284**
Airline Operational Pressure	89	89
Ground Maintenance	76	76
Ground & Ramp	61	61
Aircraft	28	28
Dispatch & Paperwork	15	15
Manuals & Charts	10	10
Communication Errors	**1**	**332**
Pilot to Pilot	174	174
Crew to External	157	157
Environmental Threats	**0**	**220**
ATC	166	166
Adverse Weather	43	43
Procedural Errors	**0**	**80**
Other Procedural	50	50
Documentation	12	12
Checklist	10	10
UAS Types	**0**	**116**
Aircraft Handling	85	85
Ground Navigation	22	22

Figure 4 can also be broken down by taxonomy for a clearer view of the coded references within each taxonomy. These are presented in Figures 5 through 8.

Figure 5

Hierarchy Chart of TEM Codes

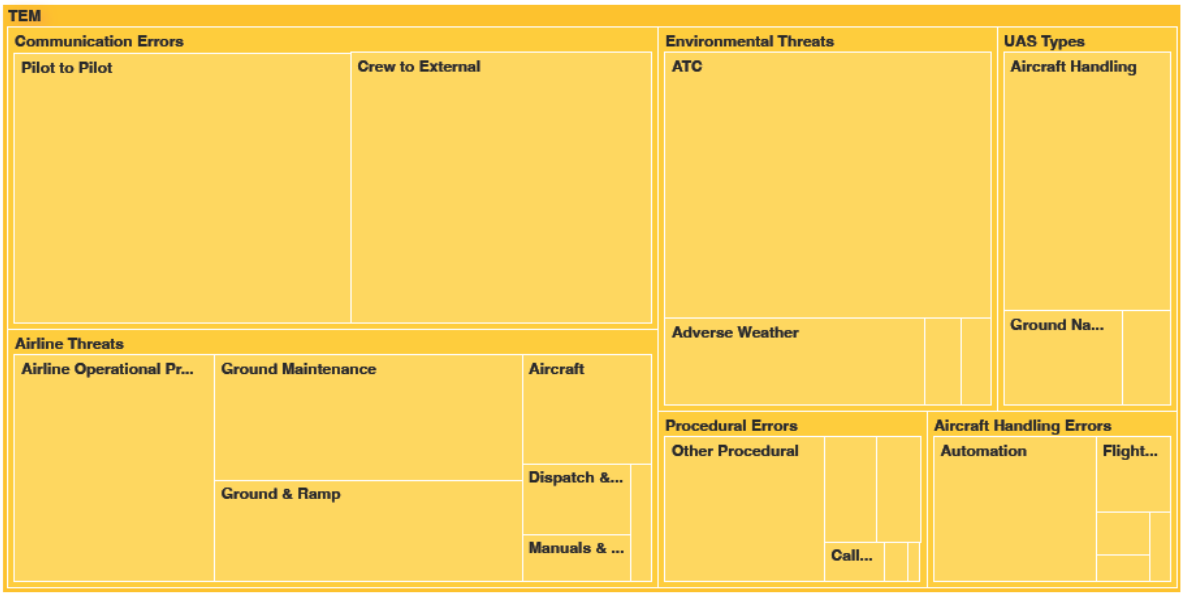


Figure 6

Hierarchy Chart of ACCERS Codes

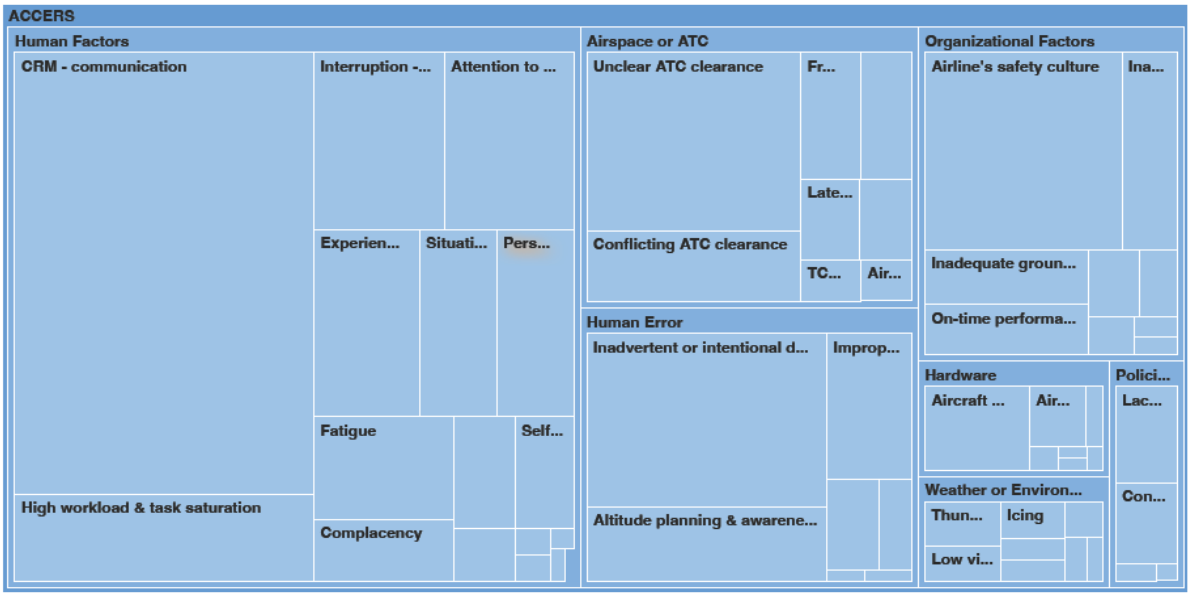


Figure 7

Hierarchy Chart of HFACS Codes

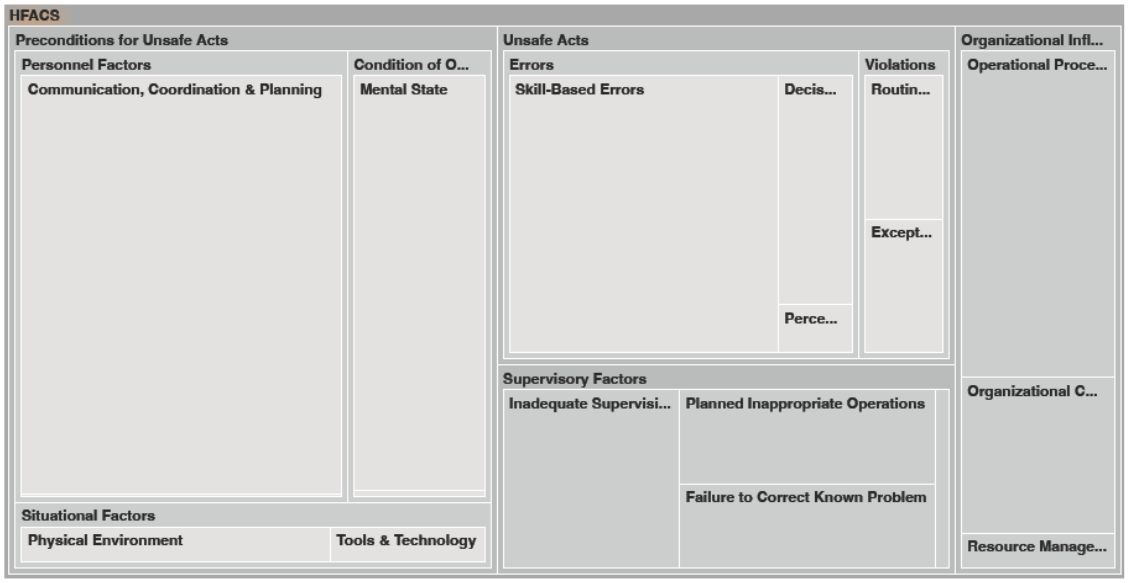
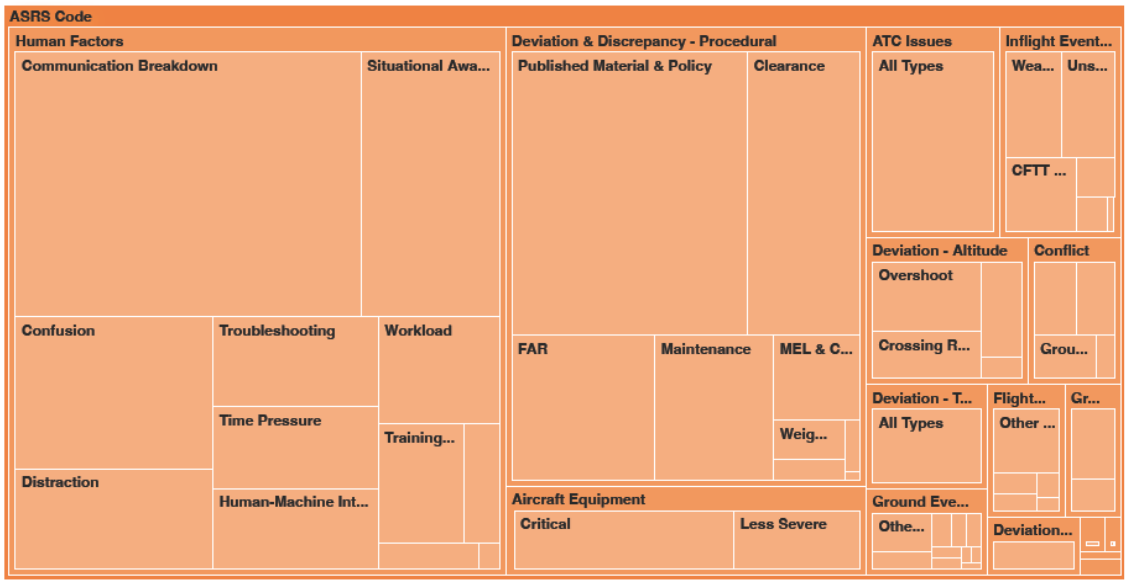


Figure 8

Hierarchy Chart of ASRS Codes



Coding Query

As the study focused on communication and coordination errors, these were explored first through coding queries. Queries were used to search for each individual communication and coordination code, found in Table 13, along with combinations of these codes based on their respective taxonomies. For example, Communication, Coordination, and Planning from HFACS was compared with the ACCERS node of CRM – Communication and the TEM node of Pilot to Pilot communication. Combinations for each pair of communication and coordination nodes, along with other commonly coded nodes, were examined. Some additional nodes included in these queries include Deviation & Discrepancy – Procedural\Published Material & Policy, Skill-Based Errors, ATC, Situational Awareness, and Condition of Operators\Mental State. These queries were used to quantify and visualize how much each taxonomy captured similar or different aspects of communication and coordination errors.

These queries identified sections of narrative text that were coded with one or more communication or coordination codes, revealing which codes most frequently overlapped within the same text segments. For the three selected taxonomies, there were 312 coded references to codes that dealt with communication and coordination across the three taxonomies. These references include the codes in Table 13.

Table 13

Communication and Coordination Codes

Taxonomy	Included Taxon
ACCERS	CRM - Communication
	CRM - Conflict Management
	CRM - Incompatible Crew Paring
	CRM - Leadership and Command
HFACS	Communication, Coordination, and Planning
TEM	Crew to External
	Pilot to Pilot

While all CRM codes for ACCERS are included, the codes primarily fall into the category of CRM – Communication. However, additional communication issues and coordination errors can fall into any of the CRM codes.

A total of 206 coded report sections were coded to ACCERS: CRM – Communication; HFACS: Communication, Coordination, and Planning; and TEM: either Pilot to Pilot or Crew to External. Crew to External communication errors were included because they can result in crew members having coordination errors, such as having different interpretations of an ATC request or different ideas on a maintenance issue.

Other commonly used codes and the number of times they were coded in the reports are reported in Table 14.

Table 14

Most Frequent Assigned Taxon, Excluding Communication Specific Taxon

Taxonomy	Taxon	Number of References
ACCERS	Inadvertent or Intentional Disregard for Policy or Procedure	97
	Airline's Safety Culture	90
	Unclear ATC Clearance	89
	High Workload and Task Saturation	57
	Attention to Detail	50
	Interruption/Distraction	50
HFACS	Skill Based Error	198
	Mental State	141
	Operational Process	117
	Inadequate Supervision	77
	Planned Inappropriate Operations	59
	Organizational Culture	56
	Failure to Correct Known Problem	53
TEM	ATC	166
	Airline Operational Pressure	89
	Aircraft Handling	85
	Ground Maintenance	76
	Ground & Ramp	61
	Automation	51
	Other Procedural	50

The taxon listed in Table 10 relate to the communication or coordination error and the conditions and state of the pilot when they made the error. For example, mental state, airline operational pressure, high workload, task saturation, and interruption/distraction are all associated with how the pilot was feeling and include external pressures that may have led to the error. Additionally, taxon such as ATC, unclear ATC clearance, operational process, airline's safety culture, or ground & ramp are factors that create confusion or pressures that impact the

pilot, leading to the error. Together with the communication codes, these make up the majority of the coded sections in the report, so they are the focus of the following analysis and framework.

While HFACS, ACCERS, and TEM use different codes and terminology, analyzing the same data with these taxonomies reveals common themes related to crew communication and coordination errors. Despite differences in structure, several important factors consistently emerged through the analysis.

A key theme is the pilot's mental state, including cognitive distractions and task saturation. ACCERS explicitly codes for high workload and task saturation, HFACS highlights mental state and cognitive failures, and TEM focuses on managing internal threats related to human factors. Additionally, external pressures such as airline scheduling demands and ATC communications appear across taxonomies as contributors to communication breakdowns. The combined use of these taxonomies uncovers overlapping themes—especially pilot cognitive state and operational pressures—that are central to understanding communication and coordination errors in aviation and developing a framework to connect these factors.

Matrix Coding and Cross-Case Analysis

To develop a better understanding of how the taxon in each taxonomy relate to the other taxonomies, matrix coding and cross-case analysis were used to see how frequently the assigned taxon codes were applied to the same text within reports.

Several distinct concepts were explored through this analysis: communication and coordination, pilot metal state and workload, external pressures, and distractions.

It was necessary to identify which codes from TEM, HFACS, and ACCERS frequently co-occur in the same coded text segments. This analysis was conducted by performing matrix coding comparisons between the complete sets of codes within each taxonomy. Specifically, all

TEM codes were compared against the full range of HFACS codes, all TEM codes against ACCERS codes, all HFACS codes against ACCERS codes, and all ASRS anomaly codes against each of the three taxonomies. This systematic approach identified areas of convergence and divergence among the taxonomies in their classification of similar human factors and error-related phenomena.

As the study focused on communication and coordination errors, this was the first area of focus, exploring how communication and coordination codes from each taxonomy relate to each other. This analysis compared each of the codes presented in Table 9 with communication and coordination codes from all four taxonomies.

Next, the two themes that emerged from the coding queries were examined. To explore the first theme, pilot mental state and distraction, analysis was conducted to examined connections between pilot mental state codes and communication errors. For example, ACCERS codes such as Situational Awareness and High Workload/Task Saturation were compared against communication codes from the four taxonomies. Similarly, HFACS codes including Failure to Correct a Known Problem, Mental State, and Decision Errors were compared to all the communication and coordination codes. TEM was not as simple as ACCERS and HFACS because there are not specific human factors codes, but codes such as Ground Maintenance, Ground/Ramp, Manuals/Charts, and Aircraft caused many of the distractions that led to pilot workload, so these were used and compared to the communication and coordination codes. This analysis identified how pilot mental state relates to communication failures and shows how these items should be connected in the new framework.

The second theme that emerged from the queries relates to external pressures and miscommunications including factors like operations pressure and ATC. The study investigated

how organizational or environmental factors relate to communication and coordination breakdowns. For this analysis, TEM codes that fell under the category Environmental threats, ACCERS codes that were in the categories of Organizational Factors and Airspace or ATC, and HFACS codes in the category of Organizational Influences were compared to the communication and coordination codes found in Table 13.

Table 15 shows this comparison for the communication-specific codes of each taxonomy. On the top row are the codes for ACCERS, HFACS, and ASRS; these are compared to the left column of TEM, HFACS, and ASRS. HFACS and ASRS communication codes are duplicated and included as both a vertical column and horizontal row, allowing codes from each taxonomy to be compared in the same chart, where HFACS and ASRS are compared to themselves in the chart "N/A" (not applicable) is listed. As discussed for Table 13, CRM – Communication is the most used code within ACCERS for communication errors. However, despite their minimal use, CRM – Conflict Management, CRM – Incompatible Crew Paring, and CRM – Leadership and Command codes are also included in Table 15 because they are communication and coordination specific codes.

Table 15

Communication and Coordination Code Comparison

	CRM-Communication (ACCERS)	CRM – Conflict Management (ACCERS)	CRM – Incompatible Crew Pairing (ACCERS)	CRM-Leadership and Command (ACCERS)	Communication, Coordination, and Planning (HFACS)	Communication Breakdown (ASRS)
Crew to External (TEM)	87	0	0	0	101	126
Pilot to Pilot (TEM)	126	0	6	10	133	131
Communication, Coordination, and Planning (HFACS)	207	0	6	8	N/A	254
Communication Breakdown (ASRS)	216	0	6	13	254	N/A

From Table 15, it can be determined that Communication Breakdown (ASRS) and Communication, Coordination, and Planning (HFACS) have the highest number of areas coded to both. However, all the major codes (excluding CRM – Conflict management, CRM – Incompatible Crew Paring, and CRM – Leadership and Command) have a significant number of overlapping coded sections within the reports. CRM – Communication (ACCERS) and Crew to External (TEM) have the fewest communication codes. Because the study focused on internal cockpit communication, it was expected that crew-to-external would be less significant than pilot-to-pilot.

While the study focused on communication and coordination errors, factors that led to these errors and external influences were also coded and considered. Table 16 depicts the number of instances that a coded section of a report for ACCERS overlaps with a coded section for ASRS. These taxonomies have significantly more taxon than what is depicted in the table, but these are the most significant overlapping sections. Highlighted in red are the taxon that had

greater than 50 overlapping coded sections. The items in red are those that had significant overlap and were selected for further analysis within the framework.

Similar to Table 17, Table 18 shows the comparison of HFACS and ASRS codes. Again, only more significant taxon are included, and red indicates more than 50 overlapping coded sections.

Table 16

Comparison of ACCERS and ASRS Codes

		ASRS Codes								
		Critical	ATC Issues	Clearance	Maintenance	Published Material & Policy	Communication Breakdown	Confusion	Distraction	Situational Awareness
ACCERS Codes	Unclear ATC clearance	2	50	54	0	33	68	41	13	32
	Inadvertent or intentional disregard for policy or procedure	18	7	17	9	78	86	33	20	38
	CRM - communication	23	68	104	11	153	216	90	57	103
	Airline's safety culture	51	2	0	51	73	83	14	5	9

Table 17

Comparison of HFACS and ASRS Codes

		ASRS Codes						
		ATC Issues	Clearance	Published Material & Policy	Communication Breakdown	Confusion	Distraction	Situational Awareness
HFACS Codes	Operational Process	40	34	73	103	32	17	35
	Mental State	34	66	92	120	41	53	62
	Communication, Coordination & Planning	80	113	186	254	98	63	112
	Failure to Correct Known Problem	4	5	42	51	5	4	10
	Inadequate Supervision	21	33	56	71	23	18	31
	Planned Inappropriate Operations	23	16	45	55	22	11	23
	Skill-Based Errors	45	81	123	159	73	47	72

Table 18 shows the same information as Tables 16 and 17, this time for TEM and ASRS codes.

Table 18

Comparison of TEM and ASRS Codes

		ASRS Codes						
		ATC Issues	Clearance	Maintenance	Published Material & Policy	Communication Breakdown	Confusion	Situational Awareness
TEM Codes	Airline Operational Pressure	2	4	41	74	82	13	14
	Ground & Ramp	2	4	0	56	56	17	28
	Ground Maintenance	0	0	54	63	70	10	6
	Crew to External	54	38	20	79	126	53	46
	Pilot to Pilot	33	77	2	102	131	54	68
	ATC	103	85	0	68	134	69	65
	Aircraft Handling	20	46	0	56	71	26	39

By viewing tables 13, 16, 17, and 18, researchers can see the communication codes from each taxonomy in context with other codes from the selected taxonomies. The communication codes in each of these tables have the greatest number of overlapping coded sections.

Coders applied codes for ACCERS, HFACS, and TEM to specific sections, but they applied the codes to ASRS reports as a whole. This causes any code (ACCERS, HFACS, or TEM) applied to any section of a report to indicate an overlap with all ASRS codes applied to the report. Conversely, only sections of a report that are coded to two or three of the selected taxonomies will indicate an overlap. For example, one section of a report may be coded with Pilot to Pilot (TEM) and a different section may be coded with CRM – Communication (ACCERS) but, if those two coded sections do not overlap, they will not show up in the above

tables. The difference in coding style for ASRS codes results in slightly inflated numbers for the number of overlapping coded sections when comparing ACCERS, HFACS, or TEM to the ASRS codes.

As noted previously, the red highlights in each table indicate more than 50 overlaps, highlighting the most common overlaps across the four taxonomies. Identifying these frequent intersections helps to construct a framework based on these connections. Each highlighted intersection shows a significant connection between the two codes that form that intersection. These directly inform the connections that are shown in the new framework. Each of these interactions was examined to ensure they are related based on their definitions and concepts.

To interpret these results, it is essential to understand that consistency in overlapping codes does not necessarily mean that taxonomies are conceptually identical, but rather that they capture related aspects of pilot behavior and human factors through different theoretical lenses. These overlaps suggest that—despite variations in terminology and taxonomy structure—the same core issues, such as degraded communication, task saturation, or situational awareness, are being identified across frameworks. This convergence supports the reliability of the outcomes, even when derived from diverse origins.

However, differences in taxonomy conceptualization and labels mean that exact one-to-one matches between codes are rare, so overlap cannot necessarily be interpreted as evidence of full agreement. Instead, these overlaps are best viewed as areas of thematic alignment, where multiple frameworks independently identify related phenomena within the same narrative. This convergence confirms that these are critical points of failure worth examining but does not suggest that one taxonomy is more accurate than another. Rather, each taxonomy offers a complementary perspective, and their integration provides a richer understanding of the

communication and coordination breakdowns observed in aviation incidents. This understanding leads to our framework presenting how these taxonomies are interconnected in their themes and definitions.

The new framework focused on communication and coordination errors can be derived from this analysis. Significant overlaps in the coded sections of the reports for these codes are applied directly to the framework. Each of the taxonomies had very clear nodes for communication errors.

The framework can also visually represent the complex associations and interactions inherent in communication and coordination errors. The foundation for this framework rests on the significant overlaps identified in coded report sections across multiple human factors taxonomies—HFACS, TEM, ACCERS—and the ASRS anomaly codes.

Each taxonomy provides distinct but often complementary perspectives, with clearly defined nodes for communication errors that serve as core ideas for the new framework. These nodes represent critical elements such as crew communication, crew resource management failures, and situational awareness lapses. Their consistent identification across taxonomies underscores the robustness of these constructs in explaining the human factors behind aviation incidents.

Beyond communication-specific nodes, the new framework incorporates other thematic areas showing significant overlap in the analysis. To come to this conclusion, pairs of nodes with high concurrence were examined (as shown in Tables 12, 13, and 14) to determine their thematic alignment. Only pairs congruent in meaning and context were integrated into the maps, to ensure coherence. This approach balances the inclusion of relevant factors with the need to avoid unrelated concepts.

Connections between TEM, ACCERS, and HFACS codes generally translated directly into the new framework because of their strong alignment in coded text. However, the ASRS anomaly codes required additional scrutiny because of their application at the report level. For example, a seemingly significant overlap between Published Material & Policy (TEM) and Aircraft Handling (ASRS) was excluded because these categories differ substantially, so including them would misrepresent the conceptual framework.

Conversely, certain node pairs with fewer than 50 overlaps were included when analysis revealed a close thematic alignment. The pairing of ATC Issues (ASRS) with Operational Process (HFACS) illustrates this point: although not among the highest frequency overlaps, these nodes share a conceptual basis. Including such nuanced connections supplements the new framework by capturing subtler but important factors influencing communication and coordination errors. The foundation for the new framework is grounded in the overlap data and supplemented with theoretical concepts and definitions to ensure meaningful representation of aviation human factors.

New Framework

Figures 9, 10, and 11 depict the taxon connections found between each of the three studied taxonomies. These connections were developed based on overlaps in each coded section and the frequency of these overlaps. Figure 9 shows the significant connections between taxon in HFACS and ACCERS. Coded sections assigned to taxon in both ACCERS and HFACS were examined, and items with the most relevant overlaps were added to the framework. The colors on the framework make the map simpler to read. Parent child node relationships are also shown with the thinner gray lines.

For example, considering Unclear ATC Clearance from the ACCERS taxonomy, there are connections going to Operational Process; Communication, Coordination, and Planning; Inadequate Supervision; and Skill-Based Error in the HFACS taxonomy. Since HFACS does not have an item for ATC errors or unclear clearances, completing the analysis of the coded sections for each of these two taxonomies, the same sections of the reports that were assigned to Unclear ATC Clearances were also assigned to one of the other four listed HFACS codes. These codes were assigned to the same report sections a significant number of times and were determined to be relevant connections through the analysis, so they were included in the framework.

Figure 10 compares ACCERS and TEM, similar to the comparison shown in Figure 9. Figure 11 compares HFACS and TEM.

Figure 9

Framework for ACCERS and HFACS

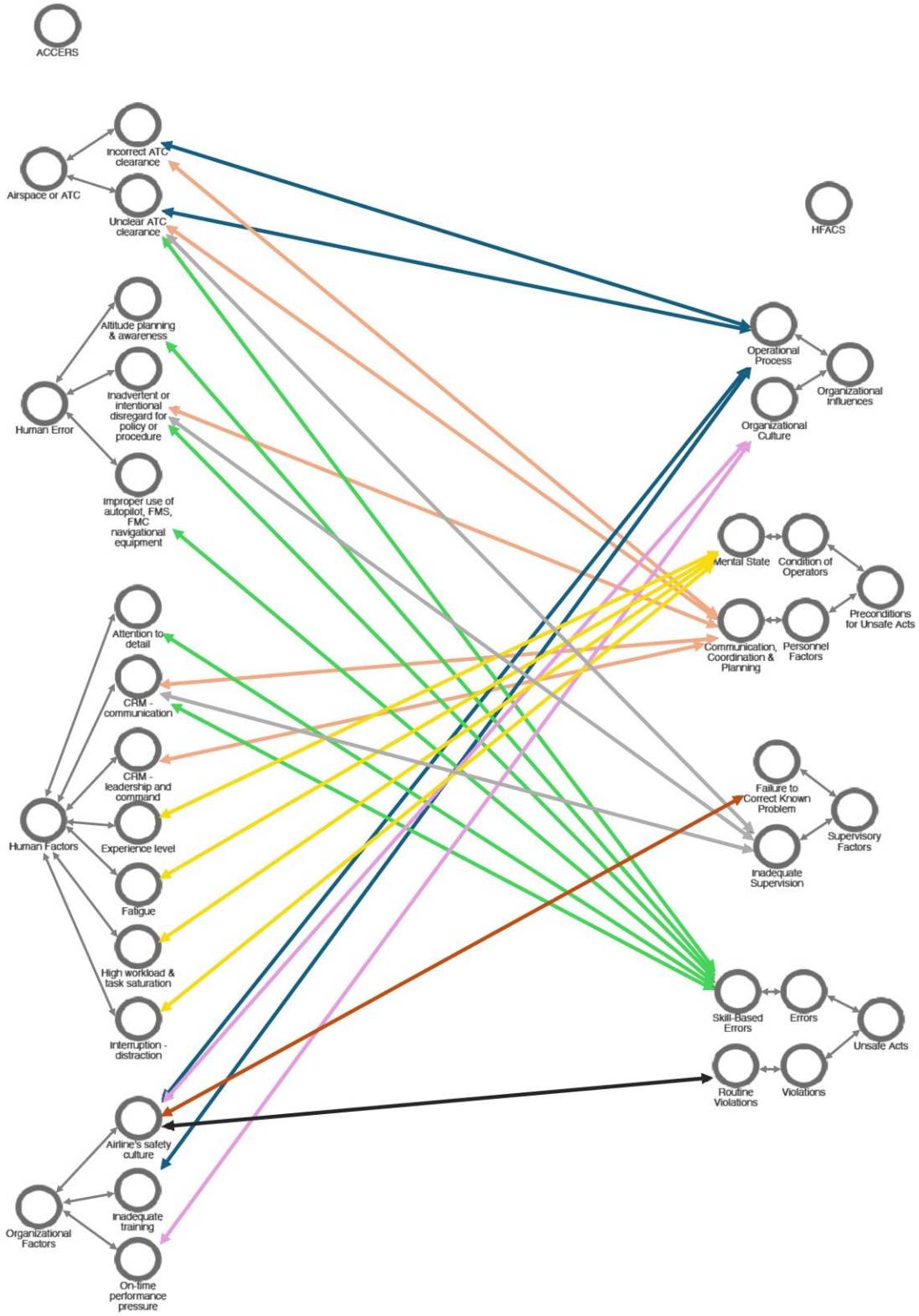


Figure 10

Framework for ACCERS and TEM

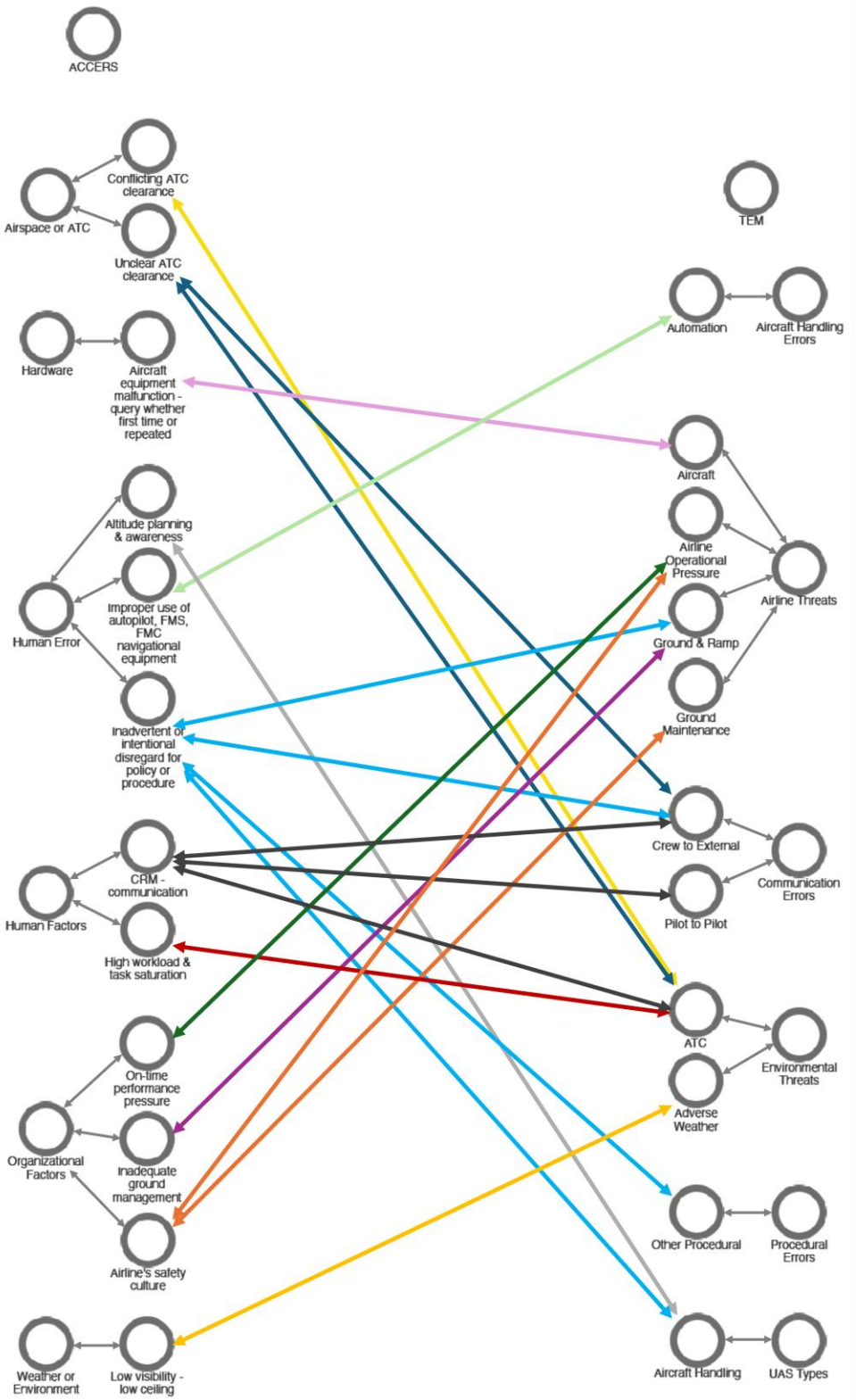
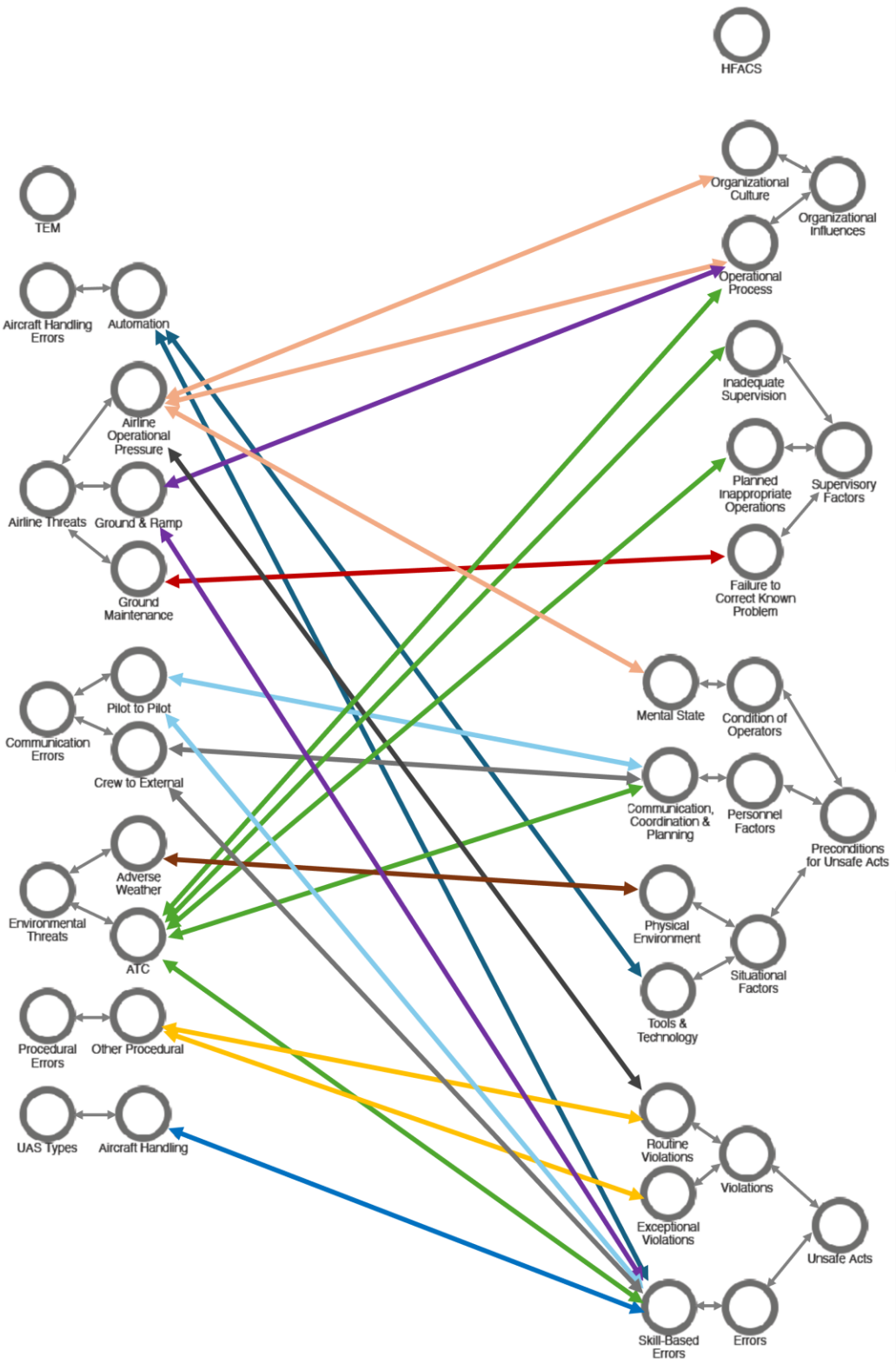


Figure 11

Framework for HFACS and TEM



Figures 9, 10, and 11 compare the three taxonomies coded through the research process as a framework. Tables 19 through 24 compare the three taxonomies to the codes given by ASRS analysts using the ASRS taxonomy.

Due to the complexities and the number of items included in these taxonomies, the framework is presented in a table rather than a diagram. Tables 19 through 24 present ASRS codes along with the ACCERS, HFACS, and TEM taxonomies. The tables present the ASRS code and the codes from the selected taxonomy, then also provide the reverse. These tables should simplify the use of the framework when comparing a database using the ASRS taxonomy and a database that uses ACCERS, HFACS, or TEM.

Table 19

Comparison Table for ASRS and ACCERS

ASRS	ACCERS
Aircraft Equipment, Critical	Airline's Safety Culture
ATC Issues	Incorrect ATC Clearance Unclear ATC Clearance CRM-Communication
Communication Breakdown	Conflicting ATC Clearance Unclear ATC Clearance Airline's Safety Culture Inadvertent or Intentional Disregard for Policy or Procedure Interruption/Distraction CRM-Communication High Workload & Task Saturation
Confusion	Unclear ATC Clearance Inadvertent or Intentional Disregard for Policy or Procedure CRM-Communication
Situational Awareness	Unclear ATC Clearance Inadvertent or Intentional Disregard for Policy or Procedure CRM-Communication
Distraction	CRM-Communication
Troubleshooting	Airline's Safety Culture
Human-Machine Interface	CRM-Communication
Workload	CRM-Communication
Clearance	Unclear ATC Clearance CRM-Communication
FAR	Airline's Safety Culture
Published Material & Policy	Airline's Safety Culture Inadvertent or Intentional Disregard for Policy or Procedure Attention to Detail CRM-Communication
Maintenance	Airline's Safety Culture

Table 20

Comparison Table for ACCERS and ASRS

ACCERS	ASRS
Incorrect ATC Clearance	ATC Issues
Conflicting ATC Clearance	Communication Breakdown
Unclear ATC Clearance	ATC Issues Communication Breakdown Confusion Situational Awareness Clearance
Airline's Safety Culture	Aircraft Equipment, Critical Communication Breakdown Troubleshooting FAR Published Material & Policy Maintenance
Inadvertent or Intentional Disregard for Policy or Procedure	Communication Breakdown Confusion Situational Awareness Published Material & Policy
Attention to Detail	Published Material & Policy
Interruption/Distraction	Communication Breakdown
CRM-Communication	ATC Issues Communication Breakdown Confusion Situational Awareness Distraction Human-Machine Interface Workload Clearance Published Material & Policy
High Workload/Task Saturation	Communication Breakdown

Table 21

Comparison Table for ASRS and HFACS

ASRS	HFACS
Deviation - Track/Heading	Skill-Based Errors
ATC Issues	Operational Process Communication, Coordination, & Planning Skill-Based Errors
Communication Breakdown	Operational Process Communication, Coordination, & Planning Inadequate Supervision Skill-Based Errors
Confusion	Mental State Communication, Coordination, & Planning Skill-Based Errors
Distraction	Mental State Communication, Coordination, & Planning
Human-Machine Interface	Skill-Based Errors
Situational Awareness	Operational Process Mental State Communication, Coordination, & Planning Failure to Correct a Known Problem Skill-Based Errors
Workload	Mental State Communication, Coordination, & Planning Skill-Based Errors
Clearance	Operational Process Mental State Communication, Coordination, & Planning Inadequate Supervision Skill-Based Errors
Published Material & Policy	Operational Process Communication, Coordination, & Planning Failure to Correct a Known Problem Planned Inappropriate Operations Skill-Based Errors

Table 22

Comparison Table for HFACS and ASRS

HFACS	ASRS
Operational Process	ATC Issues
	Communication Breakdown
	Situational Awareness
	Clearance
	Published Material & Policy
Mental State	Confusion
	Distraction
	Situational Awareness
	Workload
	Clearance
Communication, Coordination, & Planning	ATC Issues
	Communication Breakdown
	Confusion
	Distraction
	Situational Awareness
	Workload
	Clearance
	Published Material & Policy
Failure to Correct a Known Problem	Situational Awareness
	Published Material & Policy
Inadequate Supervision	Communication Breakdown
	Clearance
Planned Inappropriate Operations	Published Material & Policy
Skill-Based Errors	Deviation - Track/Heading
	ATC Issues
	Communication Breakdown
	Confusion
	Human-Machine Interface
	Situational Awareness
	Workload
	Clearance
	Published Material & Policy

Table 23

Comparison Table for ASRS and TEM

ASRS	TEM
Aircraft Equipment, Critical	Ground Maintenance
ATC Issues	Crew to External
	ATC
Published Material & Policy	Ground Maintenance
	Airline Operational Pressure
	Ground & Ramp
	Crew to External
	Pilot to Pilot
	ATC
	Other Procedural
Clearance	Crew to External
	Pilot to Pilot
	ATC
	Aircraft Handling
FAR	Ground Maintenance
	Airline Operational Pressure
Maintenance	Ground Maintenance
	Airline Operational Pressure
Distraction	Pilot to Pilot
Situational Awareness	Crew to External
	Pilot to Pilot
	ATC
Communication Breakdown	Automation
	Ground Maintenance
	Airline Operational Pressure
	Ground & Ramp
	Crew to External
	Pilot to Pilot
	ATC
	Other Procedural
	Aircraft Handling
Confusion	Crew to External
	Pilot to Pilot
	ATC
Troubleshooting	Ground Maintenance
Workload	Pilot to Pilot
Human-Machine Interface	Pilot to Pilot

Table 24

Comparison Table for TEM and ASRS

TEM	ASRS
Automation	Communication Breakdown
Ground Maintenance	Aircraft Equipment, Critical Published Material & Policy FAR Maintenance Communication Breakdown Troubleshooting
Airline Operational Pressure	Published Material & Policy FAR Maintenance Communication Breakdown
Ground & Ramp	Published Material & Policy Communication Breakdown
Crew to External	ATC Issues Published Material & Policy Clearance Situational Awareness Communication Breakdown Confusion
Pilot to Pilot	Published Material & Policy Clearance Distraction Situational Awareness Communication Breakdown Confusion Workload Human-Machine Interface
ATC	ATC Issues Published Material & Policy Clearance Situational Awareness Communication Breakdown Confusion
Other Procedural	Published Material & Policy Communication Breakdown
Aircraft Handling	Clearance Communication Breakdown

Interactive Coding Compared to Framework Assigned Codes

To answer the second research question, ASRS codes assigned by ASRS analysts were compared to the outputs of the developed framework for ACCERS, HFACS, and TEM. ASRS reports met the same criteria as the original reports and were sampled over the same time period. The framework was then used to determine what ACCERS, HFACS, and TEM codes could be assigned to the reports based on the ASRS codes. The reports were also interactively coded in the same process as the original reports.

Codes determined by using the framework were then compared to codes that were assigned through the interactive coding, in order to determine which ACCERS, HFACS, and TEM codes found thorough the framework matched the codes and which would not have been found. Knowing the differences between the assigned codes and those found through the framework allows for a determination of the effectiveness of the framework.

Results for five reports analyzed using this method is depicted in Table 25, which shows the assigned ASRS codes along with the codes given by the researcher in this study for ACCERS, HFACS, and TEM. The red codes for ACCERS, HFACS, and TEM are those that were also found through the framework using the given ASRS codes. For the overall sample of 73 reports, about 85% of the interactively assigned codes would have been found through the framework. The highest percentage of codes found through the framework that were also assigned in the interactive coding was HFACS with more than 95%. ACCERS has the lowest at 79% and TEM was calculated to be 83%.

Table 25

Comparison of Assigned ASRS Codes and Coded ACCERS, HFAS, and TEM Codes

Report	ACCERS	HFACS	TEM	ASRS
1	CRM - Communication Altitude Planning/Awareness Experience Level Unclear ATC Clearance	Communication, Coordination, and Planning Skill-Based Error Mental State	Pilot to Pilot ATC Aircraft Handling	Communication Breakdown Deviation - Altitude Crossing Restriction Not Met Deviation - Track/Heading All Types Deviation/Discrepancy - Procedural Clearance Deviation/Discrepancy - Procedural Published Material/Policy
2	CRM - Communication Unclear ATC Clearance Hear Back/Read Back	Communication, Coordination, and Planning Operational Process	Crew to External ATC	Distraction Situational Awareness Communication Breakdown Confusion ATC - Issue All Types Deviation - Altitude Crossing Restriction Not Met Deviation/Discrepancy - Procedural Clearance
3	CRM - Communication Frequency Congestion Incorrect ATC Clearance	Communication, Coordination, and Planning Operational Process	Crew to External ATC	Situational Awareness Communication Breakdown ATC Issue All Types Inflight Event/Encounter CFTT/CFIT
4	CRM - Communication Inadvertent or Intentional Disregard for Policy/Procedure Altitude Planning/Awareness Experience Level Incorrect ATC Clearance	Communication, Coordination, and Planning Skill-Based Error Mental State Planned Inappropriate Operations	Pilot to Pilot ATC Aircraft Handling	Situational Awareness Communication Breakdown Training/Qualification ATC Issue All Types Deviation - Altitude Crossing Restriction Not Met

				Deviation - Altitude Overshoot Deviation/Discrepancy - Procedural Published Material/Policy Inflight Event/Encounter CFTT/CFIT
5	CRM - Communication	Communication, Coordination, and Planning	Pilot to Pilot	Communication Breakdown
	Unclear ATC Clearance	Skill-Based Error	ATC	Confusion
	Conflicting ATC Clearance	Operational Process		Situational Awareness
		Planned Inappropriate Operations		ATC Issue All Types Deviation - Track/Heading All Types Deviation/Discrepancy - Procedural Published Material/Policy Deviation/Discrepancy - Procedural Clearance

Year Over Year Results

To answer the third research question, the codes were analyzed to examine changes by year. Each year had a different number of reports that met the qualifications of the study, so a proportional number of reports were sampled for each year, using a rate rather than a straight count of the codes. Figure 12 shows the rate of the coded taxonomies for each year. ASRS was the highest with an average of about six codes per report assigned by the ASRS analysts. The other three taxonomies all average about three codes per report. There is no significant trend in the assignment of the coded taxonomies, but this was expected because all the reports were coded with a systematic process to capture the same information from each report. Figures 13 – 16 show the number of codes assigned for each taxonomy.

Figure 12

Coded Taxonomy Rate by Year

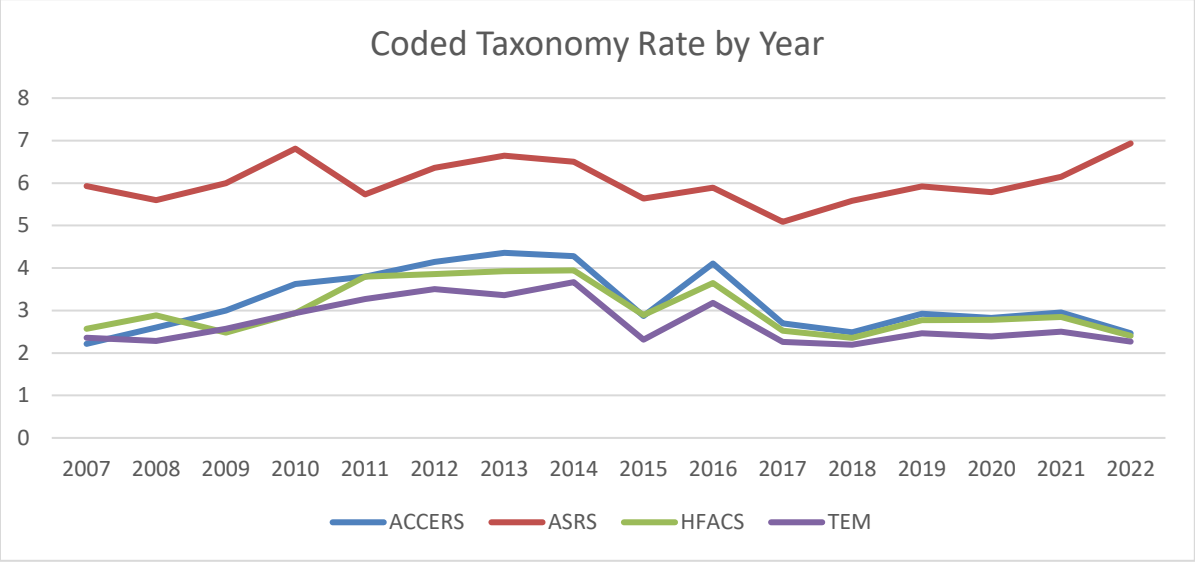


Figure 13

Coded ACCERS Categories by Year

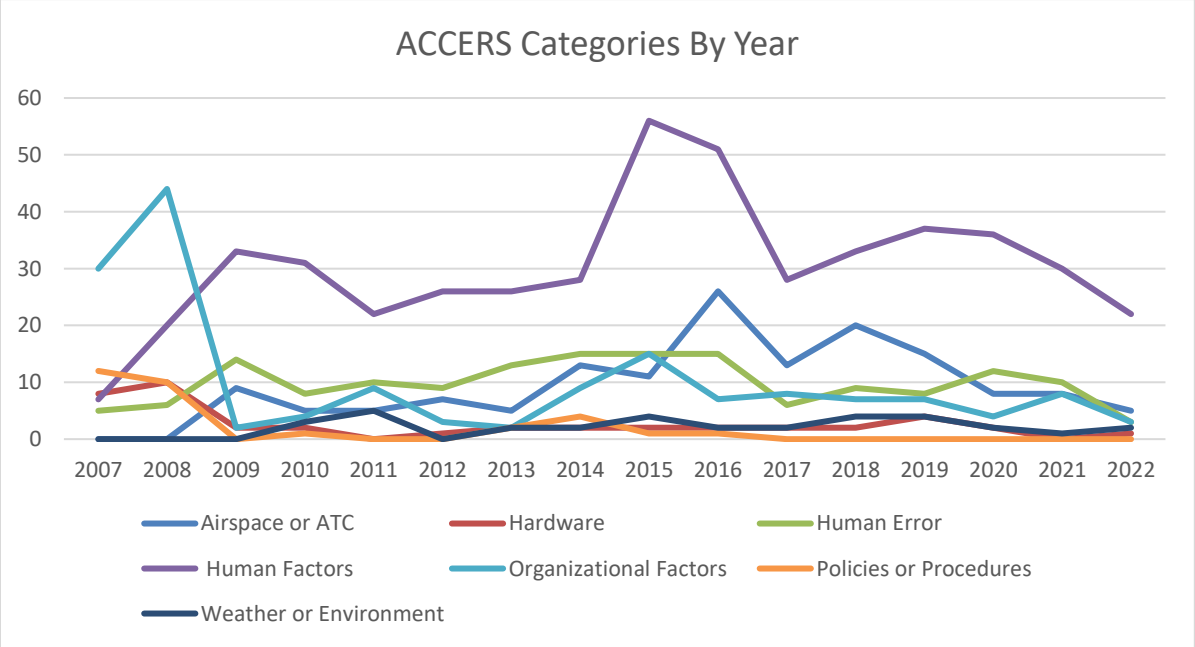


Figure 14

Coded ASRS Categories by Year

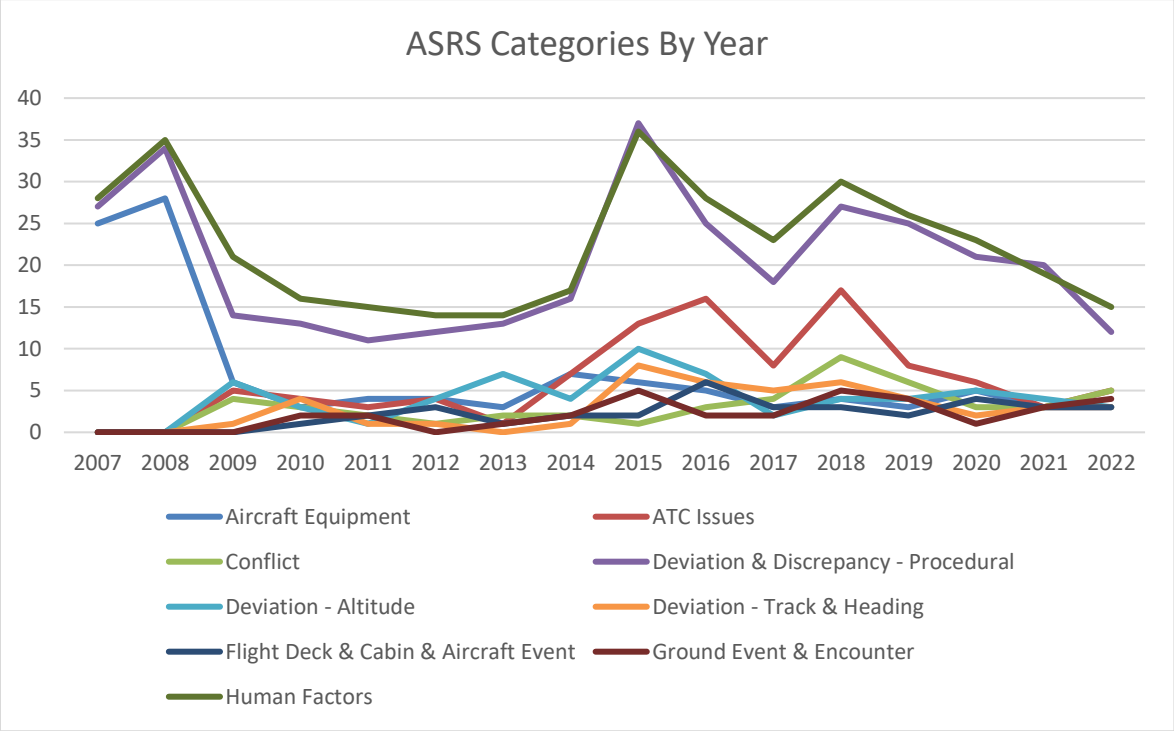


Figure 15

Coded HFACS Categories by Year

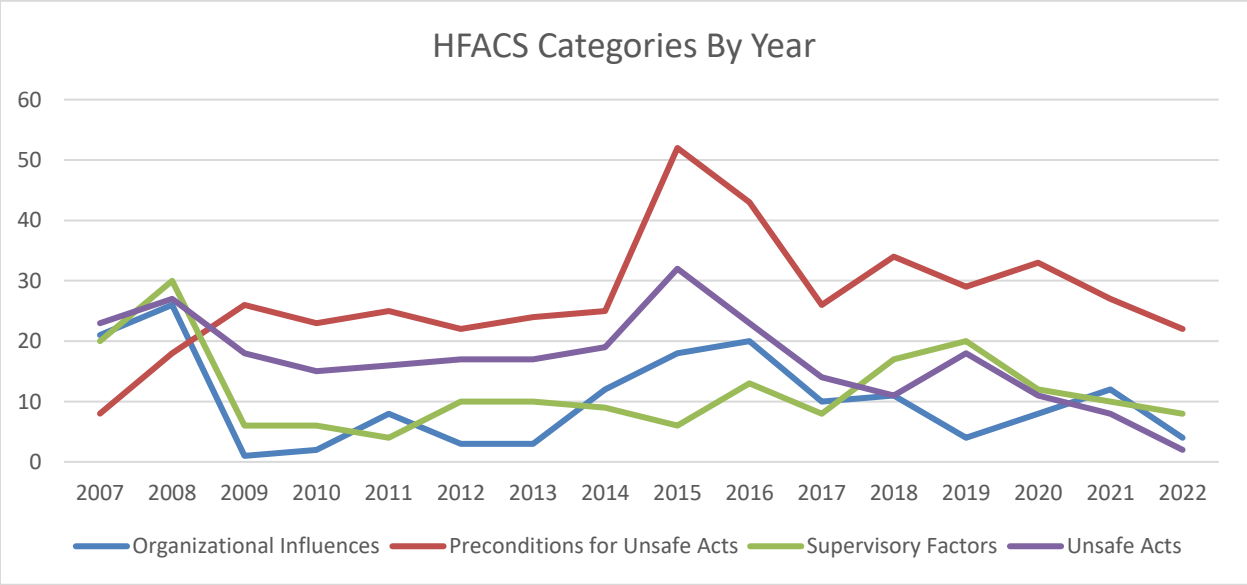
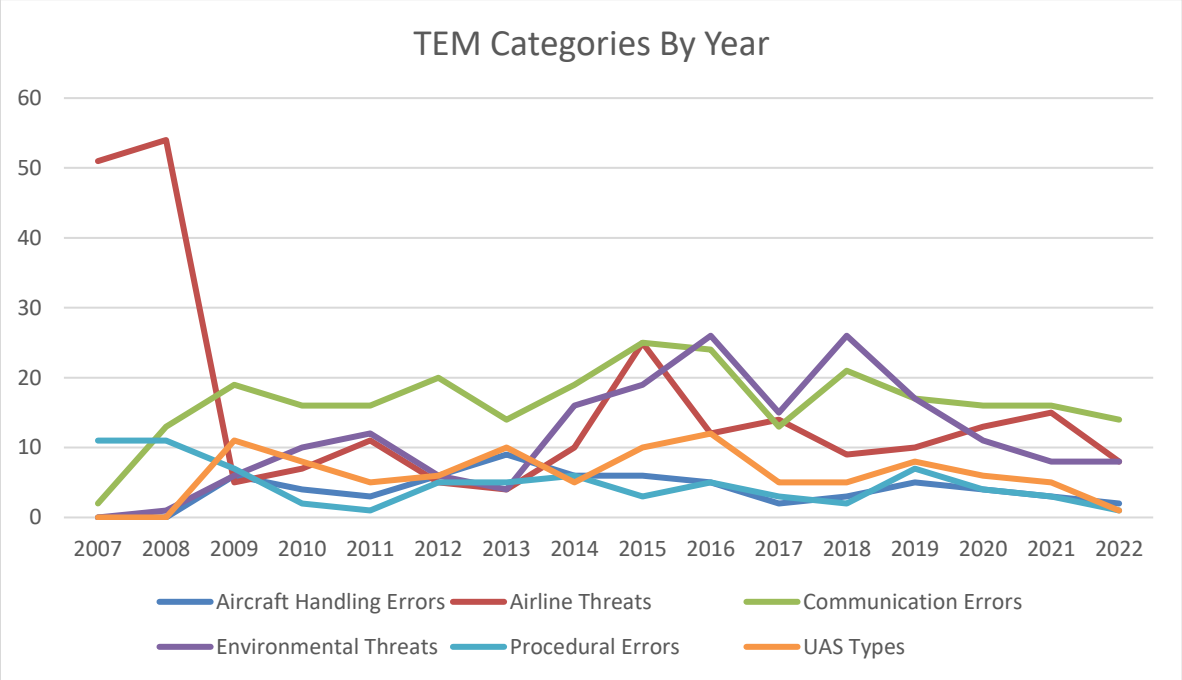


Figure 16

Coded TEM Categories by Year



While the rates of the assigned taxonomies were relatively consistent year over year, the number of communication codes showed a different trend. The ASRS codes assign just one communication code for each report, so the average is just under one each year. There were several reports each year that did not have an assigned communication code. For ACCERS, HFACS, and TEM, more than one communication code could be assigned per report, or the same communication code could be assigned to two different sections of the same report. As a result, the average number of communication codes for each report could be greater than one.

Figure 17

Rate of Communication Codes Assigned by Year

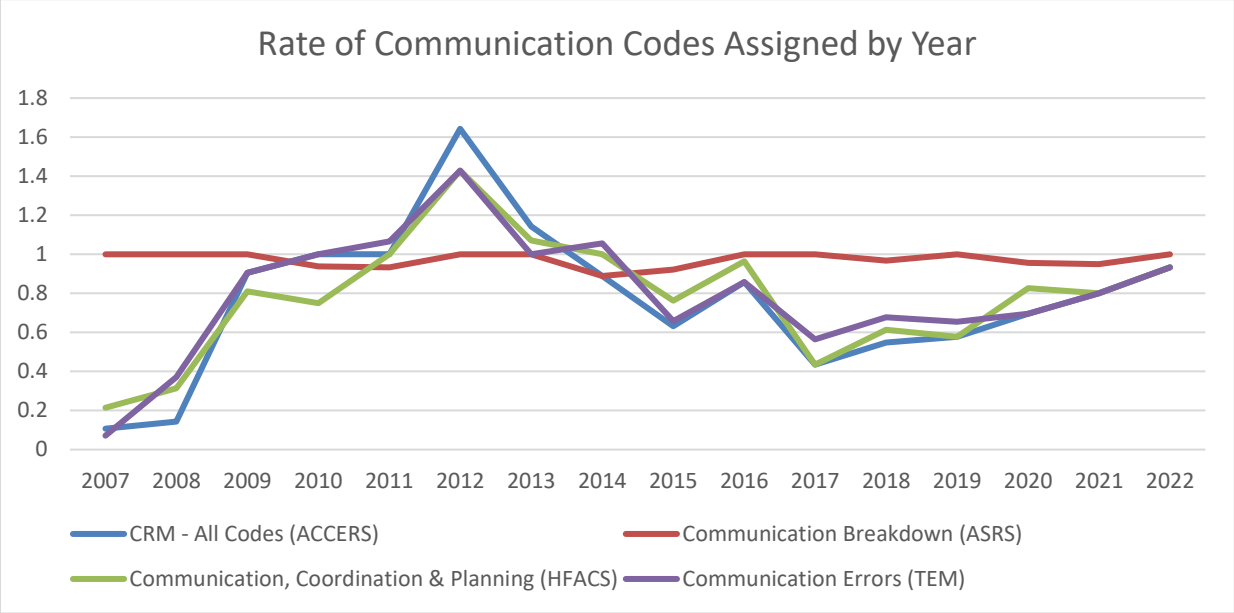


Figure 17 shows that the average for CRM-All Codes; Communication Breakdown; and Communication, Coordination, and Panning was greater than one between 2011 and 2013. Figure 17 also shows that there were significantly fewer communication codes assigned in 2007 and 2008 and again in 2017. The average rate of codes assigned per report over the time frame for ACCERS is 0.77, HFACS is 0.78, TEM is 0.80, and ASRS is 0.96.

Summary

This study examined and coded 365 reports from 2007 to 2022. Of these reports 65% were submitted by captains, with reporters relatively evenly split between Pilot Monitoring and Pilot Flying. These results align with standard airline reporting practices, which require the captain to submit internal reports when an anomaly occurs. They may also submit ASAP or ASRS reports during this time. Many ASRS reports are sourced from airline reporting systems.

The researcher coded the reports, which were then validated by two additional coders. Based on the length of the reports and the small sections coded, κ was lower than what would indicate a good agreement. Reliability and validity of the codes was also measured by agreement between coders where a value of 80% represented good agreement between the coders. For this study, there was 89.5% agreement between the two secondary coders and the primary coder, leading to the conclusion that the codes are reliable for the study.

Once all of the reports were coded, a detailed analysis was conducted to develop a framework for the taxonomies. From this analysis, it was evident that the communication codes of each taxonomy were the most commonly assigned. Other commonly assigned codes included codes that related to the mental state of the pilots or the conditions surrounding the pilot when the communication error occurred. Through the analysis, connections of the taxonomies were explored and areas where there was significant overlap between taxonomies were examined. The codes that had the most frequent overlaps and where the codes in the taxonomies were related were included in the framework.

The final part of the analysis considered how the codes were assigned year over year. While there were no significant changes in the number of codes for each taxonomy assigned to each report year over year, there were some changes in the frequency of communication-specific codes. Communication codes for ASRS remained constant for each year, while communication codes for each of the other three taxonomies showed an increase in 2012 and decreases in 2007 through 2008 and in 2017.

Chapter V: Discussion, Conclusions, and Recommendations

The purpose of this study was to develop a framework that associates anomaly codes related to communication and coordination errors with terms used in established taxonomies. Reports from the ASRS database were focused on to determine which concerns should be prioritized for aviation safety in the future. The framework developed allows for more complete analysis across databases that use different taxonomies, ensuring that critical safety data is not missed or underrepresented.

The results considered alignments between the selected taxonomies: ACCERS, HFAS, TEM, and ASRS. These alignments were used in the framework, which relates ASRS anomalies to the coded taxonomies in each report. Changes in the reporting of TEM, HFACS, and ACCERS elements in ASRS pilot reports over time were also considered.

Discussion

By understanding how the codes are applied for each taxonomy and where coded sections overlap, future researchers can use the framework to compare reports that use different taxonomies. The framework offers a detailed analysis of commonalities in the taxonomies and also explores the strengths and weaknesses of each taxonomy. Both connections and differences are discussed.

Research Question 1

"What alignments exist between the selected taxonomies (HFACS, TEM, and ACCERS) and the ASRS anomaly codes related to pilot communication and coordination errors?"

The communication codes for each of the taxonomies are addressed first, as these were the primary focus of the research.

TEM includes two primary codes relevant to communication errors: Pilot to Pilot and Crew to External. These categories provide a basic structure but do not allow for differentiation between communication failure types. In contrast, HFACS combines communication and coordination under a single category, Communication, Coordination, and Planning, which offers broader coverage but similarly lacks granularity. The ASRS taxonomy uses the code Communication Breakdown, a more general but effective label for capturing a wide range of communication-related anomalies.

ACCERS provides a more nuanced breakdown of crew communication through its CRM-related categories: CRM – Communication; CRM – Leadership and Command; CRM – Conflict Management; and CRM – Incompatible Crew Pairing. These codes offer more precise coding, but ACCERS does not include codes for communication errors that fall outside CRM. For example, some reports had communication errors with maintenance, which led to discussions or disagreements about the correct resolution. While this concept in simply coded in ASRS, HFACS, and TEM, ACCERS contains some ambiguity about the definitions of CRM and the code. The FAA defines CRM as "the effective use of all available resources: human resources, hardware, and information" (AC121-42, 2020), while the ACCERS taxonomy defines CRM – Communication as "appropriate communication protocol was not followed" (Baker & Krokos, 2007). So, while the above example fits well into codes for the other taxonomies, there is a potential gap under the ACCERS if coders are basing their work strictly on this definition. In this study, these incidents were still classified under CRM – Communication because they involved communication breakdowns between crew members and external influences. However, these cases highlight the need for clearer ACCERS distinctions to ensure more consistent and accurate application.

In addition to communication issues, ACCERS excels in detailing the conditions surrounding an event. Within the Human Factors category, the pilot's internal state can be coded with factors such as fatigue, high workload, and complacency. ACCERS also provides detailed categories for external pressures such as ATC delays and company expectations, and environmental factors like weather and hardware issues. This reflects ACCERS' origin as a tool designed to classify contributing factors in ASAP reports (Baker & Krokos, 2007). This primary goal to focus on contributing factors rather than the whole event is reflected in the name, Aviation Causal Contributors for Event Reporting Systems. As a result, the taxonomy emphasizes the surrounding influences more than the event itself, making it more suitable for identifying causes than describing what occurred.

In contrast, HFACS contains more detailed event-level codes, such as Decision Errors and Skill-Based Errors, under the Unsafe Acts category, which allow analysts to pinpoint the nature of the error more directly. TEM similarly includes codes for operational context but distinguishes itself by offering a dedicated category for Undesired Aircraft States, which identifies the nature of the error, not just the conditions surrounding it. Along with detail about the surrounding influences, ACCERS also excels in providing codes concerning the pilot's lack of knowledge or skill, including Experience Level, Inadequate Training, and Proficiency/Overreliance on Automation. These codes clarify underlying causes of skill-based errors. While HFACS includes broader categories like Skill-Based Errors and Mental State, it does not provide information about why those errors occurred, such as a lack of training or experience. Similar codes are not available in ASRS or TEM. As such, when examining the framework, there is a lack of connections between ACCERS and TEM or ASRS.

The ACCERS taxonomy also captures various aspects of the pilot's internal state, including factors such as fatigue, self-induced time pressure, high workload/task saturation, peer pressure, complacency, illness and others. ASRS codes have several of these same codes under the Human Factors Category to describe the state of the pilots. HFACS groups all these codes together under Condition of Operators, Mental State and offers no further breakdown, while TEM offers no codes related to pilots' mental or physical states except for two codes, Airline Operational Pressure and Environmental Operational Pressure, which only infer a fragment of what the other taxonomies capture.

Another area where taxonomies differ are issues that involve communications with ATC. While this study focused on communication and coordination between crew members, many occurrences also involved a communication with ATC. In some instances, this left crew members unable to meet the ATC request or clearance or caused a disagreement about the request or clearance. The ATC interaction may also cause communication errors between crew members if clearance information is not passed along to all members. HFACS has no code that captures ATC issues because it was developed to capture human errors and the barriers to preventing these errors (Wiegmann & Shappell, 2016). Since HFACS categories are broader than those in other taxonomies, it is not as useful when analyzing narrative reports as ACCERS and ASRS. ATC errors may be included in the category Organizational Influences if the ATC error captures the culture or operational process at that facility. ATC errors also can be included in the Supervisory Factors category if the pilot's error was not caught by the supervising controller. TEM has the code ATC in the category of Environmental Threats and ASRS has the code ATC Issues, which also captures issues related to ATC. ACCERS also has a category of Airspace or ATC, which captures various errors and issues surrounding ATC. ACCERS has clearance

specific codes for ATC, such as Unclear ATC Clearance, Conflicting ATC Clearance, Late ATC Clearance, or Incorrect ATC Clearance, and there are also codes for interactions between the pilot and ATC such as Hear Back/Read Back and Expected vs. Assigned ATC Clearance.

Operational pressure is another factor captured differently in the four taxonomies. TEM uses the code Airline Operational Pressure to capture these issues, while ACCERS has the code On Time Performance Pressure under the category Organizational Factors. While this code is specific to on-time performance, it effectively captured most of the issues related to operational pressure, either to keep planes moving or take planes they may are not comfortable with. The ASRS taxonomy has a code Time Pressure in the Human Factors category, but it does not distinguish between company-induced or self-induced time pressure. HFACS has categories where operational pressure could be included, such as Organizational Culture or Condition of the Operators – Mental State. However, neither of these codes are as specific or as detailed as the other taxonomies in capturing operational pressure. As such, the operational pressure issues in HFACS are coded with many other items and are difficult to separate.

Table 26 shows the associations between the taxonomies and the ASRS anomaly codes, organized according to the thematic categories developed. While the table offers a high-level comparison of the ways each taxonomy addresses key themes, it is not intended to replace the more nuanced findings presented in the accompanying framework. Instead, Table 26 depicts the variability of the taxonomies in the level of detail they provide for the themes. The table illustrates the alignment between the taxonomies and the ASRS codes across these themes. In some instances, the taxonomies provide greater detailed classifications of contributing factors than the ASRS codes, but in other cases, they reflect more generalized representations of the same themes.

Table 26

Thematic Comparative Taxonomy Coverage

Taxonomy	Communication/ Coordination	Event/Error	Pilot Skill/State	ATC Interactions	Operational Pressure
HFACS	Broad single category	Precise, Detailed Unsafe Act categories	Limited non-specific category	Difficult to interpret from Supervisory Factors	Broad Organizational Influences category
TEM	High-level, internal or external	Includes UAS Type	Not Included	Singular ATC code only	Specific Airline Operational Pressure code
ACCERS	CRM-focused only	Very minimally included	Very Detailed	Highly detailed ATC codes	Several detailed codes relating to Organizational Influences
ASRS	General single category under Human Factors	Event-focused taxonomy—very detailed	Detail included in Human Factor code	Only included in Communication breakdown code	Only broad Time pressure

The connections between the selected taxonomies and the ASRS anomaly codes should be interpreted as qualitative alignments rather than quantitative correlations. Rather than representing statistical measures of association, these connections indicate how each taxonomy corresponds to the themes and categories identified from the ASRS narratives, particularly concerning communication and coordination errors. Tables 19 through 24 and the framework acknowledge that each taxonomy was developed with different emphases and purposes, ranging from broad human factors frameworks to detailed causal contributor classifications, so each taxonomy captures aspects of the data with varying levels of granularity and scope. The alignments reflect the degree to which each taxonomy can effectively map the nuances of pilot communication and coordination anomalies to their contributing factors, as described in ASRS

reports. This approach helps clarify the strengths and limitations of each taxonomy when analyzing human factors, to allow for appropriate application rather than measuring equivalence or predictive validity.

Research Question 2

"How do the results of each taxonomy coded through the created framework relate to the anomaly codes given by the ASRS analysts?"

The second research question was answered by using a sample of reports, with the same selection criteria as the original reports, using the framework to determine ACCERS, HFACS, and TEM codes that could be assigned to reports based on the ASRS codes they were given. The sample was also interactively coded with ACCERS, HFACS, and TEM. The codes found through the framework were then compared with the interactively coded ones to identify similarities. These connections show how closely the framework's coding aligns with the given ASRS classifications.

Codes that were found in both the framework and interactive coding allow researchers to find the report when searching a database coded with one of the three selected taxonomies based on the ASRS codes. One of the purposes of this study was to provide analysts a systematic method to search a database coded with one taxonomy (ACCERS, HFACS, TEM, or ASRS) and find information coded with a different taxonomy (ACCERS, HFACS, TEM, or ASRS). When these two codes match, it can be concluded that the report would likely have been found by an analyst searching the database using the framework and ASRS codes. Table 14 provides a sample of the results of this comparative process, illustrating the matching codes that were found.

About 85% of the interactively assigned codes would have been found through the framework. HFACS had the highest percentage of codes found through the framework at just over 95%. ACCERS has the lowest at 79% and TEM was calculated to be 83%. For every report, at least two of the codes were matched and the codes not matched were more specific codes, such as “icing,” “frequency congestion,” and “dispatch/paperwork.” These might become less common as more reports and types of error are included in future studies. However, with 85% of the interactively-coded codes also found through the framework—with multiple codes for each report being repeated through the two methods—it is likely that most of these reports would be found in a database using the framework.

Research Question 3

“From 2007 to 2022, what are the changes in the reporting of TEM, HFACS, and ACCERS elements in ASRS pilot reports?”

The results for the third research question are found in Figures 12 and 17 where graphs show the change in coded taxonomy rate and rate of communication and coordination codes. From the graph showing the coded taxonomy rate from 2007 to 2022, it is evident that there are about twice as many ASRS codes assigned to reports as there are from the other three selected taxonomies. There is also an increase in the rate of coded taxonomies for all four of the taxonomies between 2012 and 2014 and ASRS has an increase in coded rate in 2022.

The two significant events of interest for this research question were the introduction of AQP in 2005 and the requirement for airlines to implement SMS by 2018. Delta Air Lines was the first to start using the AQP in 2006 for flight attendant training. Other airlines followed and now 90% of large Part 121 carriers (those with more than 1,000 pilots) use AQP for some or all of their pilot training (FAA, 2021). It is difficult to determine the exact date of AQP introduction

for each airline given the timeline that the introduction requires and the option for airlines to introduce AQP in parts. Similarly, while SMS was required to be implemented by 2018, airlines started using SMS earlier, with American Airlines beginning in 2009 (American Airlines, 2022).

When considering the dates associated with the implementation of AQP and SMS and the increases found in Figure 12, the following conclusions can be made. For the increase in all coded taxonomy rates between 2012 and 2014, many airlines were slowly implementing their AQP training along with the early adoption of SMS. While it cannot be said that there is a definitive connection, it is something that can be further studied. Other than the full adoption of SMS in 2018, there is no specific result of SMS or AQP that can account for the increase in ASRS coded rates in 2022.

Figure 17 depicts the rate of assigned communication codes, showing that about one ASRS communication codes is assigned per report. Conversely, communication codes for ACCERS, HFACS, and TEM fluctuate over this time period. Rates for the three selected taxonomies start in 2007 and 2008 around 0.3 codes per report, then increase significantly by 2012, declining to about 0.5 in 2017 and slowly increasing again through 2022. These trends may correlate to the introduction and implementation of AQP and SMS because the use of communication codes was low before AQP and SMS were introduced and before there was a significant focus on communication and coordination between pilots. As AQP starts being gradually introduced, the rates of these codes rose to the peak in 2012. The increase after 2017 could be associated with the increasing use and familiarity of SMS. While these are simply judgments of how these dates align with the coded reports, further study could look at this in more detail.

Methodology

In this study, the Cohen's κ value, calculated to assess inter-coder agreement, fell below the conventional threshold for *good* agreement. Such a result suggests some divergence in how coders applied nodes across the dataset, though it may not necessarily indicate substantive disagreement in interpretation. The lower κ values are likely attributable to a methodological decision that allowed coders to identify only a single segment that best exemplified a concept, instead of applying the code to all relevant statements. Allowing coders to select just one segment introduced variability in the precise text chosen for coding, which subsequently limited textual overlap—a key factor in NVivo's calculation of agreement. As a result, the observed kappa scores may underestimate the actual level of conceptual alignment.

If this study were to be replicated or extended, one methodological refinement would be to instruct coders to apply the node to every relevant statement in the narrative, rather than selecting only one statement that most exemplifies the concept. Under the current approach, coders often chose different illustrative segments to represent the same theme or construct, resulting in minimal or no overlap in the exact text coded. Because NVivo calculates inter-coder agreement using Cohen's kappa based on character-level overlap, even small differences in segment selection can lead to a reduced observed agreement, thereby lowering the kappa value—even when conceptual agreement exists.

Encouraging coders to code all instances rather than only the most salient one would promote greater convergence in text selection, leading to more overlap in character spans and consequently an enhanced observed agreement component of κ, producing a higher and more accurate reliability coefficient. This refinement would not only improve the statistical reliability

of the coding process but also offer a richer and more comprehensive representation of how each theme is expressed across the dataset.

Addressing this concern also mitigates a documented limitation in qualitative coding reliability—the tendency for discrepancies in coding boundaries to reflect variations in individual coding style rather than genuine theoretical divergence. Comprehensive and standardized coding guidelines would also enhance the clarity and consistency of coder decisions, providing a more accurate distinction between stylistic inconsistencies and differences in interpretation. Such refinements are likely to yield more valid reliability metrics, strengthening the methodological rigor of the analysis.

Conclusions

Through this research, a new framework was developed for each taxonomy compared to the other studied taxonomies through a detailed analysis of ASRS reports. These maps show commonalities between the codes, as well as differences in the ways events or details are captured.

This study focused on communication and coordination errors between pilots and analyzed these errors and the events and factors surrounding them. Codes for communication and coordination were coded to almost all the reports that were analyzed. Codes from each taxonomy were frequently given to the same section of the report that best exemplified the communication or coordination error, showing that those codes were used similarly in their respective taxonomies. Other codes that share significant overlaps include errors made with ATC. As noted previously, ACCERS has multiple codes that capture these errors, while other taxonomies only have one. HFACS does not have a specific ATC code, but two other codes that capture similar errors as the ATC codes found within other taxonomies. Similar items in each taxonomy with

significant overlaps in coded sections are shown above. The framework also shows areas without significant overlapping sections.

Each taxonomy has strengths based on its original intended purpose, but also some drawbacks when compared to the other taxonomies. For example, ACCERS captures influencing factors of an event very well, but the actual error can get lost. Similarly, TEM, which was developed based on accidents and incidents within LOSA observations (Merritt & Klinect, 2006), captures details about the errors, but may not capture which is most significant. ACCERS was developed to be used by ASAP analysts in classifying events and causal factors (Baker & Krokos, 2007). The ASRS taxonomy was developed by subject matter experts to classify events, so it offers a breakdown of errors found within an event but lacks detail for human factors and other contributing factors. HFACS has a greater focus on human factors than the other studied taxonomies, classifying the most significant events and contributing factors. However, it has much less detail than some of the other taxonomies, especially if there is more than one error within the event (Wiegmann & Shappell, 2001). Despite these differences, each taxonomy captures the details of the reports and has significant overlaps in coded sections of the reports.

Theoretical Contributions

This study makes a significant theoretical contribution by identifying and addressing the limitations of the current ASRS anomaly code system. While ASRS codes are useful for organizing safety reports, they lack clear definitions and are not grounded in an established theoretical framework. As a result, they often fail to capture the full scope and complexity of pilot communication and coordination errors. By comparing these codes with three well-established human factors taxonomies, HFACS, TEM, and ACCERS, this study highlights specific gaps in the ASRS classification system and shows how these gaps can be addressed

using theory-based models. This analysis reveals the need for a more structured, theory-driven approach to safety data classification.

In addition, this research contributes a framework that connects practical safety reporting with theoretical human factors concepts. Using the framework diagrams and tables, the study shows how different systems identify similar issues, such as communication failures, pilot workload, ATC communication, and operational pressures. This integration of taxonomies provides a more complete understanding of safety data and creates a foundation for improving both analysis and reporting practices. By aligning ASRS data with theoretically grounded taxonomies, the study strengthens the conceptual validity of safety research and offers a model that can be applied across other safety-critical industries as well.

Another contribution of this study is the development of a systematic framework for linking human factors taxonomies. To date, there has been no published research that effectively bridges the use of multiple taxonomies in a manner that supports integrated analysis across safety databases. This study addresses that gap by providing a method for researchers working with one taxonomy to meaningfully engage with data coded under a different taxonomy. Through the development of this framework, this research enables more structured and targeted cross-database searches, reducing the need for extensive manual review or reliance on imprecise assumptions about code equivalency. By aligning conceptual categories across taxonomies, the study enhances researchers' ability to access, interpret, and compare data from diverse sources without losing fidelity. Drawing from and extending Bazeley’s (2013) methodological principles, the study adapts framework development to a new application context, that of aviation safety reporting, linking taxonomies that were developed and are generally used independently. The resulting framework enables researchers to identify both overlaps and gaps among classification

systems, advancing theoretical understanding of how human error and communication breakdowns are conceptualized across varying analytical models.

The study also holds interdisciplinary significance by offering a model for taxonomy alignment in other safety-critical domains, such as healthcare, nuclear energy, or transportation. Many of these fields use taxonomies conceptually related to those examined in this research. The approach facilitates the cross-disciplinary transfer of knowledge by revealing how different sectors classify similar types of errors or risk factors. In doing so, it lays a theoretical foundation for comparative research across domains, where understanding how different fields conceptualize and respond to human error can enhance both the quality of analysis and the potential for shared learning across industries.

Practical Contributions

The study was designed to create a framework for analysts to use when searching databases using taxonomies different from the one the database is coded to. Across organizations and industries—including aviation—there are many different taxonomies used to code and analyze reports, which creates complications when trying to compare data from one database to another.

Analysts searching databases to find supporting information for their primary database may need to guess the keywords to search or sift through many reports to find relevant information. The framework developed through this research gives these analysts a systematic starting point to use when searching a second database. The framework enables analysts to accurately identify connections between two taxonomies, offering a theoretical framework to guide the selection of relevant terms. The ability to have an orderly search method is the most practical contribution of this study.

This research also contributed a method for using theory-based taxonomies to analyze the ASRS database, since ASRS is not based on a specific theory like the other three studied taxonomies. Analyzing the database with theory-based taxonomies provides more structure and stronger reliability in the results. While this research is still only applicable to communication and coordination errors, future studies can now analyze these errors with stronger theory-based taxonomies, along with the ASRS taxonomy. Additional research could provide a more complete framework for all codes within the ASRS taxonomy or add additional taxonomies to this framework. Once complete, this framework could be used for all database reports or for databases outside ASRS database that contain similar reports, such as those found in airline ASAP programs.

There is also an effort at work to develop a common aviation taxonomy for all airlines and databases, which would improve the quality of information and communication and allow data to be compared more effectively across databases. The results can help move this common taxonomy forward by comparing the studied taxonomies to ASRS reports and detailing the strengths of each taxonomy. These strengths can be used to inform the development of a common taxonomy.

Limitations of the Findings

This study was focused on communication and coordination errors between pilots. While these errors also encompass the environment, influencing factors, and additional report details, the scope was limited to these specific errors. Therefore, many codes were not used at all and others that were minimally used. In the framework, codes that have no connecting lines to other taxonomies are not shown, but because the study only considered communication and coordination errors, it is unclear if a code without a connecting line means the code was not used

or that it did not have a connection within the specified delimitations of the study. If this study was expanded to include all types of errors, or a greater number of errors, it could be concluded with greater certainty this situation would mean that there is no connection.

Another limiting factor of this study is that ASRS codes are given to the report as a whole, while other taxonomies are applied to specific parts of the report. When comparing what codes were assigned to each report and each coded section, ACCERS, HFACS, and TEM codes were considered based on the specific sections that were coded, but ASRS codes were given to the whole report. For example, if a report section was coded with CRM-Communication for ACCERS and Communication, Coordination, and Planning from HFACS, while the whole report was coded Communication Breakdown within ASRS, the analysis would show that all three codes overlap in that section. However, if another section of the report was coded with Communication Errors, Pilot to Pilot in TEM, it would not show as an overlapping code with ACCERS and HFACS but would still be counted with ASRS. This also occurs with codes that are not related to each other. If the same report was also coded with ASRS Aircraft Equipment, Critical, an overlap would show with all three of the communication codes even though these codes do not mean the same thing or capture the same error. Therefore, the results demonstrate a significantly greater number of connecting lines and overlapping codes when compared to the ASRS taxonomy than with the other taxonomies. When these codes were compared, greater discretion was used when confirming a connecting line in the framework between a code from ACCERS, HFACS, and TEM and an ASRS code.

Recommendations

Aviation safety is becoming increasingly proactive, relying more on data to identify weaknesses and anticipate potential future accidents. The field is also using advanced analytics

and modeling to forecast risks before they manifest, enabling more effective prevention strategies. As a result, it is vital that airline staff have effective ways to analyze data and combine it to provide a complete picture of the operation. When databases use different taxonomies to analyze their reports, it can be cumbersome to use multiple data sources to define a single error or weakness. Using multiple databases with different taxonomies can also result in data being missed or occurrences being misrepresented. This research presents a systematic approach to minimize overlooked or uncoded information, which can otherwise lead to gaps in data analysis and an incomplete understanding of the safety issues. By addressing these omissions, the methodology enhances the accuracy and completeness of data classification, supporting more robust identification of contributing factors and enabling more effective safety interventions. Based on these findings, the following recommendations are proposed to further improve data integrity and reporting practices.

Recommendations for the Target Population

The results of the study led to several recommendations for more efficient and complete analysis and future developments in the use of taxonomies in aviation safety databases.

Recommendation 1. The first recommendation is that aviation safety analysts and anyone using aviation safety databases should become familiar with the developed framework and how it can be used. Analysts can use the framework to determine what codes should be searched when comparing data from databases that use different taxonomies. The framework gives them a more effective way to search the codes to find events or errors relevant to those they are studying in their primary database. Making database search and comparison easier and more efficient will allow analysts to include more data and provide a more complete picture of events and errors. This better view allows staff to focus on ways to correct the most prevalent

errors. Reducing errors and minimizing exposure to risks and hazards leads to a more safe and reliable aviation system.

Recommendation 2. The results of this study provide a detailed analysis of how the studied taxonomies use different terms and different levels of detail to capture the same errors and events. The framework and conclusions can be used to develop a common aviation taxonomy. By understanding how taxonomies differ related to detail, developers of a common taxonomy could ensure they do not omit important codes and also include more codes that provide greater granularity and simpler codes that are more encompassing. Basing a common taxonomy on taxonomies already in use allow for a simpler transition and a gentler learning curve. Because the three studied taxonomies are also theory-based, they provide a strong foundation for a common taxonomy.

Recommendations for Future Research

This study focused on communication and coordination errors between pilots and considered only three commonly used taxonomies. These delimitations can be greatly expanded in future research to include additional types of errors and taxonomies. Adding more error types will significantly improve the depth of the framework, allowing for more errors to be studied. As more error types are added to the framework, they will become more reliable, and the connecting lines will become more meaningful because they include more data.

Similar benefits will be realized by including more taxonomies: more databases will be able to be efficiently searched for related data. Airlines can create a similar framework with their own taxonomies for ease of analysis with shared databases.

When more errors are included, there can also be future research to create an automated way of searching based on the framework. This framework can create a foundation for using

artificial intelligence and machine learning to expand and improve database searches. This again will allow for analysts to search more databases, include more relevant data, and reduce the chance that data will be missed, thereby increasing the validity of their research and improving aviation safety.

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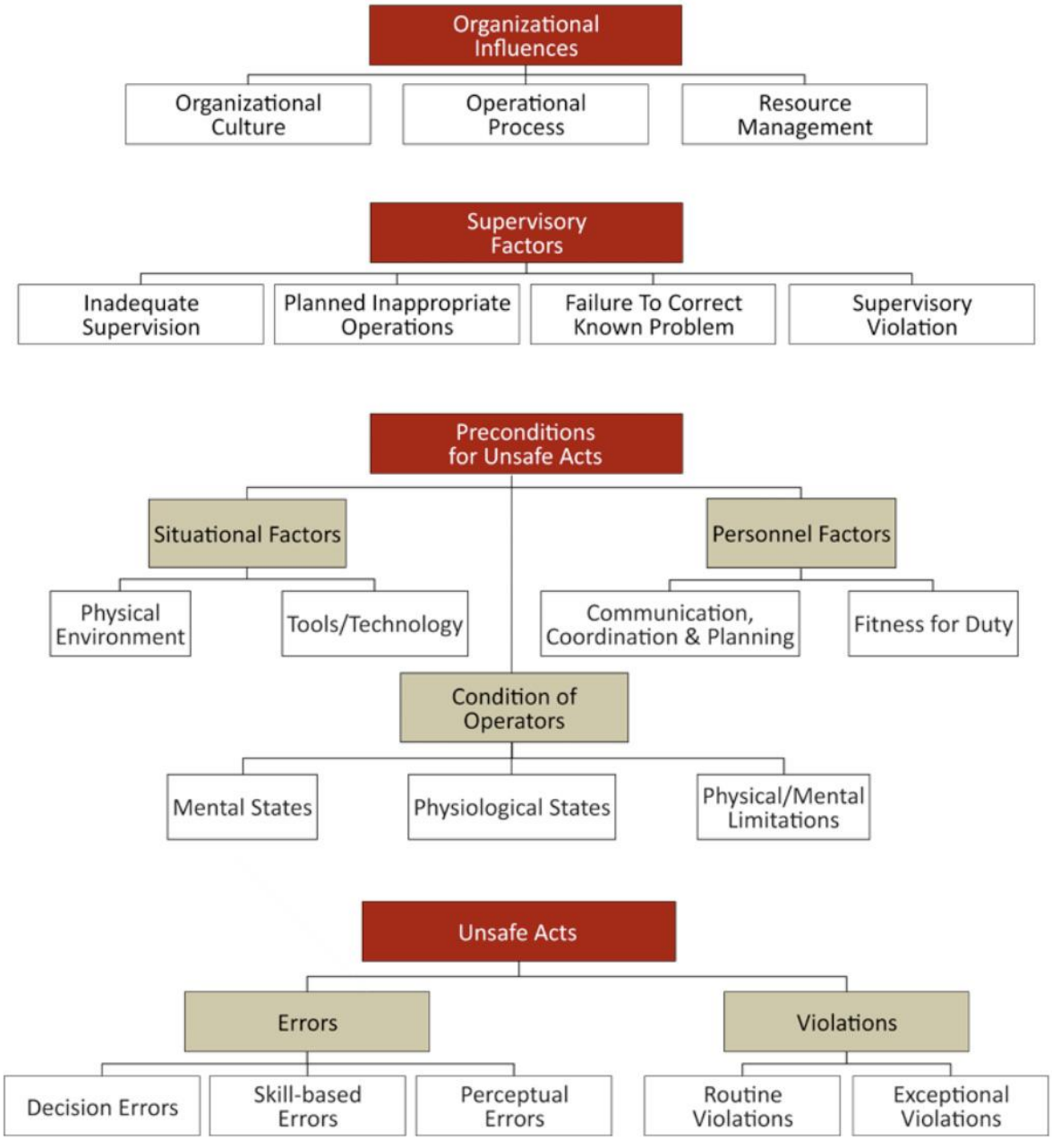
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Appendix A:

Figure A1

HFACS Model



Note: From The HFACS Framework, by HFACS.inc, 2014 (https://www.hfacs.com/hfacs-framework.html).

Figure A2

TEM Taxonomy: Threats

Threat Types with Examples

Environmental Threats	Examples
Adverse Weather	Thunderstorms, turbulence, poor visibility, wind shear, icing conditions, IMC
Airport	Poor signage, faint markings, runway/taxiway closures, INOP navigational aids, poor braking action, contaminated runways/taxiways
ATC	Tough-to-meet clearances/restrictions, reroutes, language difficulties, controller errors
Environmental Operational Pressure	Terrain, traffic, TCAS TA / RA, radio congestion
Airline Threats	**Examples**
Aircraft	Systems, engines, flight controls, or automation anomalies or malfunctions; MEL items with operational implications; other aircraft threats requiring flight crew attention
Airline Operational Pressure	On-time performance pressure, delays, late arriving aircraft or flight crew
Cabin	Cabin events, flight attendant errors, distractions, interruptions
Dispatch/Paperwork	Load sheet errors, crew scheduling events, late paperwork, changes or errors
Ground/Ramp	Aircraft loading events, fuelling errors, agent interruptions, improper ground support, de-icing
Ground Maintenance	Aircraft repairs on ground, maintenance log problems, maintenance errors
Manuals/Charts	Missing information or documentation errors

Note: From Defensive Flying for Pilots: An Introduction to Threat and Error Management by A. Merritt and J. Klinect, 2006, *The University of Texas Human Factors Research Project, The LOSA Collaborative.*

Figure A3

TEM Taxonomy: Errors

Error Types with Examples

Aircraft Handling Errors	Examples
Automation	Incorrect altitude, speed, heading, autothrottle settings, mode executed, or entries
Flight Control	Incorrect flaps, speed brake, autobrake, thrust reverser or power settings
Ground Navigation	Attempting to turn down wrong taxiway/runway Missed taxiway/runway/gate
Manual Flying	Hand flying vertical, lateral, or speed deviations Missed runway/taxiway failure to hold short, or taxi above speed limit
Systems/Radio/Instruments	Incorrect pack, altimeter, fuel switch or radio frequency settings
Procedural Errors	**Examples**
Briefings	Missed items in the brief, omitted departure, takeoff, approach, or handover briefing
Callouts	Omitted takeoff, descent, or approach callouts
Checklist	Performed checklist from memory or omitted checklist Missed items, wrong challenge and response, performed late or at wrong time
Documentation	Wrong weight and balance, fuel information, ATIS, or clearance recorded Misinterpreted items on paperwork
Pilot Flying (PF)/Pilot Not Flying (PNF) Duty	PF makes own automation changes, PNF doing PF duties, PF doing PNF duties
SOP Cross-verification	Intentional and unintentional failure to cross-verify automation inputs
Other Procedural	Other deviations from government regulations, flight manual requirements or standard operating procedures
Communication Errors	**Examples**
Crew to External	Missed calls, misinterpretation of instructions, or incorrect read-backs to ATC Wrong clearance, taxiway, gate or runway communicated
Pilot to Pilot	Within-crew miscommunication or misinterpretation

Note: From Defensive Flying for Pilots: An Introduction to Threat and Error Management by A. Merritt and J. Klinect, 2006, *The University of Texas Human Factors Research Project, The LOSA Collaborative.*

Figure A4

TEM Taxonomy: UAS

UAS Types with Examples

UAS Types	Examples
Aircraft Handling	Vertical, lateral or speed deviations Unnecessary weather penetration Unstable approach Long, floated, firm or off-centreline landings
Ground Navigation	Runway/taxiway incursions Wrong taxiway, ramp, gate, or hold spot Taxi above speed limit
Incorrect Aircraft Configuration	Automation, engine, flight control, systems, or weight/balance events

Note: From Defensive Flying for Pilots: An Introduction to Threat and Error Management by A. Merritt and J. Klinect, 2006, *The University of Texas Human Factors Research Project, The LOSA Collaborative.*

Figure A5

ACCERS Taxonomy

Category and Category Definition	Factor	Factor Definition
Policies or Procedures: includes standard operating procedures (SOPs), charts, maps, checklists, and the like that ensure that flight operations are performed in accordance with FAA and airline regulations. Factors in this category include non-existent, confusing, conflicting, or inaccurate policies or procedures, and the like.	Conflicting policies or procedures	Two or more policies or procedures provide incompatible information or guidance regarding the same operation
	Confusing policy or procedure	Policy or procedure provides information or guidance that is difficult to understand or follow
	Inaccurate policy or procedure	Policy or procedure provides incorrect or inaccurate guidance or information
	International policies or procedures	Significant difference between international policy or procedure and domestic policy or procedure for the same operation
	Lack of policy or procedure	No policy or procedure exists regarding this operation
	Old vs. new policy or procedure	Significant difference between old policy or procedure and new policy or procedure for the same operation
	Other/please provide: _____ _____	The causal factor that contributed to the incident is not provided in the taxonomy. Please enter a brief description of the factor that contributed: _____
Human Error: refers to actions that do not achieve their intended consequences, or failure to act. For example, if you incorrectly program a waypoint into the Flight Management Computer (FMC), you have made an error. Other examples include weight and balance errors, the misapplication of flight controls, inadvertent or intentional disregard of policy or procedure, improper use of the autopilot/FMS/FMC/navigation equipment, and the like.	Altitude planning/awareness	Lack of awareness of altitude or poor planning with regard to establishing or maintaining proper altitude
	Improper use of autopilot/FMS/FMC navigational equipment	Equipment used at the wrong time, by the wrong person, by inputting improper commands, or by interpreting output data incorrectly
	Inadvertent or intentional disregard for policy or procedure	Accidental or purposeful disregard of the proper policy or procedure for the operation
	MEL/AML error	Failure to review MEL/AML, failure to follow MEL/AML procedure/restrictions, or following wrong MEL/AML
	Misapplication of flight controls	Aircraft controls applied at the wrong time, by the wrong person, or by applying too little/too much directional or physical control
	Poor aircraft performance planning	Failure to consult the performance book, consulting the wrong performance book, selecting the wrong performance standard, or failing to properly apply the appropriate performance standard for a given flight condition
	Weight and balance error	Load was not distributed according to policy or procedure
	Other/please provide: _____ _____	The causal factor that contributed to the incident is not provided in the taxonomy. Please enter a brief description of the factor that contributed: _____
Human Factors: physical, psychological, and social factors that influence human performance. Examples that are likely to hinder performance include poor teamwork skills, task interruptions and distractions, complacency, stress, fatigue, task saturation, and the like. Human factors are often the precursors of errors.	Attention to detail	Lack of effort toward discovering the small elements of the situation or too much focus on the small elements of the situation
	Complacency	Lack of awareness of actual dangers or deficiencies
	CRM-communication	Appropriate communication protocol was not followed
	CRM-conflict management	Conflict between crew members was not managed appropriately
	CRM-incompatible crew pairing	Crew members do not work well together
	CRM-leadership and command	Leader failed to provide sound guidance and clear priorities
	Experience level	Crew member has too little/too much experience in this area, and this had a negative impact on the situation
	Fatigue	Crew member tired
	Fear or panic	Crew member afraid or suddenly frightened
	High workload/task saturation	Amount of work to be performed in a given time period excessive
	Illness or injury/physical health	Crew member sick or hurt
	Interruption/distraction	Interruption caused focus of attention to be shifted or diverted away from task
	Medication side effect	Physical, emotional, or cognitive reaction to medication
	Motivation	Internally or externally generated drive or influence is lacking or misdirected
	Peer pressure	Constraints or burdens generated by one's friends or coworkers
	Personal attitude(s) toward safety	Individual beliefs and feelings about safety
	Personal stress	Individual mental or physical tension
	Proficiency/overreliance on automation	Excessive dependence on mechanical aids to flight
	Self-induced time pressure	Individual personal desire to depart the gate/arrive at the destination at the scheduled time
	Situational awareness	Not knowing what is going on around you or which parameters are important, or lack of effort toward discovering important parameters
	Other/please provide: _____ _____	The causal factor that contributed to the incident is not provided in the taxonomy. Please enter a brief description of the factor that contributed: _____

Note: FMS = Flight Management System; MEL = Minimum Equipment List; AML = Approved Model List; CRM = Crew Resource Management; MELD = Mechanized Engineering and Layout for Distributing Frames; TCAS = Traffic Alert and Collision Avoidance System; RA = Resolution Alert (part of TCAS); TA = Traffic Alert (part of TCAS).

Continued on next page

Category and Category Definition	Factor	Factor Definition
Organizational Factors: airline-specific factors that affect all the pilots at the airline. Examples include the airline's safety culture, its rest/duty/flight times, flight or ground management, on-time performance pressures, and the like.	Airline's safety culture	The airline's formal and informal expectations regarding safety
	Employee staffing	Failure to provide adequate staffing for particular operation
	Inadequate flight management	Improper oversight or lack of oversight from flight management
	Inadequate ground management	Improper oversight or lack of oversight from ground management
	Inadequate training	Airline-sponsored training regarding this operation deficient in quality, quantity, or timeliness
	Lack of accountability	Chain of command unclear; no person or entity is responsible
	Mismanagement of assets	Inappropriate allocation of resources
	On-time performance pressure	Pressure to ensure that the aircraft leaves the gate and arrives at its destination on time
	Pilot selection	Pairing of crew members who are not suitable together
	Rest/duty/flight times	Work schedule allows too little rest, or schedule frequently changed at the last minute
	Other/please provide: _____ _____	The causal factor that contributed to the incident is not provided in the taxonomy. Please enter a brief description of the factor that contributed: _____
Hardware: issues concern how the airplane or ground equipment was designed, manufactured, or maintained. Examples include equipment limitations, equipment malfunctions, poor design, and the like.	Aircraft damage	Physical damage to plane
	Aircraft equipment malfunction – unable to duplicate	Equipment fails to operate correctly when in service, but maintenance personnel cannot recreate the failure
	Aircraft equipment malfunction – query whether first time or repeated	First known failure of equipment to operate correctly, or equipment fails again and again to operate correctly, despite check by maintenance personnel
	Deferred item/MELD equipment	Inoperative equipment
	Equipment limitation	Equipment lacks the capability to perform certain needed functions
	Ergonomics/poor product design, placement, or user interface	Equipment is planned, built, and/or installed in such a way that it is difficult to use
	Ground equipment inoperative or malfunctioned	Failure of ground equipment to operate correctly
	Other/please provide: _____ _____	The causal factor that contributed to the incident is not provided in the taxonomy. Please enter a brief description of the factor that contributed: _____
Weather or Environment: includes threats to safety that occur outside the airplane. Examples include storm activity, turbulence, extreme temperatures, animal/bird strike, and the like.	Animal/bird strike	A collision between the aircraft and an animal
	Excessive cold	Atmospheric temperature very cold
	Excessive heat	Atmospheric temperature very hot
	Icing	Considerable icing on the aircraft/runway
	Low visibility/low ceiling	Limited distance that one can see clearly with the naked eye
	Nighttime operations	Operation was initiated, took place, or was completed in the dark
	Perceptual illusion/spatial disorientation	Misinterpretation of visual or spatial cues
	Thunderstorm/severe weather	Extreme meteorological conditions
	Turbulence	Irregular motion of aircraft created by up-and-down currents of air
	Wind shear	An extreme shift in wind speed or direction that occurs over a very short distance
	Other/please provide: _____ _____	The causal factor that contributed to the incident is not provided in the taxonomy. Please enter a brief description of the factor that contributed: _____
Airspace or ATC: refers to characteristics of the National Airspace System (NAS) that can support or detract from safety. Examples include late, unclear, or conflicting clearances, airfield markings or layout; hear back/read back; frequency congestion; and the like.	Air traffic congestion	High-density air traffic
	Airfield markings or layout	Lack of or poor airport markings or signs
	Conflicting ATC clearance	Two or more ATC clearances provide incompatible information or guidance regarding the same operation
	Expected vs. assigned ATC clearance	ATC clearance received was different from what was anticipated
	Frequency congestion	Heavy voice traffic on radio frequency
	Hear back/read back	Failure to hear or read back the proper ATC clearance
	Incorrect ATC clearance	ATC provided inaccurate guidance or information that would have resulted in a breach of safety, policy, or procedure
	Late ATC clearance	ATC did not provide guidance or information in a timely manner
	TCAS RA/TA	A TA or an RA occurred
	Unclear ATC clearance	Communication followed proper protocol, but guidance provided by ATC did not make sense.
	Other/please provide: _____ _____	The causal factor that contributed to the incident is not provided in the taxonomy. Please enter a brief description of the factor that contributed: _____

Note: FMS = Flight Management System; MEL = Minimum Equipment List; AML = Approved Model List; CRM = Crew Resource Management; MELD = Mechanized Engineering and Layout for Distributing Frames; TCAS = Traffic Alert and Collision Avoidance System; RA = Resolution Alert (part of TCAS); TA = Traffic Alert (part of TCAS).

Note: From Development and Validation of Aviation Causal Contributors for Error Reporting Systems (ACCERS) by D. P. Baker and K. J. Krokos, 2007, *Human Factors 49*(2) p.185

Appendix B:

Table B1

Numbers for All Coded Nodes

Codes	Number of coding references	Aggregate number of coding references
ACCERS	0	1315
ACCERS\Airspace or ATC	0	190
ACCERS\Airspace or ATC\Air traffic congestion	0	0
ACCERS\Airspace or ATC\Airfield markings or layout	5	5
ACCERS\Airspace or ATC\Conflicting ATC clearance	36	36
ACCERS\Airspace or ATC\Expected vs. assigned ATC clearance	10	10
ACCERS\Airspace or ATC\Frequency congestion	18	18
ACCERS\Airspace or ATC\Hear back & read back	15	15
ACCERS\Airspace or ATC\Incorrect ATC clearance	0	0
ACCERS\Airspace or ATC\Late ATC clearance	11	11
ACCERS\Airspace or ATC\TCAS RA & TA	6	6
ACCERS\Airspace or ATC\Unclear ATC clearance	89	89
ACCERS\Hardware	0	44
ACCERS\Hardware\Aircraft damage	10	10
ACCERS\Hardware\Aircraft equipment malfunction - query whether first time or repeated	26	26
ACCERS\Hardware\Aircraft equipment malfunction - unable to duplicate	1	1
ACCERS\Hardware\Deferred item - MELD equipment	1	1
ACCERS\Hardware\Equipment limitation	2	2
ACCERS\Hardware\Ergonomics - poor product design, placement, or user interface	3	3
ACCERS\Hardware\Ground equipment inoperative or malfunctioned	1	1
ACCERS\Human Error	0	188
ACCERS\Human Error\Altitude planning & awareness	42	42
ACCERS\Human Error\Improper use of autopilot, FMS, FMC navigational equipment	29	29
ACCERS\Human Error\Inadvertent or intentional disregard for policy or procedure	97	97
ACCERS\Human Error\MEL & AML error	11	11
ACCERS\Human Error\Misapplication of flight controls	7	7
ACCERS\Human Error\Poor aircraft performance planning	1	1
ACCERS\Human Error\Weight & balance error	1	1

ACCERS\Human Factors	0	640
ACCERS\Human Factors\Attention to detail	50	50
ACCERS\Human Factors\Complacency	19	19
ACCERS\Human Factors\CRM - communication	286	286
ACCERS\Human Factors\CRM - conflict management	1	1
ACCERS\Human Factors\CRM - incompatible crew pairing	7	7
ACCERS\Human Factors\CRM - leadership and command	15	15
ACCERS\Human Factors\Experience level	43	43
ACCERS\Human Factors\Fatigue	31	31
ACCERS\Human Factors\Fear or panic	0	0
ACCERS\Human Factors\High workload & task saturation	57	57
ACCERS\Human Factors\Illness or injury - physical health	0	0
ACCERS\Human Factors\Interruption - distraction	50	50
ACCERS\Human Factors\Medication side effect	0	0
ACCERS\Human Factors\Motivation	0	0
ACCERS\Human Factors\Peer pressure	1	1
ACCERS\Human Factors\Personal attitude(s) toward safety	31	31
ACCERS\Human Factors\Personal stress	2	2
ACCERS\Human Factors\Proficiency - overreliance on automation	2	2
ACCERS\Human Factors\Self-induced time pressure	14	14
ACCERS\Human Factors\Situational awareness	31	31
ACCERS\Organizational Factors	0	177
ACCERS\Organizational Factors\Airline's safety culture	90	90
ACCERS\Organizational Factors\Employee staffing	6	6
ACCERS\Organizational Factors\Inadequate flight management	2	2
ACCERS\Organizational Factors\Inadequate ground management	21	21
ACCERS\Organizational Factors\Inadequate training	25	25
ACCERS\Organizational Factors\Lack of accountability	8	8
ACCERS\Organizational Factors\Mismanagement of assets	4	4
ACCERS\Organizational Factors\On-time performance pressure	19	19
ACCERS\Organizational Factors\Pilot selection	0	0
ACCERS\Organizational Factors\Rest, duty, flight times	2	2
ACCERS\Policies or Procedures	0	34
ACCERS\Policies or Procedures\Conflicting policies or procedures	2	2
ACCERS\Policies or Procedures\Confusing policies and procedures	14	14
ACCERS\Policies or Procedures\Inaccurate policy or procedures	1	1
ACCERS\Policies or Procedures\International policies or procedures	0	0

ACCERS\Policies or Procedures\Lack of policy or procedure	17	17
ACCERS\Policies or Procedures\Old vs. new policy or procedure	0	0
ACCERS\Weather or Environment	0	42
ACCERS\Weather or Environment\Animal or bird strike	0	0
ACCERS\Weather or Environment\Excessive cold	4	4
ACCERS\Weather or Environment\Excessive heat	0	0
ACCERS\Weather or Environment\Icing	7	7
ACCERS\Weather or Environment\Low visibility - low ceiling	8	8
ACCERS\Weather or Environment\Nighttime operations	4	4
ACCERS\Weather or Environment\Perceptual illusion - spatial disorientation	2	2
ACCERS\Weather or Environment\Thunderstorm or severe weather	10	10
ACCERS\Weather or Environment\Turbulence	3	3
ACCERS\Weather or Environment\Wind shear	4	4
ASRS Code	0	2167
ASRS Code\Aircraft Equipment	1	114
ASRS Code\Aircraft Equipment\Critical	72	72
ASRS Code\Aircraft Equipment\Less Severe	41	41
ASRS Code\Airspace Violation	0	3
ASRS Code\Airspace Violation\All Types	3	3
ASRS Code\ATC Issues	0	100
ASRS Code\ATC Issues\All Types	100	100
ASRS Code\Conflict	0	49
ASRS Code\Conflict\Airborne Conflict	15	15
ASRS Code\Conflict\Ground Conflict, Critical	16	16
ASRS Code\Conflict\Ground Conflict, Less Severe	14	14
ASRS Code\Conflict\NMAC	4	4
ASRS Code\Deviation - Altitude	0	85
ASRS Code\Deviation - Altitude\Crossing Restriction Not Met	25	25
ASRS Code\Deviation - Altitude\Excursion from Assigned Altitude	19	19
ASRS Code\Deviation - Altitude\Overshoot	37	37
ASRS Code\Deviation - Altitude\Undershoot	4	4
ASRS Code\Deviation - Speed	0	19
ASRS Code\Deviation - Speed\All Types	19	19
ASRS Code\Deviation - Track & Heading	0	45
ASRS Code\Deviation - Track & Heading\All Types	45	45
ASRS Code\Deviation & Discrepancy - Procedural	0	590
ASRS Code\Deviation & Discrepancy - Procedural\Clearance	126	126
ASRS Code\Deviation & Discrepancy - Procedural\FAR	82	82

ASRS Code\Deviation & Discrepancy - Procedural\Hazardous Material Violation	6	6
ASRS Code\Deviation & Discrepancy - Procedural\Landing without Clearance	1	1
ASRS Code\Deviation & Discrepancy - Procedural\Maintenance	68	68
ASRS Code\Deviation & Discrepancy - Procedural\MEL & CDL	29	29
ASRS Code\Deviation & Discrepancy - Procedural\Other or Unknown	3	3
ASRS Code\Deviation & Discrepancy - Procedural\Published Material & Policy	264	264
ASRS Code\Deviation & Discrepancy - Procedural\Security	0	0
ASRS Code\Deviation & Discrepancy - Procedural\Unauthorized Flight Operations (UAS)	0	0
ASRS Code\Deviation & Discrepancy - Procedural\Weight and Balance	11	11
ASRS Code\Flight Deck & Cabin & Aircraft Event	0	37
ASRS Code\Flight Deck & Cabin & Aircraft Event\Illness & Injury	5	5
ASRS Code\Flight Deck & Cabin & Aircraft Event\Other or Unknown	23	23
ASRS Code\Flight Deck & Cabin & Aircraft Event\Passenger Misconduct	4	4
ASRS Code\Flight Deck & Cabin & Aircraft Event\Passenger Electronic Device	2	2
ASRS Code\Flight Deck & Cabin & Aircraft Event\Smoke, Fire, Fumes, Odor	3	3
ASRS Code\Ground Event & Encounter	0	37
ASRS Code\Ground Event & Encounter\Aircraft	4	4
ASRS Code\Ground Event & Encounter\FOD	0	0
ASRS Code\Ground Event & Encounter\Fuel Issue	1	1
ASRS Code\Ground Event & Encounter\Gear Up Landing	0	0
ASRS Code\Ground Event & Encounter\Ground Equipment Issue	1	1
ASRS Code\Ground Event & Encounter\Ground Strike - Aircraft	3	3
ASRS Code\Ground Event & Encounter\Jet Blast	0	0
ASRS Code\Ground Event & Encounter\Loss of Aircraft Control	6	6
ASRS Code\Ground Event & Encounter\Loss of VLOS (UAS)	0	0
ASRS Code\Ground Event & Encounter\Object	2	2
ASRS Code\Ground Event & Encounter\Other or Unknown	14	14
ASRS Code\Ground Event & Encounter\Person - Animal - Bird	2	2
ASRS Code\Ground Event & Encounter\Vehicle	3	3
ASRS Code\Ground Event & Encounter\Weather & Turbulence	1	1
ASRS Code\Ground Excursion	0	1
ASRS Code\Ground Excursion\Ramp	0	0

ASRS Code\Ground Excursion\Runway	1	1
ASRS Code\Ground Excursion\Taxiway	0	0
ASRS Code\Ground Incursion	0	26
ASRS Code\Ground Incursion\Ramp	0	0
ASRS Code\Ground Incursion\Runway	18	18
ASRS Code\Ground Incursion\Taxiway	8	8
ASRS Code\Human Factors	0	968
ASRS Code\Human Factors\Communication Breakdown	355	355
ASRS Code\Human Factors\Confusion	116	116
ASRS Code\Human Factors\Distraction	76	76
ASRS Code\Human Factors\Fatigue	17	17
ASRS Code\Human Factors\Human-Machine Interface	51	51
ASRS Code\Human Factors\Other or Unknown	10	10
ASRS Code\Human Factors\Physiological - Other	3	3
ASRS Code\Human Factors\Situational Awareness	141	141
ASRS Code\Human Factors\Time Pressure	53	53
ASRS Code\Human Factors\Training and Qualification	39	39
ASRS Code\Human Factors\Troubleshooting	57	57
ASRS Code\Human Factors\Workload	50	50
ASRS Code\Inflight Event & Encounter	0	90
ASRS Code\Inflight Event & Encounter\Aircraft	0	0
ASRS Code\Inflight Event & Encounter\Bird or Animal	0	0
ASRS Code\Inflight Event & Encounter\CFTT & CFIT	24	24
ASRS Code\Inflight Event & Encounter\Fly Away (UAS)	0	0
ASRS Code\Inflight Event & Encounter\Fuel Issue	7	7
ASRS Code\Inflight Event & Encounter\Laser	0	0
ASRS Code\Inflight Event & Encounter\Loss of Aircraft Control	1	1
ASRS Code\Inflight Event & Encounter\Object	0	0
ASRS Code\Inflight Event & Encounter\Other or Unknown	5	5
ASRS Code\Inflight Event & Encounter\Unstabilized Approach	26	26
ASRS Code\Inflight Event & Encounter\VFR in IMC	0	0
ASRS Code\Inflight Event & Encounter\Wake Vortex Encounter	0	0
ASRS Code\Inflight Event & Encounter\Weather or Turbulence	27	27
ASRS Code\No Specific Anomaly Occurred	0	2
ASRS Code\No Specific Anomaly Occurred\All Types	1	1
ASRS Code\No Specific Anomaly Occurred\Unwanted Situation	1	1
ASRS Code\Other	1	1
HFACS	0	1240
HFACS\Organizational Influences	0	185
HFACS\Organizational Influences\Operational Process	117	117
HFACS\Organizational Influences\Organizational Culture	56	56

HFACS\Organizational Influences\Resource Management	12	12
HFACS\Preconditions for Unsafe Acts	0	544
HFACS\Preconditions for Unsafe Acts\Condition of Operators	0	143
HFACS\Preconditions for Unsafe Acts\Condition of Operators\Mental State	141	141
HFACS\Preconditions for Unsafe Acts\Condition of Operators\Physical & Mental Limitations	0	0
HFACS\Preconditions for Unsafe Acts\Condition of Operators\Physiological States	2	2
HFACS\Preconditions for Unsafe Acts\Personnel Factors	0	332
HFACS\Preconditions for Unsafe Acts\Personnel Factors\Communication, Coordination & Planning	330	330
HFACS\Preconditions for Unsafe Acts\Personnel Factors\Fitness for Duty	2	2
HFACS\Preconditions for Unsafe Acts\Situational Factors	0	69
HFACS\Preconditions for Unsafe Acts\Situational Factors\Physical Environment	46	46
HFACS\Preconditions for Unsafe Acts\Situational Factors\Tools & Technology	23	23
HFACS\Supervisory Factors	0	195
HFACS\Supervisory Factors\Failure to Correct Known Problem	53	53
HFACS\Supervisory Factors\Inadequate Supervision	77	77
HFACS\Supervisory Factors\Planned Inappropriate Operations	59	59
HFACS\Supervisory Factors\Supervisory Violation	6	6
HFACS\Unsafe Acts	0	316
HFACS\Unsafe Acts\Errors	0	252
HFACS\Unsafe Acts\Errors\Decision Errors	45	45
HFACS\Unsafe Acts\Errors\Perceptual Errors	9	9
HFACS\Unsafe Acts\Errors\Skill-Based Errors	198	198
HFACS\Unsafe Acts\Violations	0	64
HFACS\Unsafe Acts\Violations\Exceptional Violations	31	31
HFACS\Unsafe Acts\Violations\Routine Violations	33	33
TEM	0	1106
TEM\Aircraft Handling Errors	0	74
TEM\Aircraft Handling Errors\Automation	51	51
TEM\Aircraft Handling Errors\Flight Control	12	12
TEM\Aircraft Handling Errors\Ground Navigation	5	5
TEM\Aircraft Handling Errors\Manual Flying	3	3
TEM\Aircraft Handling Errors\Systems & Radio & Instruments	3	3
TEM\Airline Threats	0	284
TEM\Airline Threats\Aircraft	28	28
TEM\Airline Threats\Airline Operational Pressure	89	89
TEM\Airline Threats\Cabin	5	5

TEM\Airline Threats\Dispatch & Paperwork	15	15
TEM\Airline Threats\Ground & Ramp	61	61
TEM\Airline Threats\Ground Maintenance	76	76
TEM\Airline Threats\Manuals & Charts	10	10
TEM\Communication Errors	1	332
TEM\Communication Errors\Crew to External	157	157
TEM\Communication Errors\Pilot to Pilot	174	174
TEM\Environmental Threats	0	220
TEM\Environmental Threats\Adverse Weather	43	43
TEM\Environmental Threats\Airport	6	6
TEM\Environmental Threats\ATC	166	166
TEM\Environmental Threats\Environmental Operational Pressure	5	5
TEM\Procedural Errors	0	80
TEM\Procedural Errors\Briefings	2	2
TEM\Procedural Errors\Callouts	5	5
TEM\Procedural Errors\Checklist	10	10
TEM\Procedural Errors\Documentation	12	12
TEM\Procedural Errors\Other Procedural	50	50
TEM\Procedural Errors\Pilot Flying (PF) & Pilot Not Flying (PNF) Duties	1	1
TEM\Procedural Errors\SOP Cross-verification	0	0
TEM\UAS Types	0	116
TEM\UAS Types\Aircraft Handling	85	85
TEM\UAS Types\Ground Navigation	22	22
TEM\UAS Types\Incorrect Aircraft Configuration	9	9

Appendix C:

Table C1

Percentage Agreement for All Coded Nodes

Name	Agreement	Disagreement
Codes\\ACCERS	89.43	10.58
Codes\\ACCERS : Communication Sample for NVIVO	89.43	10.58
Codes\\TEM\Environmental Threats\Adverse Weather	98.95	1.06
Codes\\TEM\Environmental Threats\Adverse Weather : Communication Sample for NVIVO	98.95	1.06
Codes\\ACCERS\Airspace or ATC\Air traffic congestion	100	0
Codes\\ACCERS\Airspace or ATC\Air traffic congestion : Communication Sample for NVIVO	100	0
Codes\\TEM\Airline Threats\Aircraft	99.14	0.86
Codes\\TEM\Airline Threats\Aircraft : Communication Sample for NVIVO	99.14	0.86
Codes\\ACCERS\Hardware\Aircraft damage	99.7	0.3
Codes\\ACCERS\Hardware\Aircraft damage : Communication Sample for NVIVO	99.7	0.3
Codes\\ACCERS\Hardware\Aircraft equipment malfunction - query whether first time or repeated	99.22	0.79
Codes\\ACCERS\Hardware\Aircraft equipment malfunction - query whether first time or repeated : Communication Sample for NVIVO	99.22	0.79
Codes\\TEM\UAS Types\Aircraft Handling	97.94	2.06
Codes\\TEM\UAS Types\Aircraft Handling : Communication Sample for NVIVO	97.94	2.06
Codes\\ACCERS\Airspace or ATC\Airfield markings or layout	99.89	0.12
Codes\\ACCERS\Airspace or ATC\Airfield markings or layout : Communication Sample for NVIVO	99.89	0.12
Codes\\TEM\Airline Threats\Airline Operational Pressure	97.67	2.33
Codes\\TEM\Airline Threats\Airline Operational Pressure : Communication Sample for NVIVO	97.67	2.33
Codes\\ACCERS\Organizational Factors\Airline's safety culture	97.75	2.25
Codes\\ACCERS\Organizational Factors\Airline's safety culture : Communication Sample for NVIVO	97.75	2.25
Codes\\TEM\Environmental Threats\Airport	99.85	0.15
Codes\\TEM\Environmental Threats\Airport : Communication Sample for NVIVO	99.85	0.15
Codes\\ACCERS\Human Error\Altitude planning & awareness	98.95	1.06
Codes\\ACCERS\Human Error\Altitude planning & awareness : Communication Sample for NVIVO	98.95	1.06
Codes\\ACCERS\Weather or Environment\Animal or bird strike	100	0
Codes\\ACCERS\Weather or Environment\Animal or bird strike : Communication Sample for NVIVO	100	0
Codes\\TEM\Environmental Threats\ATC	96.58	3.41
Codes\\TEM\Environmental Threats\ATC : Communication Sample for NVIVO	96.58	3.41
Codes\\ACCERS\Human Factors\Attention to detail	98.54	1.46

Codes\\ACCERS\Human Factors\Attention to detail : Communication Sample for NVIVO	98.54	1.46
Codes\\TEM\Aircraft Handling Errors\Automation	98.91	1.09
Codes\\TEM\Aircraft Handling Errors\Automation : Communication Sample for NVIVO	98.91	1.09
Codes\\TEM\Airline Threats\Cabin	99.89	0.11
Codes\\TEM\Airline Threats\Cabin : Communication Sample for NVIVO	99.89	0.11
Codes\\TEM\Procedural Errors\Checklist	99.78	0.23
Codes\\TEM\Procedural Errors\Checklist : Communication Sample for NVIVO	99.78	0.23
Codes\\HFACS\Preconditions for Unsafe Acts\Personnel Factors\Communication, Coordination & Planning	93.7	6.3
Codes\\HFACS\Preconditions for Unsafe Acts\Personnel Factors\Communication, Coordination & Planning : Communication Sample for NVIVO	93.7	6.3
Codes\\ACCERS\Human Factors\Complacency	99.44	0.57
Codes\\ACCERS\Human Factors\Complacency : Communication Sample for NVIVO	99.44	0.57
Codes\\ACCERS\Airspace or ATC\Conflicting ATC clearance	98.87	1.13
Codes\\ACCERS\Airspace or ATC\Conflicting ATC clearance : Communication Sample for NVIVO	98.87	1.13
Codes\\ACCERS\Policies or Procedures\Confusing policies and procedures	99.55	0.45
Codes\\ACCERS\Policies or Procedures\Confusing policies and procedures : Communication Sample for NVIVO	99.55	0.45
Codes\\TEM\Communication Errors\Crew to External	96.63	3.38
Codes\\TEM\Communication Errors\Crew to External : Communication Sample for NVIVO	96.63	3.38
Codes\\ACCERS\Human Factors\CRM - communication	94.45	5.55
Codes\\ACCERS\Human Factors\CRM - communication : Communication Sample for NVIVO	94.45	5.55
Codes\\ACCERS\Human Factors\CRM - incompatible crew pairing	99.89	0.11
Codes\\ACCERS\Human Factors\CRM - incompatible crew pairing : Communication Sample for NVIVO	99.89	0.11
Codes\\ACCERS\Human Factors\CRM - leadership and command	99.59	0.42
Codes\\ACCERS\Human Factors\CRM - leadership and command : Communication Sample for NVIVO	99.59	0.42
Codes\\HFACS\Unsafe Acts\Errors\Decision Errors	98.91	1.09
Codes\\HFACS\Unsafe Acts\Errors\Decision Errors : Communication Sample for NVIVO	98.91	1.09
Codes\\TEM\Airline Threats\Dispatch & Paperwork	99.59	0.42
Codes\\TEM\Airline Threats\Dispatch & Paperwork : Communication Sample for NVIVO	99.59	0.42
Codes\\TEM\Procedural Errors\Documentation	99.63	0.38
Codes\\TEM\Procedural Errors\Documentation : Communication Sample for NVIVO	99.63	0.38
Codes\\HFACS\Unsafe Acts\Violations\Exceptional Violations	98.99	1.01
Codes\\HFACS\Unsafe Acts\Violations\Exceptional Violations : Communication Sample for NVIVO	98.99	1.01
Codes\\ACCERS\Weather or Environment\Excessive cold	99.93	0.08

Codes\\ACCERS\Weather or Environment\Excessive cold : Communication Sample for NVIVO	99.93	0.08
Codes\\ACCERS\Weather or Environment\Excessive heat	100	0
Codes\\ACCERS\Weather or Environment\Excessive heat : Communication Sample for NVIVO	100	0
Codes\\ACCERS\Human Factors\Experience level	98.84	1.16
Codes\\ACCERS\Human Factors\Experience level : Communication Sample for NVIVO	98.84	1.16
Codes\\HFACS\Supervisory Factors\Failure to Correct Known Problem	98.2	1.81
Codes\\HFACS\Supervisory Factors\Failure to Correct Known Problem : Communication Sample for NVIVO	98.2	1.81
Codes\\ACCERS\Human Factors\Fatigue	99.29	0.71
Codes\\ACCERS\Human Factors\Fatigue : Communication Sample for NVIVO	99.29	0.71
Codes\\ACCERS\Human Factors\Fear or panic	100	0
Codes\\ACCERS\Human Factors\Fear or panic : Communication Sample for NVIVO	100	0
Codes\\ACCERS\Airspace or ATC\Frequency congestion	99.62	0.38
Codes\\ACCERS\Airspace or ATC\Frequency congestion : Communication Sample for NVIVO	99.62	0.38
Codes\\TEM\Airline Threats\Ground & Ramp	98.5	1.5
Codes\\TEM\Airline Threats\Ground & Ramp : Communication Sample for NVIVO	98.5	1.5
Codes\\TEM\Airline Threats\Ground Maintenance	97.82	2.18
Codes\\TEM\Airline Threats\Ground Maintenance : Communication Sample for NVIVO	97.82	2.18
Codes\\TEM\UAS Types\Ground Navigation	99.63	0.38
Codes\\TEM\UAS Types\Ground Navigation : Communication Sample for NVIVO	99.63	0.38
Codes\\ACCERS\Airspace or ATC\Hear back & read back	99.66	0.34
Codes\\ACCERS\Airspace or ATC\Hear back & read back : Communication Sample for NVIVO	99.66	0.34
Codes\\HFACS	89.54	10.47
Codes\\HFACS : Communication Sample for NVIVO	89.54	10.47
Codes\\ACCERS\Human Factors\High workload & task saturation	98.92	1.09
Codes\\ACCERS\Human Factors\High workload & task saturation : Communication Sample for NVIVO	98.92	1.09
Codes\\ACCERS\Weather or Environment\Icing	99.81	0.19
Codes\\ACCERS\Weather or Environment\Icing : Communication Sample for NVIVO	99.81	0.19
Codes\\ACCERS\Human Factors\Illness or injury - physical health	100	0
Codes\\ACCERS\Human Factors\Illness or injury - physical health : Communication Sample for NVIVO	100	0
Codes\\ACCERS\Human Error\Improper use of autopilot, FMS, FMC navigational equipment	99.4	0.6
Codes\\ACCERS\Human Error\Improper use of autopilot, FMS, FMC navigational equipment : Communication Sample for NVIVO	99.4	0.6
Codes\\HFACS\Supervisory Factors\Inadequate Supervision	97.53	2.48
Codes\\HFACS\Supervisory Factors\Inadequate Supervision : Communication Sample for NVIVO	97.53	2.48

Codes\\ACCERS\Organizational Factors\Inadequate training	99.36	0.64
Codes\\ACCERS\Organizational Factors\Inadequate training : Communication Sample for NVIVO	99.36	0.64
Codes\\ACCERS\Human Error\Inadvertent or intentional disregard for policy or procedure	97.45	2.56
Codes\\ACCERS\Human Error\Inadvertent or intentional disregard for policy or procedure : Communication Sample for NVIVO	97.45	2.56
Codes\\TEM\UAS Types\Incorrect Aircraft Configuration	99.82	0.19
Codes\\TEM\UAS Types\Incorrect Aircraft Configuration : Communication Sample for NVIVO	99.82	0.19
Codes\\ACCERS\Policies or Procedures\International policies or procedures	100	0
Codes\\ACCERS\Policies or Procedures\International policies or procedures : Communication Sample for NVIVO	100	0
Codes\\ACCERS\Human Factors\Interruption - distraction	98.68	1.31
Codes\\ACCERS\Human Factors\Interruption - distraction : Communication Sample for NVIVO	98.68	1.31
Codes\\ACCERS\Policies or Procedures\Lack of policy or procedure	99.52	0.49
Codes\\ACCERS\Policies or Procedures\Lack of policy or procedure : Communication Sample for NVIVO	99.52	0.49
Codes\\ACCERS\Airspace or ATC\Late ATC clearance	99.89	0.12
Codes\\ACCERS\Airspace or ATC\Late ATC clearance : Communication Sample for NVIVO	99.89	0.12
Codes\\ACCERS\Weather or Environment\Low visibility - low ceiling	99.78	0.23
Codes\\ACCERS\Weather or Environment\Low visibility - low ceiling : Communication Sample for NVIVO	99.78	0.23
Codes\\TEM\Aircraft Handling Errors\Manual Flying	99.96	0.04
Codes\\TEM\Aircraft Handling Errors\Manual Flying : Communication Sample for NVIVO	99.96	0.04
Codes\\TEM\Airline Threats\Manuals & Charts	99.74	0.27
Codes\\TEM\Airline Threats\Manuals & Charts : Communication Sample for NVIVO	99.74	0.27
Codes\\ACCERS\Human Factors\Medication side effect	100	0
Codes\\ACCERS\Human Factors\Medication side effect : Communication Sample for NVIVO	100	0
Codes\\ACCERS\Human Error\MEL & AML error	99.66	0.34
Codes\\ACCERS\Human Error\MEL & AML error : Communication Sample for NVIVO	99.66	0.34
Codes\\HFACS\Preconditions for Unsafe Acts\Condition of Operators\Mental State	96.62	3.37
Codes\\HFACS\Preconditions for Unsafe Acts\Condition of Operators\Mental State : Communication Sample for NVIVO	96.62	3.37
Codes\\ACCERS\Human Error\Misapplication of flight controls	99.89	0.11
Codes\\ACCERS\Human Error\Misapplication of flight controls : Communication Sample for NVIVO	99.89	0.11
Codes\\ACCERS\Organizational Factors\Mismanagement of assets	99.93	0.08
Codes\\ACCERS\Organizational Factors\Mismanagement of assets : Communication Sample for NVIVO	99.93	0.08
Codes\\ACCERS\Human Factors\Motivation	100	0

Codes\\ACCERS\Human Factors\Motivation : Communication Sample for NVIVO	100	0
Codes\\ACCERS\Weather or Environment\Nighttime operations	99.93	0.08
Codes\\ACCERS\Weather or Environment\Nighttime operations : Communication Sample for NVIVO	99.93	0.08
Codes\\ACCERS\Policies or Procedures\Old vs. new policy or procedure	100	0
Codes\\ACCERS\Policies or Procedures\Old vs. new policy or procedure : Communication Sample for NVIVO	100	0
Codes\\ACCERS\Organizational Factors\On-time performance pressure	99.48	0.53
Codes\\ACCERS\Organizational Factors\On-time performance pressure : Communication Sample for NVIVO	99.48	0.53
Codes\\HFACS\Organizational Influences\Operational Process	96.85	3.15
Codes\\HFACS\Organizational Influences\Operational Process : Communication Sample for NVIVO	96.85	3.15
Codes\\HFACS\Organizational Influences\Organizational Culture	98.54	1.46
Codes\\HFACS\Organizational Influences\Organizational Culture : Communication Sample for NVIVO	98.54	1.46
Codes\\TEM\Procedural Errors\Other Procedural	98.42	1.57
Codes\\TEM\Procedural Errors\Other Procedural : Communication Sample for NVIVO	98.42	1.57
Codes\\ACCERS\Weather or Environment\Perceptual illusion - spatial disorientation	100	0
Codes\\ACCERS\Weather or Environment\Perceptual illusion - spatial disorientation : Communication Sample for NVIVO	100	0
Codes\\ACCERS\Human Factors\Personal attitude(s) toward safety	99.51	0.49
Codes\\ACCERS\Human Factors\Personal attitude(s) toward safety : Communication Sample for NVIVO	99.51	0.49
Codes\\HFACS\Preconditions for Unsafe Acts\Condition of Operators\Physical & Mental Limitations	100	0
Codes\\HFACS\Preconditions for Unsafe Acts\Condition of Operators\Physical & Mental Limitations : Communication Sample for NVIVO	100	0
Codes\\ACCERS\Organizational Factors\Pilot selection	100	0
Codes\\ACCERS\Organizational Factors\Pilot selection : Communication Sample for NVIVO	100	0
Codes\\TEM\Communication Errors\Pilot to Pilot	96.59	3.42
Codes\\TEM\Communication Errors\Pilot to Pilot : Communication Sample for NVIVO	96.59	3.42
Codes\\HFACS\Supervisory Factors\Planned Inappropriate Operations	98.09	1.92
Codes\\HFACS\Supervisory Factors\Planned Inappropriate Operations : Communication Sample for NVIVO	98.09	1.92
Codes\\ACCERS\Organizational Factors\Rest, duty, flight times	100	0
Codes\\ACCERS\Organizational Factors\Rest, duty, flight times : Communication Sample for NVIVO	100	0
Codes\\HFACS\Unsafe Acts\Violations\Routine Violations	98.98	1.02
Codes\\HFACS\Unsafe Acts\Violations\Routine Violations : Communication Sample for NVIVO	98.98	1.02
Codes\\ACCERS\Human Factors\Self-induced time pressure	99.63	0.38
Codes\\ACCERS\Human Factors\Self-induced time pressure : Communication Sample for NVIVO	99.63	0.38

Codes\\ACCERS\Human Factors\Situational awareness	98.99	1.02
Codes\\ACCERS\Human Factors\Situational awareness : Communication Sample for NVIVO	98.99	1.02
Codes\\HFACS\Unsafe Acts\Errors\Skill-Based Errors	95.84	4.17
Codes\\HFACS\Unsafe Acts\Errors\Skill-Based Errors : Communication Sample for NVIVO	95.84	4.17
Codes\\TEM\Procedural Errors\SOP Cross-verification	100	0
Codes\\TEM\Procedural Errors\SOP Cross-verification : Communication Sample for NVIVO	100	0
Codes\\ACCERS\Airspace or ATC\TCAS RA & TA	99.85	0.15
Codes\\ACCERS\Airspace or ATC\TCAS RA & TA : Communication Sample for NVIVO	99.85	0.15
Codes\\TEM	89.46	10.54
Codes\\TEM : Communication Sample for NVIVO	89.46	10.54
Codes\\ACCERS\Weather or Environment\Thunderstorm or severe weather	99.7	0.3
Codes\\ACCERS\Weather or Environment\Thunderstorm or severe weather : Communication Sample for NVIVO	99.7	0.3
Codes\\ACCERS\Weather or Environment\Turbulence	99.96	0.04
Codes\\ACCERS\Weather or Environment\Turbulence : Communication Sample for NVIVO	99.96	0.04
Codes\\ACCERS\Airspace or ATC\Unclear ATC clearance	98.35	1.66
Codes\\ACCERS\Airspace or ATC\Unclear ATC clearance : Communication Sample for NVIVO	98.35	1.66
Codes\\ACCERS\Weather or Environment\Wind shear	99.93	0.08
Codes\\ACCERS\Weather or Environment\Wind shear : Communication Sample for NVIVO	99.93	0.08